The Sweep-Rule Strikes Back

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October 19, 2026

Abstract

We suggest a new proof of quantum fault tolerance based on the alternative proofs of Toom's theorem [BS88] (see also [Gac21]). The fault tolerance construction is relative to the circuit model, subjected to local stochastic noise, where at each time unit only a constant depth of non-noisy classical computation is allowed. Our main error correction code is the (d, d')-Toric code, namely qubits are set on the d'-dimensional faces of the d-dimensional Toric. As byproducts of the study, we provide and prove the following:

1. An explicit description of the sweep-rule —a generalization of Toom's rule for topological codes—for any dimension greater than two. In doing so, it answers affirmatively to the conjecture made by Kubica and Preskill [KP19].
2. A fault tolerant construction in the circuit model that both uses the (d, d')-Toric code and has a constant-depth correction cycle, where $d', d - d' \geq 2$. Until now, the use of a topological code as the primary encoding and a constant-depth correction cycle were only achieved separately; this is the first work to achieve them simultaneously.
3. A proof that the sweep-rule is an effective decoder against locally stochastic noise. Concurrently, denote by n the side length of the d-dimensional Toric. If the noise rate is below a certain threshold, then an application of $\Theta(n^{2d+1})$ ideal sweep-cycle iterations erases the error with high probability. As far as we know, until now rigorous proofs for the sweep-rule were given only in the i.i.d setting [KP19].

Specifically, all these results hold for the 4D-Toric code.

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1 Introduction

1.1 Background

Fault-tolerant constructions take ideal quantum circuits and establish their logical equivalence such that they endure fault occurrences. Their efficiency is typically measured by the amount of additional resources – qubits, computation time, range connectivity, etc. – required by the resulting circuit, beyond the scope of theoretical discussion; those parameters translate into the machine complexity and energy cost required to run the circuits. Minimizing overheads brings quantum computing applications closer to the near future.

Since the introduction of the quantum threshold theorem [AB99], which showed that quantum fault tolerance is in principle feasible, the main effort in subsequent work has focused on reducing the additional qubit overhead. A line of work has led to a constant-factor overhead in qubits, first in the instant-classical computation model [Got14], then in the standard circuit model [Gro19], [FGL18], [NP25]. Yet, although the best qubit overhead was reached, the depth overhead was only slightly improved from poly-logarithmic to 'tilde'-logarithmic [NP25]. In fact, it seems that the same means that are used to reduce the qubit overhead are the ones that challenge reducing the depth overhead.

A key concept for reducing qubit overhead is to encode qubits with a high-rate quantum LDPC code. For those codes, except for the control-not, there are no known gates that can be realized in constant depth; furthermore, since the structure of the codes is relatively complicated – based on the skeletons of expander graphs - it seems hard to look for realizations of their logical operators. Without a constant-depth realization of logical operators, any construction that doesn't interfere with the order of the logical operations would admit a non constant-depth overhead.

Contrary to high-rate LDPC codes, topological error codes have a geometrical image that is well understood and, as a result, a wide range of works suggest efficient realization of their logical operators. For example, consider the 3D-color code [Bom15], fold-transversal gates on the surface code [BB24], and realization of the Clifford group on the 2D Toric [GV24].

Another branch of work, whose primary goal was to show that the embedding dimension of the computation can be reduced to 2D [BDL26], and more recently [DST26], uses topological codes as the main error correction code. Those constructions simplify the hardware realization; they suggest that quantum computers can be implemented by nearest-neighbor interactions automata - what is believed to be easier to develop. Yet, as in the threshold theorem [AB99], and in contrast to the works that use high-rate LDPC codes, they rely on a hierarchical correction cycle whose execution time grows with system size. Hence, even though those directions have the potential to overcome the non-constant depth overhead per gate, it seems that the correction cycle imposes a barrier for them to come up with a constant factor in depth-overhead construction.

1.2 Stating Our Results

In this work, we present a fault-tolerance construction that uses the (d, d')-Toric code and has a constant-depth 'correction' cycle, for any $d', d - d' \geq 2$. In other words, the construction uses a relatively simple code - the 4D Toric code, yet doesn't incur the cost of a hierarchical correction cycle.

The computation model we consider is quantum circuits subject to *local-stochastic noise*, where only qubits are subjected to the noise, while classical bits are non-noisy. Classical computation is allowed, and the classical depth matters. Namely, at each time unit, only a constant depth of classical computation can be done. Local stochastic channel is defined to act such that the probability of any subset of locations S (space-time coordinates), which is picked prior to the channel's action, being affected decays exponentially with the subset size.

Definition 1.1 (Local stochastic noise channel)**.** *Let $p \in (0, 1)$. We say that the noise channel $\mathcal{N}_\mathcal{E}$ is local stochastic with rate p if for every finite set of locations $S \subset \mathbb{N} \times \mathbb{N}$,*

$$\mathbf{Pr}_{\mathcal{E} \sim \mathcal{N}_\mathcal{E}} \left[S \subset \mathrm{Loc}(\mathcal{E}) \right] \leq p^{|S|}.$$

For circuits $\tilde{C}$ and C, we say that $\tilde{C}$ simulates C with probability p if, when one stops a noisy run of $\tilde{C}$ at a midpoint, one can, with probability p, recover by non-noisy operation an 'image' of a midpoint in the running of C, and if stopping at the last time point of $\tilde{C}$ enables recovery of the 'image' of C at the end of its run. We state our main theorem as follows:

Theorem (Informal version of Theorem 6.3)**.** *There exists a constant $p_0 \in (0, 1)$ such that for any local stochastic noise channel $\mathcal{N}_\mathcal{E}$ with rate $p < p_0$, a quantum circuit C with depth μ and width w, and a big enough integer n, there exists a 'logical equivalence' circuit $\tilde{C}$ with depth $\Theta(\mu n^5)$ and width $\Theta((w + \mu) n^8 \mathit{Polylog}(n))$ that simulates C with probability:*

$$1 - O\left(\mathit{poly}(w, \mu, n)\right) p^{\Omega(n)}$$

Furthermore, $\tilde{C}$ has the following two properties:

1. *After a short initialization stage (at depth $\Theta(\mathit{Poly}(\log n))$), each of the qubits is encoded in the n-length 4D Toric code.*

2. *An error correction cycle takes $O(1)$ depth, and the gates in each correction cycle don't depend on time. In particular, if C is the identity circuit, then after the initialization stage, the layers in $\tilde{C}$ are the same constant-depth error reduction procedure.*

As mentioned earlier, both code simplicity and error correction cycles at constant depth have already been achieved: the former as part of the quantum automation line of work, and the latter through the use of high-rate error correction codes. However, this is the first construction to achieve both simultaneously.

Sweep-rule in High Dimensions. The correction cycle in our construction consists of local operations done in parallel. Concurrently, any vertex of the Toric is thought of as a computation unit that has to decide on a correction to be applied over its local neighborhood. The specific procedure it runs is the sweep-rule, a generalization of Toom's rule invented by Kubica and Preskill [KP19]. The sweep-rule can be sketched informally as follows: The vertex looks for a subset of bits to flip on its 'first quarter'[1], such that the flipping erases the syndrome that is seen on the 'first quarter' and all the bits to be flipped are supported on vertices that support the local syndrome. If there is such a subset of bits, then the vertex flips the bits. In [KP19], it is argued that for small dimensions, any vertex that sees only syndrome on its 'first quarter' has a subset of bits to be flipped that satisfy the sweep-rule requirement. Moreover, the authors conjectured that this is also true for higher dimensions. We prove their conjecture; furthermore, we give an explicit algorithm for finding the subset. We refer to the outline section for an elaborated discussion.

Sweep-rule as Ideal Decoder Against Local Stochastic Noise. The sweep rule was initially introduced as a local decoder with a provable threshold [KP19] against independent noise. Our final contribution is extending that result to local stochastic noise. We prove the following theorem:

Theorem (Informal version of Theorem 7.5). *There exists a constant $p_0 \in (0,1)$ such that for any local stochastic noise channel $\mathcal{N}_\mathcal{E}$ with rate $p < p_0$, and a quantum state $\rho = \mathcal{N}_\mathcal{E}(|\psi\rangle)$, where $|\psi\rangle$ is a codeword of the (d,d')-Toric code, the application of $2dn^{2d}$ (ideal) sweep-cycles on ρ results in $|\psi\rangle$ with probability at least:*

$$1 - O\left(poly(n)\right) p^{\Omega(n)}$$

1.3 Discussions, Applications, and Further Directions

Our result can be understood as follow: when the qubit overhead is not a consideration, it means that the 4D Toric code is might be as good as the good quantum LDPC / Hyperproduct code. That translates to the two following applications:

1. New approach for quantum fault tolerance: Our proof is inherently different from prior works, describes in higher resolution the expected error, and might be used as a springboard for a constant depth overhead construction.

2. Memory/Storage components: Protecting a 4D Toric encoded state for an exponentially long time is achieved by performing the same operation at each time unit, suggesting that the 4D Toric code is a promising candidate for memory in the circuit model. Combined with the fact that the 4D Toric code also functions as memory in thermal noise models [Ali+08], and given its relative simplicity, it appears to be a natural choice for real-world memory applications. We emphasize that an experimental usage of codes in four dimensions has already been demonstrated [Aas+25], [Ber+24], [Rei+24].

[1] or the positive direction

2 Outline of the Proof

In studying fault tolerance, the typical method is to first distinguish between configurations that can be tolerated and those that cannot. For example, in von Neumann-style proofs [Neu56], it is common to track the number of flipped bits; if fewer than half are flipped, the logical states can be recovered. In the work of Aharonov and Ben-Or [AB99], a logical bit at the n level admits a logical error if at least a constant fraction of the $n-1$ level logical bits that compose it admit an error. In the work of [Gro19], a tolerated configuration is one in which all encoded registers can be considered as distributed according to sufficiently weak local stochastic noise. The next step is to show that from a tolerated configuration, we move with high probability to another tolerated configuration. For our goals, each approach presents a challenge: The concatenation method, by definition, requires non-trivial depth computation at each step. Tracking the number of flipped bits fails almost immediately, as topological error correction codes have sublinear distance when a linear number of bits is expected to be flipped. And preserving the local stochastic noise distribution seems to heavily rely on the expansion properties [Gro19] of the underlying structure, where for topological codes, we expect to forge a 'spatial' dependence.

In contrast, Berman and Simon provided a proof that, instead of tracking the dynamics, considers the entire run as a possible outcome, ruling out improbable runs only at the end [BS88]. We outline their idea informally as follows: a logical bit is represented by $n \times n$ bits set on the vertices of the grid. In each correction cycle, the bits determine their value by taking the majority of their current value, and their north and east neighbors values, a rule known as Toom's rule. For simplicity, let's ignore the logical operations. Now, given a run, and in particular the set of space-time locations of faults, consider the following directed graph $\mathcal{H} := (V_{\mathcal{H}}, E_{\mathcal{H}})$ induced by it:

1. If at time t, the bit at vertex v diverges relative to the ideal run, then add (v, t) to the vertices.

2. If at time t, vertex v 'leads' one of its neighbors u to diverge, then connect (v, t) and $(u, t+1)$ by an 'arrow'.

3. Connect any two vertices in $V_{\mathcal{H}}$, which are connected to a third by 'arrows', by a 'fork'.

Notice that the vertices of $\mathcal{H}$ with no incoming arrows correspond to faults; let's call them leaves. Thus, the probability of each running can be bounded by the number of leaves in the induced graph. Their key lemma states that a cluster of flipped bits with a 'perimeter' s, connected by arrows, induces a subgraph with at least $\Theta(s)$ leaves[2]. In other words, the probability of a connected component (by arrows) being flipped decays exponentially with its perimeter.

The graph $\mathcal{H}$ was later named Toom's contours, a term that, to the best of our knowledge, was coined in [SST26]. We chose to follow their approach. The rest of the section reviews the challenges that we encountered on the way to generalize it and the ideas to overcome them.

[2] And number of edges $\leq \Theta$(leaves).

The Toric Code, the Sweep-Rule, and the Infinite Plane. The first issue was finding an analog for Toom's rule. We decided to adopt the sweep rule invented by Preskill and Kubica in [KP19], which they introduced as a local rule for any CSS code induced by a simplicial Toric, equipped with a notion of 'positive' and 'negative' directions (in their paper 'future' and 'past'). They also provided a rigorous proof that the iterative non-noisy applications of the sweep rule succeed in decoding faults that were sampled independently. For our purposes, we had to extend their proof for cubical Toric[3]. The main reason for this is that the alignment of the positive/negative directions with the axes direction of $\mathbb{R}^d$ space allows quantifying the 'perimeter' of error clusters. Yet, we are sure that our proof can be adjusted to support simplicial Toric complexes as well, and we leave it for further work.

In complete contrast to typical works in topology, where global properties are studied and local characterization is left as general as possible, here we need notations to describe the local view of each vertex of the complex. We needed it for defining the sweep-rule action and stating claims about how it shrinks errors. That led us to the $\theta_{\mathbf{v},\mathbf{S}'}$ notation defined in detail in Definition 4.1. In general, we denote by S the generator set of the Toric, and for a subset of generators $S' \subset S$, we define $\theta_{\mathbf{v},\mathbf{S}'}$ to be the face whose 'bottom-left' corner is the vertex v, and sides of the face correspond to edges on paths obtained by stepping through subsets of S' (for example, if $S' = \{g_1, g_2, g_3\}$ then $\{g_2 g_1 v, g_3 g_2 g_1 v\}$ is an edge of $\theta_{\mathbf{v},\mathbf{S}'}$). Figure 1 shows how the 2D-Toric complex is characterized via $\theta_{\mathbf{v},\mathbf{S}'}$ notation.

[3]The standard Toric codes.

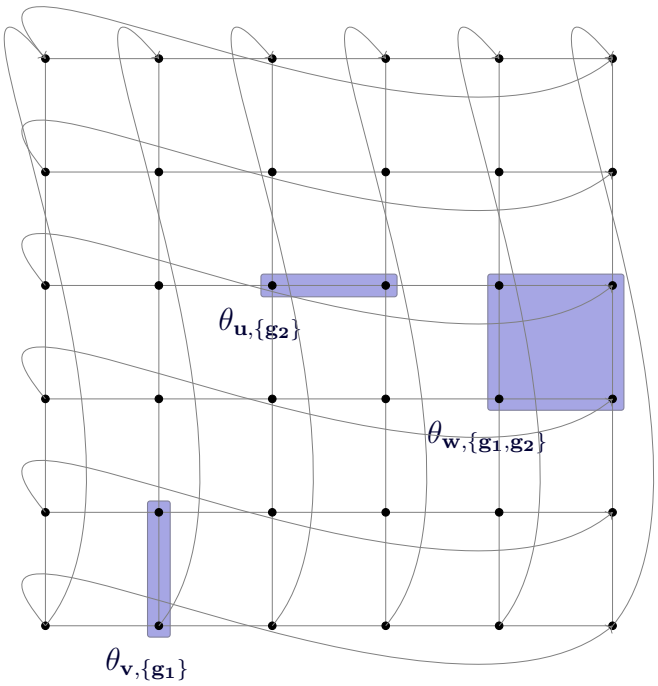


Figure 1: Here we consider a 2D Toric complex. We name the set of generators $S = \{g_1, g_2\}$, where g_1 is the generator that sends us one step in the north direction, and g_2 sends us in the east direction. For vertices v, u, w, we mark in blue three faces that are rooted at them.

The 4D-Toric code is defined as the binary assignment over the 2D-faces (plaquettes), where the Z-stabilizers (parity checks) are set on the edges, namely restrict the Z product of the faces adjacent to each edge, and the X-stabilizers (phase checks) are set on the 4D-faces (hypercubes). Similarly, the (d, d')-Toric code is defined by associating the d'-dimensional faces of the d-dimensional Toric code with the qubits, while associating the $d' \pm 1$-dimensional faces with the X/Z-stabilizers.

To present the sweep-rule, we need to have a definition of directions. For this section, we will be satisfied with the following hand-waving: For a vertex v, the bits and checks on the first 'quarter' of it (high-dimensional quarter) will be named the positive bits and checks of v.

Definition 2.1 (Sweep Rule (Inforaml))**.** *Let z be a binary assignment on the Toric faces representing 'error'. The sweep rule decoding procedure at a vertex v is defined as follows:*

Measure the positive checks of v and denote the outcome by σ. Next, look for an assignment $\tilde{z}$ on the positive bits of v such that the vertex support of $\tilde{z}$ is contained in

the vertex support of σ and that the positive checks of v assign to $\tilde{z}$ the same syndrome as σ. If there is such, flip $\tilde{z}$.

See Definition 4.6 for the formal version. A full error correction cycle, or a sweep-cycle, consists of applying the sweep-rule to each vertex and then doing the same in the dual space. Figure 2 provides an example of the result of the sweep rule given a syndrome.

To see the effectiveness of the sweep rule, consider an error that does not close a non-trivial loop on the Toric. Such errors have 'poles', and in particular, one can think of the most 'negative pole'. In the 2D case, it is the most 'bottom-left' vertex that supports the error. For simplicity, let's call that vertex the corner of the error. In [KP19], it was argued that, at the end of a sweep-rule execution, the corner always suggests a non-trivial correction. Thereby advancing the boundary of the error toward shrinking. That argument is easy to verify for low dimensions; for higher dimensions, the authors left it as a conjecture. In this work, we prove that conjecture and, from the proof, derive an algorithm for finding the correction. Nevertheless, the proof doesn't distinguish between simplicial or cubical complexes.

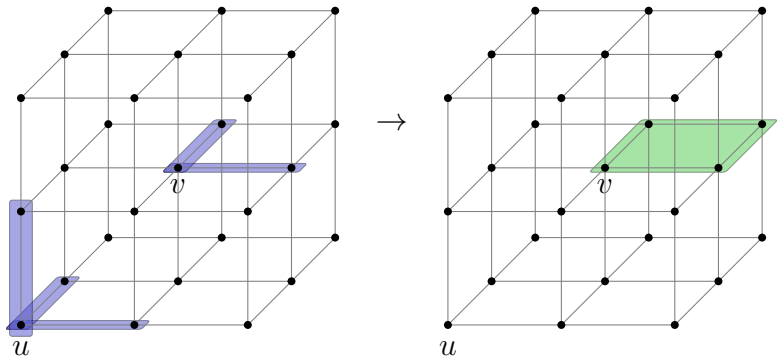


Figure 2: An example of applying the sweep rule within the $(3, 2)$-Toric code. In the $(3, 2)$-Toric code, bits are placed on the squares and Z-stabilizers (parity checks) on the edges. The view is restricted to two vertices, u and v. On the left, the unsatisfied positive checks of u and v are highlighted in blue. On the right, the bits to be flipped are marked in green. Given the syndrome, vertex v flips one of its positive faces, whereas for u, there is no subset of positive faces that, when flipped, decreases the number of local unsatisfied checks. Therefore, u flips no bit.

The way we quantify the 'perimeter' of an error cluster is by considering the (hyper)triangle that bounds the vertices supporting the cluster. Now, the poles of an error cluster are the vertices that lie on the sides of the triangle. We followed [BS88], and by embedding the Toric in the infinite plane, we were able to identify the poles as the max/min points of linear functions which we call *potential walls*. Then, we show that the sweep rule advances only one pole - matching the corner mentioned earlier. And hence, advances one side, forward the others. As a consequence, a sweep-cycle decreases the sides of the bounding triangle, and hence the 'perimeter'.

The embedding restricts periodicity conditions on the assignments. We encounter this twice: first, an assignment on the Toric's faces that closes a non-trivial cycle

translates to an infinite assignment in the infinite plane. In that case, no triangle bounds the error, which agrees with the fact that a logical error—in that context, an error that changes the logical states—should be 'decoded' to a different codeword. From a technical perspective, it comes into effect as the claims about shrinking assume that a given assignment is a connected finite assignment (or a collection of connected components such that each of them is finite). The second time we have to deal with the mapping onto the infinite plane is in the calculation of the probability of a given Toom's contour to occur. Since different leaves (faults) of Toom's contour might be associated with the same qubit[4] and hence dependent, to bypass this, we took from [Gac21] the claim that states any such graph has a subgraph whose leaves are independent, and their number is at least a constant factor of the original leaves.

[4]Actually, the leaves are vertices that support the face associated with a faulty qubit.

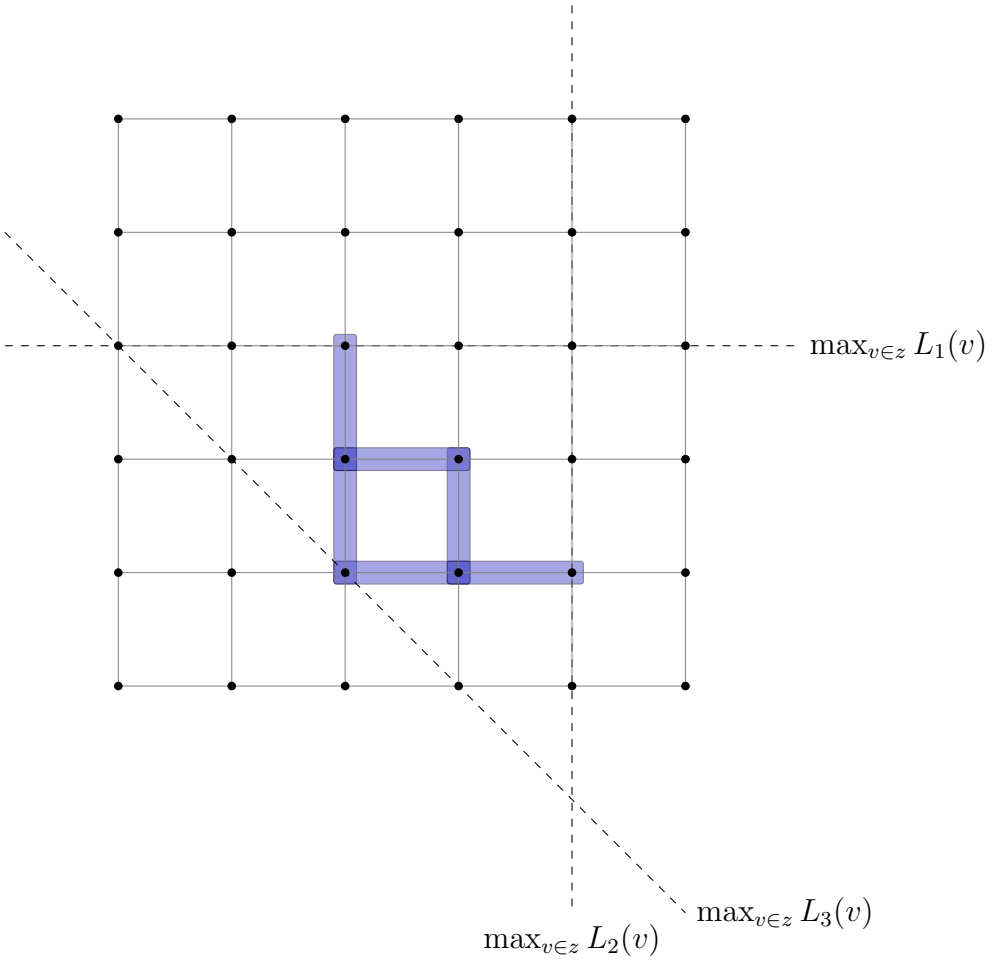


Figure 3: A graphical example of the maximal values of the potential walls. Here we consider the $(2,1)$-Toric code, namely the bits are set on the edges where the Z-stabilizers (parity checks) are set on the vertices. The grid is a finite 6×6 patch of the infinite 2D plane. The edges marked in blue correspond to a bits assignment z. Each of the dashed lines marks the line of vertices whose L_i value converges with the maximal value reached by a vertex in z. Moving into the infinite plane enables us to determine those lines and draw the triangle that bounds z.

See Section 4.2 for embedding into the infinite plane, Section 4.4 for the proof of shrinking, and Section 8.2 for the claim about the existence of a sufficient subgraph.

The Intermediate Argument. Now that we have the code and decoder, the next difference we have to address is how to classify an error. Unlike the classical regime, flipping a stabilizer's qubits doesn't count as a fault. Moreover, the sweep-rule discussed above doesn't shrink stabilizers (as it acts trivially on them). Therefore, we need to redefine what is considered a divergence from the ideal computation. To overcome it, we define the reduced form of an error to be its translation by stabilizer that minimizes the 'perimeter' of the bounding triangle. Then we prove that the poles

of the error in it's reduced coalesce with the poles of the syndrome it induces. In particular, the proof that the sweep-rule shrinks an error (Section 4.4) assumes that the given error is in its reduced form. As a result, instead of defining the vertices of Toom's contour graph to be the flipped vertices as in [BS88], we define them to be qubits[5] in the reduced form of the error.

We then face another challenge: if we analyze Toom's contour, which is induced by the single time unit running consisting of flipping all the qubits at once, then in the layer following the fault cycle, even though the error contains many qubits, its reduced form is empty. In the language of Toom's contour, as will be defined later, the Toom's contour is empty, and no bounds on the event's probability to occur can be inferred using it. More generally, there is a threshold, such that for errors exceeding that threshold, one cannot restore the history of the error. In fact, the threshold is far lower than flipping all the qubits and matching errors that close a non-trivial cycle.

We address this by introducing an analysis framework where all computation, including faults, is imagined to occur sequentially — in steps. Within this framework, we can prove an *intermediate claim* stating that on the path to an error configuration closing a non-trivial cycle, one must pass through a step with an error where the 'perimeter' its reduced form is large. Thus, a sufficient condition for a logical error to occur in some time unit is the existence of a step with a reduced error that is large. Moreover, having such a step implies the existence of a step for which it happens for the first time, meaning that prior to that, no error has closed a non-trivial cycle, the sweep-rule worked as it should, and a Toom's contour describing the history of the error can be constructed. Thus, an upper bound on the probability of the condition can be derived. In the proof of the theorem, we show that this condition happens with negligible probability. To elaborate on the issue, we first need to present the computation model.

We model a noisy running of a given circuit C as follows: first, we sample space-time locations for the faults and denote the outcome by $\mathcal{E}$. Then, we impose layers of computation alternately, such that the even and odd layers are taken from $\mathcal{E}$ and C according to their order. The combination of the circuit C with a subset of faults $\mathcal{E}$ defines a possible running, and we call it a subjection of C to $\mathcal{E}$ and denote it by $C[\mathcal{E}]$. Furthermore, we showed in **??** that for a local-stochastic noise model, unbalancing partition to noisy and logical layers, for example placing a noisy layer after every three logical layers of C, is equivalent to odd-even alternations up to a rescaling of the noise rate (by a constant). We use that since the error correction cycle of the sweep rule takes more than a single layer.

As mentioned, we are going to analyze the likelihood of getting logical errors by looking back in time and asking what a run that has led to them is supposed to look like. The sweep rule ensures that an error cluster below a certain size, sufficiently low to not closing a non-trivial cycle, shrinks. The problem is that a single fault layer (i.e., a layer whose source is in $\mathcal{E}$) might increase the error cluster above that size. At first glance, one might be tempted to show that the expansion of error clusters beyond a fraction of the code distance happens with negligible probability, but that calculation

[5] In fact: vertices on the qubits' support, see Section 5.2.

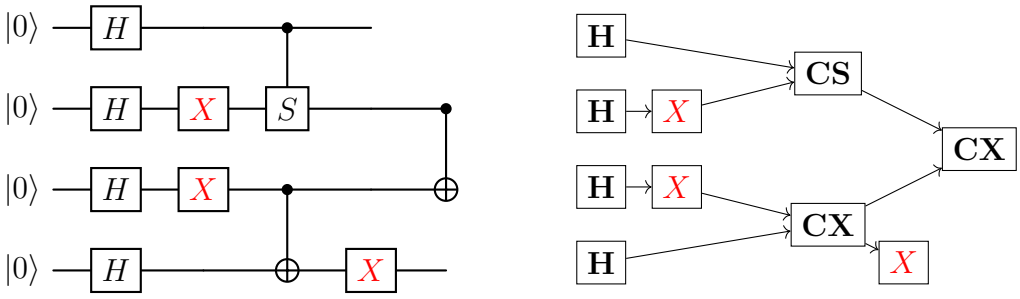


Figure 4: Example of a subjected quantum circuit: the even layers (0, 2, 4) contain the original gates, while the faults are imposed on the odd layers and marked in red.

soon becomes complicated when considering the merging of disjoint clusters. Here comes the splitting of the computation into steps to advantage. In each step, only one gate or one fault is applied, and steps are ordered such that they agree with the computation. Figure 4 illustrates a quantum circuit and a graph of steps corresponding to it.

Since the gates and the faults have a bounded fan-in/out[6] In each step, the maximal size of an error cluster can increase by at most a constant factor, with the maximal increase occurring when multiple clusters are merged. Equipped with that insight, we are able to infer the intermediate argument.

Section 3 formalize the computation model and the computation graph presentation (splitting to steps). In addition, see **??** for the formal proof of the intermediate argument.

Description of the Construction. Similarly to other fault tolerance approaches, we use the concatenation construction [AB99] in order to initialize zeros and resource states encoded in the 4D-Toric code, as referred to in [Gro19], [Got14]. From there, we assume that all the quantum registers are encoded in the 4D-Toric code, and we name each of the encoded block a quantum register. The resources states are spread across two registers.

Then, after the initialization step, the logical cycles are divided into error correction, which is a single sweep-cycle, and 'logical operation' cycle. For our universal logical gate set, we choose $\mathbf{X}, \mathbf{Z}, \mathbf{CX}$ (controlled NOT), $\mathbf{H}$, $\mathbf{S}$, and $\mathbf{T}$ gates. Except for the Paulis and the 'Controlled NOT' gate, which are realized by transversal Paulis and transversal wire-wise physical 'Controlled NOT' gates, all the others are implemented via gate teleportation [ZLC00].

The general realization by gate teleportation is done as follows: first, we XOR the target state into a resource state via $\mathbf{CX}$, measure the input state and the first register of the resource state with a Bell measurement, the input register in the X basis, and the first register of the resource state in the Z basis. Then, on the classical outcomes,

[6]The faults are bounded-fan because we can assume that in the time units representation they are Pauli products and therefore in the steps presentation they are decomposed into single qubit Pauli matrices.

run a classical decoding procedure to extract the logical codeword. Meanwhile, on the second register of the resource, which is intended to contain the result of the action, perform a sweep cycle. At the end of the decoding, depending on the result, we apply either a transversal Pauli operator or the **S** gate (again by teleportation) depending on the original gate that was asked to simulate. The gate teleportation lasts for multiple logical cycles, depending on the depth required by the classical decoding procedure.

We emphasize that for measuring the input register in the X-basis, we apply the Hadamard through its fold-transversal realization [BB24], so the measurement is done in constant depth. However, the reason we don't implement the Hadamard by fold-transversal from the beginning is that it introduces an expansion in the bounded-triangle by the fold-transversal realization. Since it guarantees that after the measurement the register will not observe more faults, we can tolerate a one-time blowup in the size of the error and still be able to decode the logical state in the register correctly. Yet, for the Clifford-free depth overhead, we do use the fold-transversal implementation followed by sufficiently many followed sweep-cycles. As a result, we stretch the depth of the basic logical layer, which in turn comes into effect by a lower threshold for the noise rate.

In Section 4.5, we discuss in detail the structures of the logical codeword of the (d, d')-Toric code and the realization of its logical gates.

Toom's Contours. A sufficiently simple definition of Toom's contours in [BS88] would be a directed graph that describes a noisy run with the additional marking of diverged bits. In our case —the quantum case— the vertices of Toom's contour are neither bits nor checks; instead, they are the vertices on the Torices that 'diverged'. A proper conceptual definition would be:

> *A graph that describes a noisy run from the Torices' perspective. In particular, the vertices of the graph are the vertices of the Torices, equipped with a time coordinate, that witness a faulty qubit in a reduced form.*

We adjust their construction as follows: Denote the graph by $\mathcal{H} := (V_{\mathcal{H}}, E_{\mathcal{H}})$ and by $P_1, .., P_t$ a *Pauli-path*, such that P_i is the propagated error up to time i (including faults that occur at that time), and each 'cluster' of error in P_i is in its reduced form. Then,

1. If at time t, a vertex v on one of the Toric complexes[7] corresponding to one of the quantum registers supports a face -qubit- that belongs to a reduced error, then add (v, t) to the vertices.

2. If at time t, vertex v belongs to a reduced error, and at time $t+1$ its neighbor u belongs to a reduced error, then we think of the situation as 'v "leads" to u' and connect (v, t) and $(u, t+1)$ with an "arrow." The vertices v and u might belong to different registers (which matches the case that at time t we apply an operator on both registers). For convenience, we don't require u at time t to belong only to non-faulty qubits.

[7] Actually the infinite grid.

3. As in the classical case, connect any two vertices in $V_{\mathcal{H}}$ that are connected to a third by 'arrows' with a 'fork'.

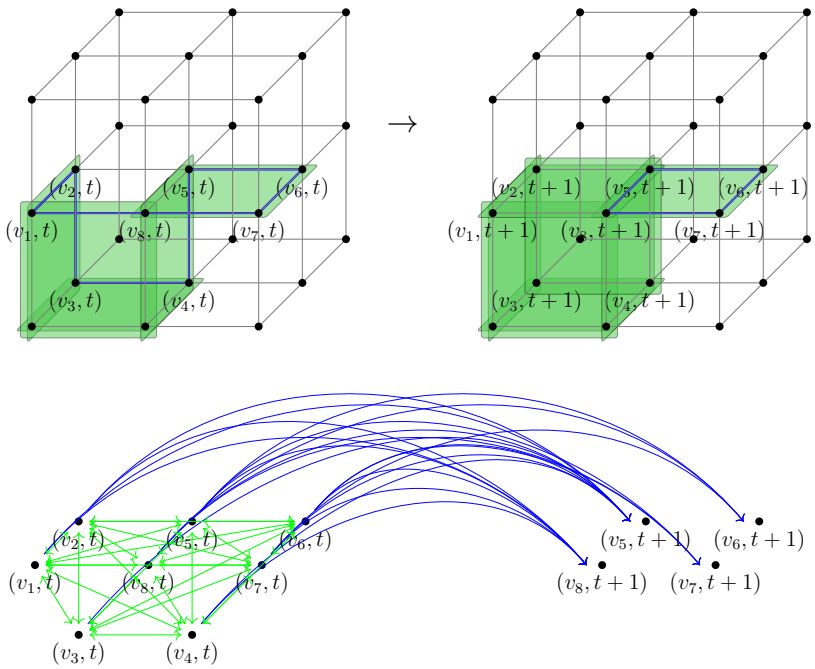


Figure 5: Above: The application of a sweep-cycle, where the green faces represent the bits assignment z, and the blue edges correspond to unsatisfied checks $\partial^- z$. The vertices in the reduced form are labeled as $v_1, v_2, \ldots, v_8$. Notice that the leftmost bottom-inner vertex is not in the reduced form even though it belongs to z. Below: The Toom's contour derived from the single computation layer that matches the cycle. Arrows are colored in blue, while forks are colored in green. Note that $(v_1, t+1)$ doesn't belong to the Toom's contour (since v_1 doesn't belong to a reduced error cluster at time $t+1$).

Later, we incorporate the 'splitting into turns' idea into the formation of Toom's counter $\mathcal{H}$-graphs, by gluing the layers corresponding to the first $t-1$ layers with a final layer that matches a partial execution of the tth time unit. The layer encodes the situation of the running up to a certain step. Then, combining with the intermediate theorem, we have that any running that admits a logical error derives a Toom's contour, with sufficiently large vertices at final layer.

The rest of the analysis follows [BS88]. We define *slices* at time t to be connected (by arrows) components of vertices at time t. Then, in the merging of two slices into one, it is guaranteed that faults induced by each are disjoint; otherwise, those two slices will be connected by arrows, namely - the same slice. For a *slice*, we define its size to be the perimeter of the triangle that would bound the vertices of the slice if they were projected on the same Toric. Then, by induction, we prove that Toom's contour that induces at time t a slice with size s has a spanning tree with at most and at least a constant times s edges and leaves (and therefore faults). That allows us

to bound the probability of such a slice occurring by counting the number of possible spanning trees and assigning them the probability that the minimal number of faults occurs.

One technical change that we made and haven't discussed yet is that in the gluing of the last layer, which was defined to capture a partial layer execution up to a turn, we connect by arrows any two vertices that were added to $V_{\mathcal{H}}$ and share a check. By doing so, we ensure that connected components in the bits-to-checks graph would also be connected by arrows in Toom's contours. That is necessary since, without the last gluing, an error cluster that closes a non-trivial loop on the Toric might consist of two 'connected by arrows' error clusters which are tangent. We emphasize that such edges connection can be made only in the last stage, as it doesn't pass the induction step. However, the final slice that is formed passes the desired inequalities only by constants and is therefore sufficiently good to derive the probability bound.

Section 5.2 defines Toom's contours, Section 5.3 defines the slices and states a claim connecting a slice to its predecessors, and Section 5.4 provides the induction proofs.

3 Notations and Preliminaries

We follow the standard notations commonly used in the literature. For an integer n, we denote by $[n]$ the set of integers $\{1, 2, \ldots, n\}$. To present a countable set, which might or might not be finite, where the number of elements does not matter, we use curly brackets with a subscript denoting the first index of the set. For example, $\{u_i\}_{i=1}$ represents the set $u_1, u_2, \ldots$. For finite sets, we might add a superscript to present the size of the set. If the first index is clear from the context, then we might omit the assignment. We use a colon inside curly brackets to specify a set of elements satisfying a condition, i.e., for sets A, B, the notation $\{a : a \in A, a \in B\}$ equals $A \cap B$. Curly brackets without subscripts or conditions refer to singletons. For example, for a given integer i, the set $\{u_i\}$ is the set that contains only the element u_i. The size of a set A is denoted by $|A|$. For an integer i, we denote by $\mathbf{1}_i$ the vector consisting of one at the ith coordinate and zero elsewhere. The char $|$ symbolizes a restriction, for example $x|_y$, is the restriction of the 'entity x' to 'y'. The particular behavior of the restriction would change depending on the target type.

When not otherwise mentioned, we consider quantum states over qubits, each lying in a two-dimensional Hilbert space $\mathcal{H}_2$, and denote by $\mathcal{H}_{2^n}$ the Hilbert space of an n-qubit system. The $\otimes$ operator stands for tensor product, for example: $\mathcal{H}_{2^n} = \mathcal{H}_2^{\otimes n}$. The space of linear operators on the Hilbert space is denoted by $L(\mathcal{H})$. Pure quantum states and their dual presentations are denoted by $|\psi\rangle, \langle\psi|$, and in the matrix representation, we use $|\psi\rangle\langle\psi|$ to denote its density matrix. We will use ρ to denote a mixed state. The action of an operator A on a mixed state is written as $A \circ \rho$. For example, if A is a unitary operator, then $A \circ \rho = A\rho A^\dagger$. The letters, X and Z, will usually denote the single qubit X and Z Pauli operators. Yet, where the Pauli matrices are out of context, we use these letters for other variables. The Pauli group itself, over n qubits, is denoted by $\mathcal{P}^{\otimes n}$, and when the number of qubits is either clear or doesn't matter,

we omit the superscript. For a Pauli $P \in \mathcal{P}$, we denote by $X(P)$ and $Z(P)$ the X- and the Z-components of P such that $P = X(P)Z(P)$. The gates 'Controlled Not' and 'Controlled Z' are denoted by **CX** and **CZ**, and the gates Hadamard, S, and T are denoted by:

$$\mathbf{H} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}, \mathbf{S} = \begin{bmatrix}1 & 0\\ 0 & i\end{bmatrix}, \mathbf{T} = \begin{bmatrix}1 & 0\\ 0 & e^{i\frac{\pi}{4}}\end{bmatrix}$$

For a quantum gate U, we denote by $|U\rangle$ the resource state that can be used to simulate U. Notice that, in general, the resource state is not unique. In this work, we address specific states that will be discussed later. The function $\mathbf{Pr}\big[\,\cdot\,\big]$ is the probability function. In the language of groups, for a group $\mathcal{G}$ and elements $g_1, \ldots, g_l$, we denote by $\langle\{g_1, \ldots, g_l\}\rangle$ the subgroup generated by $g_1, \ldots, g_l$.

Throughout the paper, we will deal with CSS codes [CS96], [Ste96]. For a linear subspace A, we use $|A\rangle$ to denote the uniform superposition over the vectors in A. A CSS code will usually be denoted by the pairs $\mathcal{C}_X, \mathcal{C}_Z$ such that the Z-stabilizers match the codewords of $\mathcal{C}_Z^{\perp}$, and the logical codewords in the computational basis are the elements in $\mathcal{C}_X/\mathcal{C}_Z^{\perp}$. The logical zero codeword is $\big|\mathcal{C}_Z^{\perp}\big\rangle$. We denote the equivalence relation up to Z-stabilizer addition by $\sim_{\mathcal{C}_X}$. Namely, for $x_1, x_2 \in \mathbb{F}_2^n$, we have $x_1 \sim_{\mathcal{C}_X} x_2$ if $x_1 + x_2 \in \mathcal{C}_Z^{\perp}$. By the same way we define $\sim_{\mathcal{C}_Z}$.

The Computation Model. Informally, a quantum circuit is a placement of a collection of quantum gates, taken from a finite set, at different space-time locations. Each such location records when the gate is applied and on which qubits it acts. Formally:

Definition 3.1 (Quantum Circuit)**.** *A* space-time location *is a pair $l = (t, S)$, where t is a time coordinate and $S = (s_1, \ldots, s_\Delta)$ is an ordered tuple of qubit indices. For a finite collection of quantum gates*

$$\mathcal{G} \subset \Big\{g\colon L(\mathcal{H}_2^{\otimes\Delta}) \to L(\mathcal{H}_2^{\otimes\Delta})\Big\},$$

and integers $w, d \in \mathbb{N}$, a quantum circuit C of depth d and width w is a collection of tuples

$$C = \{u_i = (g_i, l_i)\}_i,$$

where each $g_i \in \mathcal{G}$ and each location $l_i = (l_{i,1}, l_{i,2})$ belongs to $[d] \times [w]^\Delta$. We call the tuples u_i the gates of C. They must satisfy the following compatibility condition: if two gates have the same time coordinate, then their supports are disjoint. Namely, for $u_i = (g_i, l_i), u_j = (g_j, l_j) \in C$, the equality $l_{i,1} = l_{j,1}$ implies that the sets of qubit indices appearing in $l_{i,2}$ and $l_{j,2}$ are disjoint. We denote by

$$\mathrm{Loc}(C) := \{l_i : u_i = (g_i, l_i) \in C\}$$

the set of locations occupied by the gates of C. We define the t-layer of C to be its restriction to gates at time t:

$$C|_t := \{u_i = (g_i, l_i)\colon u_i \in C \text{ and } l_{i,1} = t\}.$$

Later, we will restrict $\mathcal{G}$ to contain $\mathbf{X}, \mathbf{Z}, \mathbf{CX}, \mathbf{H}, \mathbf{S}, \mathbf{T}$, and measurements. To define the action of a quantum circuit C on a given mixed state, we first associate C with a directed acyclic graph.

Definition 3.2 (Associated Computation DAG)**.** *Let $C = \{u_i\}_i$ be a quantum circuit with depth d and width w. We define the associated computation DAG of C, denoted by π_C, to be the directed acyclic graph $\pi_C = (\mathcal{U}, E)$ as follows:*

1. *The vertex set is the gate set of C, namely $\mathcal{U} = \{u_i\}_i$.*
2. *For $u_i = (g_i, l_i)$ and $u_j = (g_j, l_j)$, there is a directed edge $(u_i, u_j) \in E$ if and only if j is one of the minimizers of*

$$\big\{l_{k,1} : (g_k, l_k) \in \mathcal{U},\ l_{k,1} > l_{i,1},\ l_{k,2} \cap l_{i,2} \neq \emptyset\big\}.$$

   Equivalently, u_j is one of the earliest later gates whose support intersects the support of u_i. There may be more than one such minimizer, so the out-degree may exceed one.

Suppose that $J = (j_1, j_2, \ldots, j_k)$ is an ordering of the gate indices that agrees with a topological order induced by π_C. Then we define the associated partial circuits $C|_J = \{C|_{J,\to i}\}_{i=1}^k$ by

1. *$C|_{J,\to 1} = \{u_{j_1}\}$.*
2. *For $1 \leq i < k$, set $C|_{J,\to i+1} = C|_{J,\to i} \cup \{u_{j_{i+1}}\}$.*

We call $C|_{J,\to i}$ the partial computation circuit up to step i. When the order itself does not play a role, except for being time-respecting (namely, a gate u that is applied before gate u' has a lower step than u''s step in J), we will omit J from the subscript and briefly write $C|_{\to i}$ to describe a partial circuit up to that step.

Definition 3.3. *Let ρ be a density matrix over w qubits, and let $C = \{u_i\}_{i=1}$ be a quantum circuit with depth d and width w. For any density matrix σ, quantum gate $g \in \mathcal{G}$, and location l, the application of the gate-location tuple (g, l) on σ, denoted by $(g, l) \circ \sigma$, is the application of g on σ where wires are ordered such that the first qubit in the input to g is $l_{2,1}$, the second is $l_{2,2}$, and so on. The result of the application of C on ρ, denoted by $C \circ \rho$, is given by iterative applications of the gates $\{u_i\}_{i=1}$ on ρ, according to a topological order induced by π_C.*

We say that $\rho_1, \rho_2, \ldots, \rho_k$ is a computation prefix if $\rho_1 = \rho$ and $\rho_{i+1} = u_i' \circ \rho_i$, where $\{u_i'\}_{i=1}$ is a prefix of a topological ordering of $\{u_i\}_{i=1}$. If $\{u_i'\}_{i=1}$ contains all the gates in $\{u_i\}_{i=1}$, then we call $\rho_1, \rho_2, \ldots, \rho_k$ a running.

For a quantum circuit C, we would like to discuss its action in the presence of different outcomes of faults. The following definition encodes the running in the presence of a specific set of faults $\mathcal{E}$.

Definition 3.4 (C subjected to $\mathcal{E}$ every m computational layers)**.** *Let $C = \{u_i\}_i$ be a quantum circuit with depth d and width w, let $m \in \mathbb{N}$, and set*

$$d_m := \left\lceil \frac{d}{m} \right\rceil .$$

Let $\mathcal{L} \subset [d_m] \times [w]$ be a set of locations, and let $\mathcal{E}$ be a quantum circuit, such $Loc(\mathcal{E}) = \mathcal{L}$. We name $\mathcal{E}$ subsets of faults, and we name the layers of $\mathcal{E}$ fault layers. The circuit C subjected to $\mathcal{E}$ every m computational layers, denoted by $C[\mathcal{E}]_m$, is obtained by setting the first layer to be the first layer of $\mathcal{E}$, then placing in order the layers of C and inserting one fault layer after every m consecutive C's layers. Thus the second fault layer is inserted after the layers $1, \ldots, m$ of C, the third is inserted after the layers $m+1, \ldots, 2m$ of C, and so on. Concretely, each gate $u_i = (g_i, (t_i, S_i))$ of C is moved to the time coordinate

$$t_i + \left\lfloor \frac{t_i - 1}{m} \right\rfloor ,$$

while each fault gate $(g, (t, S)) \in \mathcal{E}$ is placed at time coordinate

$$t(m+1).$$

When $m = 1$, the gates occupy the even time coordinates, and the fault layers occupy the odd time coordinates. When the subscript parameter is omitted, we mean that the noise layers are injected between any two logical layers, and the logical layers are clear from the context. We denote the associated computation DAG of $C[\mathcal{E}]_m$ by $\pi_{C[\mathcal{E}]}$.

Remark 3.5. *Formally, a fault circuit is just a particular quantum circuit whose gates are interpreted as faults rather than as computational operations. Thus the distinction between a quantum circuit and a fault circuit is semantic rather than structural: both are circuits in the sense of Definition 3.1, but they play different roles in the analysis.*

Example 3.6. *When $m = 3$, the 'computational' layers are sent to the times*

$$2, 3, 4, 6, 7, 8, 10, 11, 12, \ldots ,$$

while the fault layers are sent to the times

$$1, 5, 9, 13, \ldots .$$

Later, all the quantum registers will be encoded via stabilizer codes. Then, following the common approach in the literature, we define the fault layers to end with stabilizer measurements, thereby forcing the faults to be Paulis (see [Got97]). Hence, we confine ourselves to noise channels of only this type.

Definition 3.7 (Noise channel and local stochastic noise channel)**.** *A noise channel $\mathcal{N}_\mathcal{E}$ is a distribution over sets of faults and their locations. Let $p \in (0, 1)$, we say that the noise channel $\mathcal{N}_\mathcal{E}$ is local stochastic with rate p if for every finite set of locations $S \subset \mathbb{N} \times \mathbb{N}$,*

$$\mathbf{Pr}_{\mathcal{E} \sim \mathcal{N}_\mathcal{E}} \left[S \subset \mathrm{Loc}(\mathcal{E}) \right] \leq p^{|S|}.$$

Through this work, the sampled faults will be assumed to be Pauli.

So, $N_{\mathcal{E}}$ defines a probability space. The set of faults $\mathcal{E}$ is an atomic outcome, and for a given circuit C, the subjected circuit $C[\mathcal{E}]$ is a random variable. Equipped with this notation, we are allowed to assume having constant depth cycles that don't observe faults. The following definition considers the effective circuit when the running is restricted such that the outcomes of the measurement are fixed.

Definition 3.8. *Let C be a quantum circuit that contains l measurements. For possible outcomes $\alpha \in \mathbb{F}_2^l$, let $C[,\alpha]$ be the circuit obtained by conditioning that $\alpha_1, .., \alpha_l$ were measured. We extend the notation for the subjected circuit by denoting $C[\mathcal{E}, \alpha]$ as the subjected circuit where the measurements are assumed to outcome $\alpha_1, .., \alpha_l$.*

There might be multiple different ways to present a conditioned circuit with a collection of gates and their locations. Since we would like to keep the step-order invariant to a conditioning of the measurement outcome, we choose a representation that substitutes the original gates with gates that act on the same location. The following remark explains how to do that explicitly.

Remark 3.9. *Consider the presentation of a conditioned circuit as a subset of gates acting on locations. For an outcome $\alpha \in \mathbb{F}_2^{(\cdot)}$ for which a conditional gate (for example, a* **CCZ***, where the control bits are conditioned on measurement outcomes) in the original circuit C is mapped to a Pauli in $C[,\alpha]$, we substitute this gate with the Pauli on the target tensored with identity acting on the control locations. Thus, the order of the steps, the locations of each gate, and the computation graph of $C[,\alpha]$ are the same as for C (and independent of the outcomes). We summarize this by briefly saying that the locations are invariant to the conditioning of measurements.*

In our case, the conditioned circuit $C[\mathcal{E}, \alpha]$ is a Clifford + projections circuit; in particular, it induces a 'structured trajectory' tracking on the accumulated error. This brings us to define, for a given set of faults $\mathcal{E}$ and outcomes α, a 'propagation' that describes the evolution of the faults through the circuit.

Definition 3.10 (Propagation)**.** *Let $C[\mathcal{E}]$ be a subjected quantum circuit at depth μ, and let $\alpha \in \mathbb{F}_2^{(\cdot)}$ be a possible outcome vector for $C[\mathcal{E}]$. Suppose that C is a Clifford+measurements circuit, so $C[\mathcal{E}, \alpha]$ is a Clifford + projections circuit. We define the propagation through the conditioned circuit $C[\mathcal{E}, \alpha]$ to be the tuple $(\mathbf{P}, \mathbf{a})$ such that $\mathbf{P}$ consists of $\mu + 1$ Paulis $\mathbf{P} := P_1, \ldots, P_{\mu+1}$ and $\mathbf{a}$ consists of $\mu + 1$ outcome measurements $\mathbf{a} := \alpha^1, \alpha^2, \ldots, \alpha^{\mu+1}$ satisfying that $P_1 = \mathcal{E}|_1, \alpha^1 = \alpha$ and for any $i \in [2, \mu + 1]$:*

$$P_i C[,\alpha^i]|_{\to i} = C[\mathcal{E}, \alpha]|_i P_{i-1} C[,\alpha^{i-1}]|_{\to i-1} = C[\mathcal{E}, \alpha]|_{\to i}$$

We extend the notation to step-propagation by considering the propagation of the circuit obtained from $C[\mathcal{E}, \alpha]$ by flattening the circuit such that each gate is applied at the time that equals its step.

Remark 3.11. *Notice that, for two different measurement outcomes α, α', the conditioned $C[,\alpha]$ and $C[,\alpha']$ might differ also by their Clifford gates (and not only by the projections).*

4 The (d, d')-Dimensional Toric Code

The main error correction code that we will use is the Toric code invented by Kitaev. In the 2D case, the code is obtained by taking a Toric, tiling it with squares, setting qubits on the edges, and checks as follows: For every vertex, the parity of the qubits adjoining it is even, and for any square, the parity of the qubits lying on its sides, in the phase basis, is even. A high-dimensional version can be obtained by taking a high-dimensional Toric, encoded as a cubical complex, and then setting the X-checks, the qubits, and the Z-checks to be faces at sequenced dimensions. Throughout this paper, qubits will be set on d'-dimensional faces, and the X-checks and Z-checks on the $d' \mp 1$ dimensional faces. For a review, we refer to [Kit97], [Den+02].

For defining a decoder and proving its 'correctness,' we will have to overload notation. We derive the Toric complex from the skeleton of the Cayley graph of the cyclic group, using the generating subset as a collection of positive directions. The $\theta_{\mathbf{v},\mathbf{S}'}$ encodes a face, one vertex v that serves as its 'root' and a subset of directions S'. Formally:

Definition 4.1 (d-Dimensional Toric Complex)**.** *For integers d and n, denote the group resulting from the d-directed power of the cyclic group of order n by $\mathcal{G}_n^d = \mathbb{Z}_n^{\times d}$, and denote the standard generating set of $\mathcal{G}_n^d$ by $S = \{1_i \colon i \in [d]\}$. We define the d-dimensional Toric complex to be the hypergraph obtained from the Cayley graph $Cay(\mathcal{G}_n^d, S)$ by taking its vertices as the 0-faces and for any $i \in [d]$ we define the i-faces as follow: First, for a vertex $v \in \mathcal{G}_n^d$ and for a subset of generators $S' \subset S$ of size $|S'| = i$, the i-face rooted at v and spanned by S', denoted by $\theta_{v,S'}$, is:*

$$\theta_{v,S'} := \bigcup_{I \subset S'} \left\{ \left(\prod_{g \in I} g \right) v \right\}$$

The i-faces of the complex are the collection of all the possible rooted spanned i-faces induced by $Cay(\mathcal{G}_n^d)$. We will denote them by $\mathcal{F}_i$. Additionally, let us define the multiplication of the generators subset $S' \subset S$ by a vertex $v \in V(\mathcal{G})$ as follows:

$$S'v = \left(\prod_{g \in S'} g' \right) v$$

Finally, we name n and d, the side length and the dimension of the complex.

Next, we formalize binary assignments on the Toric complex and introduce boundary operators. These boundary operators are the same as standard boundary operators in the chain complex formalism.

Definition 4.2 (Assignments on a Toric complex)**.** *Let $\mathcal{T}$ be a d-dimensional Toric complex. For each i, the set of i-faces induces a vector space $\mathbb{F}_2^{|\mathcal{F}_i|}$. An element $z \in \mathbb{F}_2^{|\mathcal{F}_i|}$ is called an assignment on the i-faces of $\mathcal{T}$.*

For a vertex v and a subset of generators $S' \subset S$ of size $|S'| = i$, let $\theta_{\mathbf{v},\mathbf{S}'}$ denote the standard basis vector corresponding to the i-face $\theta_{v,S'}$. Thus the family

$$\left\{ \theta_{\mathbf{v},\mathbf{S}'} : v \in \mathcal{F}_0,\ S' \subset S,\ |S'| = i \right\}$$

is the standard coordinate basis of $\mathbb{F}_2^{|\mathcal{F}_i|}$.

Let z *be an assignment on the* i*-faces. Let* $v_1, \ldots, v_k \in \mathcal{F}_0$ *and* $S'_1, \ldots, S'_k \subset S$ *be such that* $\theta_{v_1,S'_1}, \theta_{v_2,S'_2}, \ldots, \theta_{v_k,S'_k}$ *are precisely the* i*-faces in the support of* z*, namely:*

$$z = \sum_{i=1}^{k} \theta_{\mathbf{v_i},\mathbf{S'_i}}$$

We refer to $\theta_{v_1,S'_1}, \theta_{v_2,S'_2}, \ldots, \theta_{v_k,S'_k}$ *as the collection of faces that form* z*, or briefly, the faces form* z*. For* $i > 0$*, we define the linear map* $\partial^- : \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_{i-1}|}$ *such that*

$$\partial^- \theta_{\mathbf{v},\mathbf{S'}} := \sum_{g \in S'} \left(\theta_{\mathbf{v},\mathbf{S'/g}} + \theta_{\mathbf{gv},\mathbf{S'/g}} \right)$$

For $i < d$*, we define the linear map* $\partial^+ : \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_{i+1}|}$ *such that*

$$\partial^+ \theta_{\mathbf{v},\mathbf{S'}} := \sum_{g \in S/S'} \left(\theta_{\mathbf{v},\mathbf{S'\cup g}} + \theta_{\mathbf{g^{-1}v},\mathbf{S'\cup g}} \right)$$

We define the restriction of z *at vertex* v *to be the linear map* $|_v \colon \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_i|}$ *such that:*

$$\theta_{\mathbf{u},\mathbf{S'}}|_v := \begin{cases} \theta_{\mathbf{u},\mathbf{S'}} & v \in \theta_{u,S'} \\ 0 & else \end{cases}$$

We refer to the faces rooted at v *as the positive faces of* v*, and define the positive restriction at vertex* v *to be the linear map* $|_{v+} \colon \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_i|}$ *such that:*

$$\theta_{\mathbf{u},\mathbf{S'}}|_{v+} := \begin{cases} \theta_{\mathbf{u},\mathbf{S'}} & u = v \\ 0 & else \end{cases}$$

Two i*th dimensional faces will be said to be connected if their intersection is* $i-1$*th dimensional face. So, an assignment* z *is connected if the graph that is induced by taking the faces that support* z *as vertices and defining the edges to be the adjacency relation between them is connected. We define the connected components of* z *to be its subsets such that each of them is a connected assignment. For a* j*-face* f*, we say that* $f \in z$ *if* z *has a support on* $\theta_{\mathbf{v},\mathbf{S'}}$ *such that* $f \in \theta_{v,S'}$*. Consider an assignment* $z \in \mathbb{F}_2^{\mathcal{F}_i}$ *and a vertex* $v \in \mathcal{G}_n^d$*. We say that* z *is a rooted assignment at vertex* v *if there are subsets* $S'_1, \ldots, S'_k \subset S^{\times k}$ *such that each of* S'_j *is at size* i *and* z *is decomposed as:*

$$z = \sum_{j}^{k} \theta_{\mathbf{v},\mathbf{S'_j}}$$

Put differently, z *is the result of taking a collection of* i*-faces that are rooted at* v*.*

We are ready to give a formal definition of the (d, d')-Toric code.

Definition 4.3 ((d, d') Toric Code)**.** *Let d, d' be integers such that $1 \leq d' \leq d$. For any integer n, we construct the nth instance of the quantum code family (d, d') Toric Code as follows: Let $G = (\mathcal{F}_0, \ldots, \mathcal{F}_d)$ be the d-dimensional Toric complex obtained from $\mathcal{G}_n^d$ as defined in Definition 4.1. We associate the (q)bits with the $\mathcal{F}_{d'}$, the X-checks with the $\mathcal{F}_{d'-1}$, and the Z-checks with the $\mathcal{F}_{d'+1}$. An X-check, c_x, is satisfied if the parity of (q)bits set on the d'-face containing c_x is even. Similarly, a Z-check, c_z, is satisfied if the parity, in the Hadamard basis, of the (q)bits set on the d'-faces contained in c_z is even.*

We argue that the (d, d')-Toric code is indeed a CSS, and prove that by showing that boundary operators $\partial^{\pm} : \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_{i\pm1}|}$ induce a chain complex.

Claim 4.4 (CSS structure)**.** *The (d, d')-Toric code is a CSS code.*

Proof. We will show that $\mathbb{F}_2^{|\mathcal{F}_i|} \xrightarrow{\partial^{\pm}} \mathbb{F}_2^{|\mathcal{F}_{i\pm1}|}$ are a chain complex and its dual chain. First, we show that $(\partial^+)^\top = \partial^-$. Consider vertices $v, u \in V$ and two generator subsets $A, B \subset S$. Consider the basis elements defined by them: $\theta_{\mathbf{v},\mathbf{A}}$ and $\theta_{\mathbf{u},\mathbf{B}}$. Suppose that:

$$(\partial^+ \theta_{\mathbf{v},\mathbf{A}})\theta_{\mathbf{u},\mathbf{B}} = 1$$

Means that there exists $g \notin A$ such that $A \cup \{g\} = B$ and either $v = u$ or $g^{-1}v = u$. But that is equivalent to saying that there is $g \in B$ such that $B \setminus \{g\} = A$ and either $v = u$ or $v = g^{-1}u$. In other words:

$$\begin{aligned} &(\partial^+ \theta_{\mathbf{v},\mathbf{A}})\theta_{\mathbf{u},\mathbf{B}} = \theta_{\mathbf{v},\mathbf{A}}(\partial^- \theta_{\mathbf{u},\mathbf{B}}) \\ \Rightarrow &\left(\partial^+\right)^\top = \partial^- \end{aligned}$$

So it's left to show that the boundary of a boundary is zero. Consider a basis vector $\theta_{\mathbf{v},\mathbf{S}'}$:

$$\begin{aligned} \partial^-\left(\partial^- \theta_{\mathbf{v},\mathbf{S}'}\right) &= \partial^- \left(\sum_{g \in S'} \theta_{\mathbf{v},\mathbf{S}'/\mathbf{g}} + \sum_{g \in S'} \theta_{\mathbf{gv},\mathbf{S}'/\mathbf{g}} \right) \\ &= \sum_{\substack{g', g \in S' \\ g' \neq g}} \left(\theta_{\mathbf{v},\mathbf{S}'/\{\mathbf{g},\mathbf{g}'\}} + \theta_{\mathbf{g}'\mathbf{v},\mathbf{S}'/\{\mathbf{g},\mathbf{g}'\}} + \theta_{\mathbf{gv},\mathbf{S}'/\{\mathbf{g},\mathbf{g}'\}} + \theta_{\mathbf{g}'\mathbf{gv},\mathbf{S}'/\{\mathbf{g},\mathbf{g}'\}} \right) \\ &= 0 \end{aligned}$$

Where we used that the term in the closure is invariant to sweeping between g and g', and therefore any term in the sum appears twice. Similarly:

$$\begin{aligned} \partial^+\left(\partial^+ \theta_{\mathbf{v},\mathbf{S}'}\right) &= \partial^+ \left(\sum_{g \in S/S'} \theta_{\mathbf{v},\mathbf{S}' \cup \mathbf{g}} + \sum_{g \in S/S'} \theta_{\mathbf{g}^{-1}\mathbf{v},\mathbf{S}' \cup \mathbf{g}} \right) \\ &= \sum_{\substack{g', g \in S/S' \\ g' \neq g}} \left(\theta_{\mathbf{v},\mathbf{S}' \cup \{\mathbf{g},\mathbf{g}'\}} + \theta_{\mathbf{g}'^{-1}\mathbf{v},\mathbf{S}' \cup \{\mathbf{g},\mathbf{g}'\}} + \theta_{\mathbf{g}^{-1}\mathbf{v},\mathbf{S}' \cup \{\mathbf{g},\mathbf{g}'\}} + \theta_{\mathbf{g}'^{-1}\mathbf{g}^{-1}\mathbf{v},\mathbf{S}' \cup \{\mathbf{g},\mathbf{g}'\}} \right) \\ &= 0 \end{aligned}$$

The conditions $(\partial^{\pm})^{\circ 2} = 0$ imply that $\mathbb{F}_2^{|\mathcal{F}_i|} \xrightarrow{\partial^{\pm}} \mathbb{F}_2^{|\mathcal{F}_{i\pm1}|}$ are chain complexes. Therefore, for any d', every $d'-1$ face and every $d'+1$ face overlap on an even number of d'-faces, so the corresponding X- and Z-checks commute. This is exactly the CSS condition. □

The fact that $\partial^{\pm} : \mathbb{F}_2^{|\mathcal{F}_i|} \to \mathbb{F}_2^{|\mathcal{F}_{i\pm1}|}$ induces a chain complex motivates referring to the collection of qubits/checks/vertices as chains of various dimensions. Next, we give a definition of the 'positive' vertices that a vertex v sees, which lie on a chain z.

Definition 4.5 (Positive Neighborhood)**.** *For a vertex v and a chain z, we use the notation $\boldsymbol{Str}_{0+}(v,z)$ to denote the vertices in z that are connected to v through a stepping via generator, namely all the vertices that support a face in z of the form gv for some $g \in S$. We think of $\boldsymbol{Str}_{0+}(v,z)$ as the positive neighborhood of v in z.*

4.1 Sweep Rule

The sweep-rule is the core of our construction. A decoding cycle begins with each vertex v reading the syndrome on the X-checks rooted at v (the positive X-checks of v) and, based on what it sees, applying a Pauli correction supported on the qubits rooted at v. The sweep-rule defines the correction to be over the positive qubits such that it will turn off the positive syndrome. It's not hard to see that for the $(2,2)$-Toric complex, the Toom's and the sweep rules are the same. Compared to other correction cycles on topological codes, the main advantage of the sweep is that it is independent of time. The invention of the sweep-rule is attributed to Aleksander Kubica and John Preskill [KP19]. Here we extend their work to cubical complexes.

Definition 4.6 (Sweep Rule)**.** *Let σ be a $d'-1$-dimensional chain. The sweep rule decoding procedure at a vertex $v \in \mathcal{G}_n^d$ is defined as follows: Measure the checks rooted at v and look for a rooted chain $\tilde{z}$ at v such that:*

1. *The vertices in $\tilde{z}$ are contained in the vertices of σ, namely: $\mathbf{Str}_{0+}(v,\tilde{z}) \subset \mathbf{Str}_{0+}(v,\sigma)$*
2. *The boundary of $\tilde{z}$ agrees with σ on the checks rooted at v, namely: $(\partial^{-}\tilde{z})|_{v+} = \sigma|_{v+}$.*

If such an assignment exists, flip $\tilde{z}$.

The first property is analogous to the second property of 'causal structure' defined in [KP19][8]. There, they conjectured that vertices that see a syndrome only on their positive checks can suggest a correction that is 'supported' on the vertices of their unsatisfied checks. For lower dimensions, they mentioned that the property can be easily checked. Here, we provide a proof for any dimensions where the face dimension is greater than one. We use the language of cubical complexes; however, since the 'positive' local view at a vertex is identical for both cubical and simplicial complexes, generalizing the proof for simplicial complexes is a matter of adjusting notations. By

[8] In their language: $\diamondsuit(\tilde{z}) = \diamondsuit(\sigma|_v)$.

that, we affirmatively answer their conjecture. Moreover, the proof will provide us with an algorithm to find $\tilde{z}$, which is significantly better than running all options.

Claim 4.7. *Consider a nontrivial $d'-1$-dimensional chain σ. If there is a d'-dimensional chain z that is rooted at v and whose boundary agrees with $\sigma|_{v+}$, then there is a d'-dimensional chain $\tilde{z}$ rooted at v that satisfies the sweep-rule conditions.*

Proof. We prove it by induction on d'. Without loss of generality, we can assume that σ has support only over the positive faces of v, namely $\sigma|_{v+} = \sigma$.

1. Base. Consider $d' = 2$ and let d be any integer equal to or greater than 3. We argue that the support size of σ has to be even. To see that, denote the faces in z by $\theta_{\mathbf{v},\mathbf{S}'_\mathbf{i}}$, where $S'_i = \{g_{i,1}, g_{i,2}\}$. Using the fact that σ is the boundary of z at v, we have:

$$\sigma = (\partial^-(z))|_v = \sum_{\theta_{\mathbf{v},\mathbf{S}'} \in z} \sum_{g \in S'} \theta_{\mathbf{v},\mathbf{S}'/\mathbf{g}} = \sum_i \theta_{\mathbf{v},\{\mathbf{g_{i,1}}\}} + \theta_{\mathbf{v},\{\mathbf{g_{i,2}}\}}$$

   So the number of terms in the sum is even, and since a term is zeroed out only if it appears an even number of times, we have that after excluding the vanishing terms, σ has an even number of faces. Since the number of faces is even, there is a perfect matching between them. It's easy to see that the 2-chain $\tilde{z}$, whose each face corresponds to the union of the two edges in a match, satisfies the sweep-rule conditions.

2. Assume the correctness of the claim for every face dimension lower than d'. Now for a face dimension d' and an arbitrary generator $g \in S$ such $gv \in \sigma$, divide σ into a sum of two chains, the first contain the faces of σ that supported on the vertex gv where the second contains the compliment. Denote them by $\sigma|_{gv}$ and $\sigma|_{/gv}$, so $\sigma = \sigma|_{gv} + \sigma|_{/gv}$.

   Now let σ^* the chain obtained from $\sigma|_{gv}$ by removing from any face of it the vertex gv, thus σ^* is a $d'-2$-dimensional chain. Moreover, let z^* be the $d'-1$-dimensional chain complex obtained from $z|_{gv}$ by excluding gv from its faces. Observes that

$$(\partial^- z^*)|_v = \sum_{\theta_{\mathbf{v},\mathbf{S}'} \in z|_{gv}} \sum_{g' \in S'/g} \theta_{\mathbf{v},\mathbf{S}'/\{\mathbf{g},\mathbf{g}'\}} = \sum_{\theta_{\mathbf{v},\mathbf{S}'} \in (\partial^- z)|_{gv}} \theta_{\mathbf{v},\mathbf{S}'/\{\mathbf{g}\}} = \sigma^*$$

   Thus, by the induction assumption, there is a $d'-1$-dimensional chain $\tilde{z}^*$ such that:

   (a) $\mathbf{Str}_{0+}(v, \tilde{z}^*) \subset \mathbf{Str}_{0+}(v, \sigma^*)$

   (b) $(\partial^- \tilde{z}^*)|_{v+} = \sigma^*|_{v+}$

   The first condition implies that $gv \notin \tilde{z}^*$. Hence, we can define the d'-dimensional chain $\tilde{z}$ by adding the vertex gv to each face of $\tilde{z}^*$. From the second it follows

that:

$$(\partial^- \tilde{z})|_v = \big(\partial^-\big(\sum_{\theta_{\mathbf{v},\mathbf{S}'} \in \tilde{z}^*} \theta_{\mathbf{v},\mathbf{S}'\cup\{\mathbf{g}\}}\big)\big)|_v = \sum_{\theta_{\mathbf{v},\mathbf{S}'} \in \tilde{z}^*} \big(\sum_{g' \in S'/g} \theta_{\mathbf{v},\mathbf{S}'/\{\mathbf{g}'\}\cup\{\mathbf{g}\}}\big) + \theta_{\mathbf{v},\mathbf{S}'}$$
$$= \sum_{\theta_{\mathbf{v},\mathbf{S}'} \in (\partial^- \tilde{z}^*)|_v} \theta_{\mathbf{v},\mathbf{S}'\cup\{\mathbf{g}\}} + \tilde{z}^* = \sigma|_{gv} + \tilde{z}^*$$

Therefore,

$$\sigma + (\partial^- \tilde{z})|_v = \sigma|_{gv} + \sigma|_{/gv} + \sigma|_{gv} + \tilde{z}^*$$
$$= \sigma|_{/gv} + \tilde{z}^*$$

Thus we reduce the question to handle the $d'-1$-dimensional chain $\sigma + (\partial^- \tilde{z})|_v$. However, since $gv \notin \sigma + (\partial^- \tilde{z})|_v$ it follows that:

$$|\mathbf{Str}_{0+}(v, \sigma + (\partial^- \tilde{z})|_v)| \leq |\mathbf{Str}_{0+}(v, \sigma)| - 1$$

Thus, we can repeat recursively until the resulting $d'-1$ chain is 'supported' on fewer than d' vertices. In that case, the chain must be trivial.

□

The algorithm given in the proof of Claim 4.7 for finding $\tilde{z}$ is as follows: given σ, pick a $g \in S$ such that $gv \in \sigma$. Then, define σ^* by removing gv from each face of $\sigma|_{gv}$. Let $\tilde{z}^*$ be the result of the recursive call on σ^*, and return '$\tilde{z}^* \cup \{gv\}$' plus the recursive call on $\sigma|_{/gv} + \tilde{z}^*$.

We stop the recursion whenever either $d' = 2$ or $|\mathbf{Str}_{0+}(v, \sigma)| < d'$. The work in each recursion step consists of decomposing σ into $\sigma|_{gv}$ and $\sigma|_{/gv}$ and summing $\binom{d}{d'}$-length vectors. We summarize by introducing the following claim which gives the running time of the algorithm in the RAM computation model.

Claim 4.8. *Let $T(d, d', k) : \cdot \to \mathbb{R}$ be the upper bound function for the running time of the algorithm given in Claim 4.7, where d is the dimension of the skeleton, d' is the face dimension, and k is the number of vertices supporting the local view, namely $k = |\mathbf{Str}_{0+}(v, \sigma)|$. Then:*

$$T(k, d') = O\big(2^k \binom{d}{d'}\big)$$

Proof. In each recursive call, the vertex support of σ matches a different subset of the vertices in $\mathbf{Str}_{0+}(v, \sigma)$; thus, the number of recursive calls is at most the number of subsets, namely 2^k. The decomposition of σ into $\sigma|_{gv}$ and $\sigma|_{/gv}$, and the summation $\sigma|_{/gv} + \tilde{z}^*$ are linear in the vector's length, namely linear in $\binom{d}{d'}$. So we bound the total work by the number of recursive steps multiplied by the maximal work at each recursive step. That becomes at most:

$$T(d, d', k) \leq O\big(2^k \binom{d}{d'}\big)$$

□

Later (see Definition 4.31 and Definition 4.32), we will see that an isomorphism between the (d, d')-Toric code and its dual, which is the $(d, d - d')$-Toric code, can be realized by transversal Hadamards followed by the following permutation over the qubits.

$$\phi(\theta_{\mathbf{v},\mathbf{S}'}) \to \theta_{\mathbf{S}'\mathbf{v},\mathbf{S}/\mathbf{S}'}$$

A full decoding cycle consists of four steps: first, applying the sweep rule regarding the X-syndrome; next, moving to the dual code; then applying the sweep rule again; and finally getting back to the code.

Definition 4.9 (Sweep cycle)**.** *We define the sweep cycle to be the following procedure:*

1. *For each vertex, perform the sweep rule.*
2. *Apply the Hadamard transform* $\mathbf{H}^{\otimes \mathcal{F}^n_{d'}}$ *followed by the* ϕ*-transpose.*
3. *Again, apply the sweep rule at each vertex.*
4. *Reverse stage 2 by reordering the qubits according to the* ϕ^{-1}*-transpose, and then apply the Hadamard transform* $\mathbf{H}^{\otimes \mathcal{F}^n_{d'}}$*.*

4.2 Infinite Sides Torics

The Toric skeleton provides a notation for positive directions. However, to demonstrate that an 'error cluster' shrinks as a result of the decoding cycle, or at least does not expand too much during the operation of the logical operator, we will need a way to mark extremities (for example, the northernmost point). To achieve this, we will analyze the view over the finite Toric as a projection derived from a view over the infinite plane. For this purpose, given our initial (subjected) quantum circuit, we construct a representation in which any gate that originally acted on a quantum register, namely on the finite Toric, is mapped to infinite translations of it, each acting on qubits on the infinite plane. Using the Wrapping Lemma, proved in [SW04], and combining that the mapping respects a periodic condition, the finite and the infinite presentations are isomorphic.

The following definition extends the notion of location to differentiate between a coordinate that encodes a register and a coordinate of the location inside the register. This will be useful, as in the infinite plane representation, each register will be mapped to a unique infinite register.

Definition 4.10 (Toric register coordinate system)**.** *Let* $C = \{u_i\}_i$ *be a quantum circuit of width* w *with registers* $\mathcal{R} = R_1, \ldots, R_k$*, each encoded in the* (d, d')*-Toric code, induced by a* d*-dimensional Toric with a side length* n*. We define the 'Toric register coordinate system' to be the coordinate system that encodes spatial location with two coordinates as follows: In the first coordinate, the register number is recorded, and in the second, the qubit is encoded by the face associated with it. We call the first the register coordinate and the second the inner coordinate.*

We will need two types of gate translation. The first is the standard spatial translation, which means that any of the inner coordinates of the gate are advanced by the same vector. The second is 'rooted translation.' Here, we assume that the locations of the gate are given in a relative-to-root fashion, meaning that there is a root vertex, and all the locations are given by positive paths from the root. Then, we first translate the root and then advance the inner coordinate of any of the other locations to be on the path from the translated root. When considering a product gate, the translations act the same.

Definition 4.11 (Translation by vertex v)**.** *Let u be a gate, and denote the inner coordinates of u's locations by $\theta_{v_1,J_1},\ldots,\theta_{v_\Delta,J_\Delta}$. The translation of u by vertex v is defined to be the gate obtained from u by replacing the inner coordinates of its locations by: $\theta_{vv_1,J_1},\ldots,\theta_{vv_\Delta,J_\Delta}$, where the multiplication $v\cdot v_i$ is with respect to the infinity cover $\mathcal{G}^d$. Furthermore, for a d-dimensional Toric complex with side length n, we define a translation by an integer vector $a\in\mathbb{Z}^d$ as a translation by the vertex:*

$$(g_1^n)^{a_1},(g_2^n)^{a_2},..,(g_d^n)^{a_d}$$

We denote the translation by an integer vector a as $a*u$.

Definition 4.12 (Rooted gate at vertex v)**.** *Let u be a quantum gate that acts on a register whose qubits are associated with a Toric complex at dimension d. We say that a gate u is rooted at vertex v if the inner coordinate of its location can be encoded by lists of pairs $(I_1,J_1),(I_2,J_2),\ldots,(I_l,J_l)$ such that $I_i\in S^d, J_i\subset S$, and the ith qubit on which it acts is in the face θ_{I_iv,J_i}.*

Definition 4.13 (Root translation in Register System)**.** *For a gate u that is a rooted gate at vertex v, we define its translation by a vertex v' to be the gate obtained by copying u and replacing its inner location encoding the root by $v'v$.*

Similarly to the standard translation by an integer vector, we extend the action to a root translation by an integer vector $a\in\mathbb{Z}^d$, as the root translation by the vertex:

$$(g_1^n)^{a_1},(g_2^n)^{a_2},..,(g_d^n)^{a_d}$$

We denote the root translation by an integer vector a as $a*_r u$.

Now we are ready to give an explicit description of the mapping. Given a quantum circuit C, we name its presentation over the infinite plane the ∞-*lift of* C.

Definition 4.14. *Let n be an integer, and let $C=\{u_i\}_{i=1}$ be a quantum circuit with width w and quantum registers $\mathcal{R}=R_1,\ldots,R_k$ such that each register R_i encodes a $(d,d')-$Toric code with side at length n. We define the ∞-lift of C, denoted by C^∞, as follows:*

1. *Each register R_i in $\mathcal{R}$ is mapped to a register R_i', which it's qubits are associated with the d'-dimensional faces of the infinite d-dimensional Toric.*

2. *The gate $u_i \in C$ is mapped to an infinite collection of gates, each being a rooted translation by an integer vector, as follows:*

$$u_i \mapsto \bigcup_{a \in \mathbb{Z}^d} \{a *_r u_i\}$$

The restriction of the qubits in C^∞'s registers to finite Torices with side length n will be denoted by $\cdot|_{[w]}$, and it includes all the faces whose root vertex has 'powers' (the powers of $g_1, \ldots, g_d$) in the range $[0, n-1]$. For a set of faults $\mathcal{E} = \{f_i\}_{i=1}$, and measurements outcomes $\alpha \in \mathbb{F}_2^{(\cdot)}$, we denote by $C^\infty[\mathcal{E}, \alpha]$ the ∞-lift of $C[\mathcal{E}, \alpha]$.

We define the ∞-lift of a Pauli operator $P \in \mathcal{P}^{(\cdot)}$ be the result of lifting the circuit composed only of P. Denote it by P^∞. Finally, the ∞-lifting of a chain $x \in (\mathcal{F}_{d'}^n; \mathbb{F}_2)$ is the chain in the infinite complex $(\mathcal{F}_{d'}; \mathbb{F}_2)$ obtained by taking all the translations of x by integer vectors.

The infinite presentation of a circuit agrees with it in terms of action. We give, without proof, an adaptation of the Wrapping Lemma, proved in [SW04].

Claim 4.15 (Adaptation of [SW04])**.** *The ∞-lifting is an isomorphism. In particular, consider a conditioned subjected quantum circuit $C[\mathcal{E}, \alpha]$, such that it has a propagation $P_1, P_2, \ldots$, it holds that:*

$$P_i^\infty = (C[\mathcal{E}, \alpha]|_i P_{i-1})^\infty = C[\mathcal{E}, \alpha]|_i^\infty P_{i-1}^\infty$$

The computation DAG lifts naturally to the ∞-lift gates that arise as translates of the same local gate act on disjoint translated supports inside different periods, so they may be executed simultaneously. The same periodic lifting applies to the underlying geometric objects, passing from the finite Toric complex to its infinite cover.

We close that subsection by introducing the *potential walls*. Analyzing the infinite presentation enables us to bound a subset of errors by 'triangles'. A potential wall is a linear function $\mathbb{R}^d \to \mathbb{R}$. For the subset of vertices (that might be the subset that supports some error), the maximal value of the potential wall on it is achieved at the point of tangency between the subsets and its bounding triangle. With the assistance of potential walls, we can discuss the endpoint of a bunch of qubits and the 'size' of the (hyper)triangle that bounds them. Soon, the qubits will be associated with the support of an error, and showing that an action decreases the maximums of the potential walls implies that the error shrinks. The next definition explains how the embedding of vertices of the infinite in $\mathbb{R}^d$ is handled.

Definition 4.16 (The standard embedding)**.** *Consider the d-dimensional infinite grid $\mathcal{G}^d$. The standard embedding $\mathcal{G}^d \to \mathbb{R}^d$ is given by fixing a vertex in $\mathcal{G}^d$ to be sent to the origin and then mapping the generators of the grid to the standard basis. We extend notation as follows: for a vertex $v \in \mathcal{G}^d$, a generator $g \in S$, and a coordinate $i \in [d]$, such that under the embedding $g \to i$, we denote by $v_i = v_g$ the i'th coordinate of the image of v.*

Now we are ready to define the *potential walls*:

Definition 4.17 (Potential Walls)**.** *We define the potential walls to be the $d+1$ functions $L_1, L_2, .., L_{d+1}$, from the vertices of $\mathcal{G}^d$ to the reals as follow:*

1. *For any coordinate $i \in [d]$, we define $L_i \colon \mathcal{G}^d \to \mathbb{R}$ as $L_i(v) := v_i$. In addition, for the generator $g \in S$ which is mapped to i, i.e $g \to i$, we identify $L_g = L_i$.*
2. *We define $L_{d+1} \colon \mathcal{G}^d \to \mathbb{R}$ by $L_{d+1}(v) := -\sum_{i=1}^{d} v_i$.*

Consider an assignment z over the faces. For a vertex $v \in z$, and a potential wall $l \in \{L_1, .., L_{d+1}\}$, we say that v is a local maximum of l in z if for any neighbor u of v, such that $u \in z$, we have $l(v) \geq l(u)$. When the assignment z is clear from the context, we might omit it and briefly say that v is a local maximum of l. In addition we define the function $\mathbf{Size}(\cdot) : \mathcal{G}^d \to \mathbb{R}$, to act on a subset of points $A \subset \mathcal{G}^d$, as follows:

$$\mathbf{Size}(A) := \sum_{g \in S} \max_{v \in A} L_g(v) + \max_{v \in A} L_{d+1}(v)$$

and extend the function to act over assignment on faces z, by taking its action on the set of vertices that support z.

The following claim shows that size has a similar behavior to a metric:

Claim 4.18. *Let $A, B \subset \mathbb{R}^d$ such that there $A \cap B \neq \emptyset$ then:*

$$\mathbf{Size}(A \cup B) \leq \mathbf{Size}(A) + \mathbf{Size}(B)$$

Proof. Let p be a point in $A \cap B$. Then:

$$\begin{aligned}
\mathbf{Size}(A \cup B) &= \sum_{i \in [d+1]} \max_{v \in A \cup B} L_i(v) \\
&= \sum_{i \in [d+1]} \max\big\{\max_{v \in A} L_i(v), \max_{v \in B} L_i(v)\big\} \\
&= \sum_{i \in [d+1]} \max_{v \in A} L_i(v) + \max_{v \in B} L_i(v) - \min\{\max_{v \in A} L_i(v), \max_{v \in B} L_i(v)\} \\
&\leq \sum_{i \in [d+1]} \max_{v \in A} L_i(v) + \max_{v \in B} L_i(v) - L_i(p) \\
&\leq \mathbf{Size}(A) + \mathbf{Size}(B) - \sum_{i \in [d+1]} L_i(p) = \mathbf{Size}(A) + \mathbf{Size}(B)
\end{aligned}$$

Where in the last pass, we used that the size of a single point is zero. □

4.3 Reduced Error

Errors can be thought of as assignments on the d'-dimensional faces that have non-vanishing boundaries. In particular, X-errors can be characterized as d'-dimensional chains $z \in (\mathcal{F}_{d'}; \mathbb{F}_2)$ such that $\partial^- z \neq 0$. In this section, we assume dealing with the ∞-lift of a circuit (see Definition 4.14), so the cubical complex is the d-dimensional grid with infinite side length. The *reduced* form of an error is its representative, a translation by a stabilizer, which minimizes its size. Here we prove that the sizes of a reduced error and its boundary (derived from ∂^-) are the same[9]. In particular, they share poles of the potential walls. That will come into use in analyzing the behavior of the sweep-rule, as it is defined to decide on correction upon the syndrome it sees.

For doing so we present two codes that describe local views of vertices. They are the *stabilizers code at vertex* v and the *small code at vertex* v. The first corresponds to stabilizers rooted at the vertex, while the second is the collection of local views on faces rooted at v that satisfy the checks rooted at v. Then, in Claim 4.21, we consider a vertex whose X-checks don't admit a syndrome, meaning that its local view is in its *small code*, and show that there is a codeword of its *stabilizers code* that agrees with its positive faces. Addition by that stabilizer yields a new chain excluding v. If v were a unique maximum of L_{d+1}, then the vertices on the yielded chain would have a lower value of L_{d+1}. Subsequently, since the exclusion decreases the size, it follows that the 'corner'—the vertex achieving the maximal L_{d+1} value—of a reduced error sees a syndrome.

Definition 4.19 (Stabilizers Code at Vertex v)**.** *We define stabilizer code at vertex v, denoted by $Stab_v$, as the restriction of the (d, d')-Toric code to the d'-dimensional faces such that their vertices are at a distance of at most one positive step in each direction from v. That is equivalent to the union of all the d'-dimensional faces rooted at v with the d-dimensional faces of each of its sibling gv, excluding faces containing g as their generator. Namely:*

$$\begin{aligned} Stab_v := \big\{ z : \partial^- z = 0 \ \textit{and} \\ z \in Span\big(\{\theta_{\mathbf{v},\mathbf{S'}} : S' \subset S, |S'| = d'\} \\ \bigcup_{g \in S} \{\theta_{\mathbf{gv},\mathbf{S'/g}} : S' \subset S/g, |S'| = d'\} \big) \big\} \end{aligned}$$

The checks of the code, are of the following three types:

1. Checks that are rooted at v, each identified by choosing $d-1$ generators from S. Namely:

$$\{\theta_{\mathbf{v},\mathbf{S'}} : S' \subset S, |S'| = d' - 1\}$$

   For each check, the bits that connect to it, are given by restricting the coboundary of the check to bits domain of Stab_v:

$$(\partial^+ \theta_{\mathbf{v},\mathbf{S'}})|_{\text{Stab}_v} = \big(\sum_{g \in S/S'} \theta_{\mathbf{v},\mathbf{S'} \cup \mathbf{g}} + \sum_{g \in S'} \theta_{\mathbf{g^{-1}v},\mathbf{S'} \cup \mathbf{g}} \big)|_{\text{Stab}_v} = \sum_{g \in S/S'} \theta_{\mathbf{v},\mathbf{S'} \cup \mathbf{g}}$$

[9] In fact, the hypertriangles bounding them are the same.

2. Checks are rooted at the positive sibling of v. Consider such a sibling, let's say gv (namely the vertex that is given by stepping in the direction $g \in S$ from v). Notice that any face rooted at gv that contains ggv cannot be contained in faces rooted at v. Thus, the checks of gv correspond to choosing subsets from S/g. Namely:

$$\bigcup_{g\in S} \left\{\theta_{\mathbf{gv},\mathbf{S'}} : S' \subset S/g, |S'| = d' - 1\right\}$$

And each check is supported by:

$$\begin{aligned}\left(\partial^+ \theta_{\mathbf{gv},\mathbf{S'}}\right)|_{\mathrm{Stab}_v} &= \Big(\sum_{g'\in S/S'} \theta_{\mathbf{gv},\mathbf{S'}\cup\mathbf{g'}} + \sum_{g'\in S'} \theta_{\mathbf{g'^{-1}gv},\mathbf{S'}\cup\mathbf{g'}}\Big)|_{\mathrm{Stab}_v} \\ &= \Big(\sum_{\substack{g'\in S/S' \\ g'\neq g}} \theta_{\mathbf{gv},\mathbf{S'}\cup\mathbf{g'}}\Big) + \theta_{\mathbf{v},\mathbf{S'}\cup\mathbf{g}}\end{aligned}$$

3. Checks that their roots are at a two-step distance from v, namely vertices of the form $g'gv$ where $g, g' \in S$, and $g \neq g'$. By the arguments presented in the second type, the checks are the $d' - 1$ faces which do not have the generators on the path leading from v to them, i.e., g, g', in their generator set.

$$\bigcup_{g,g'\in S'} \left\{\theta_{\mathbf{g'gv},\mathbf{S'}} : S' \subset S/\{g, g'\}, |S'| = d' - 1\right\}$$

And for each check, its support is:

$$\begin{aligned}\left(\partial^+ \theta_{\mathbf{g'gv},\mathbf{S'}}\right)|_{\mathrm{Stab}_v} &= \Big(\sum_{g''\in S/S'} \theta_{\mathbf{gg'v},\mathbf{S'}\cup\mathbf{g''}} + \sum_{g\in S'} \theta_{\mathbf{g''^{-1}gg'v},\mathbf{S'}\cup\mathbf{g''}}\Big)|_{\mathrm{Stab}_v} \\ &= \theta_{\mathbf{gv},\mathbf{S'}\cup\mathbf{g'}} + \theta_{\mathbf{g'v},\mathbf{S'}\cup\mathbf{g}}\end{aligned}$$

Definition 4.20 (Small Code at Vertex v)**.** *For a vertex v, we denote by $\mathcal{C}_v$ the small code at vertex v, which consists of all the d'-dimensional chains rooted at v that satisfy the checks rooted at v, namely:*

$$\begin{aligned}\mathcal{C}_v := \big\{ z : (\partial^- z)|_{v+} = 0 \ \textit{and} \\ z \in \textit{Span}\big(\left\{\theta_{\mathbf{v},\mathbf{S'}} : S' \subset S, |S'| = d'\right\}\big)\big\}\end{aligned}$$

The checks of the code are exactly the first type checks of Stab_v.

Claim 4.21. *Let v be a vertex. For any $d' \geq 2$, $d \geq d' + 1$, and any $z \in \mathcal{C}_v$, there is a stabilizer $w \in \mathrm{Stab}_v$ such that $z = w|_v$; namely, z and w agree on the faces rooted at v. Moreover, the vertex support of $w|_v$ is contained in the vertex support of $z|_v$, namely:*

$$\boldsymbol{Str}_{0+}(v, w) \subset \boldsymbol{Str}_{0+}(v, z)$$

Proof. By induction on the dimensions d, d'. We extend the notation of the stabilizer and the small codes at vertex v to $\mathcal{C}_v^J, \mathrm{Stab}_v^J$, where J is a subset of S, to refer to the codes derived from the skeleton generated by J. Showing that the belonging $z \in \mathcal{C}_v^J$ implies the existence of $w \in \mathrm{Stab}_v^J$ such that $w|_v = z|_v$ proves the claim.

1. Base. For the base cases, let d be any number greater than or equal to three, and let d' be two. First, observe that, in the case $d' = 2$, the code $\mathcal{C}_v^J$ is a Tanner code, where the bits are set on edges of the form $\{vg_i, vg_j\}$ and checks are associated with vertices of the form $\{vg_i\}$. Therefore, any codeword is a cycle. Let:

$$\{vg_1, vg_2\}, \{vg_2, vg_3\}, .., \{vg_k, vg_1\}$$

Be the cycle corresponding to z. Namely:

$$z = \theta_{\mathbf{v},\{\mathbf{g_1},\mathbf{g_2}\}} + \theta_{\mathbf{v},\{\mathbf{g_2},\mathbf{g_3}\}} + .. + \theta_{\mathbf{v},\{\mathbf{vg_k},\mathbf{vg_1}\}}$$

And consider the stabilizer w defined as:

$$w = \partial^- \left(\theta_{\mathbf{v},\{\mathbf{g_1},\mathbf{g_2},\mathbf{g_3}\}} + \theta_{\mathbf{v},\{\mathbf{g_1},\mathbf{g_3},\mathbf{g_4}\}} + .. + \theta_{\mathbf{v},\{\mathbf{g_1},\mathbf{g_i},\mathbf{g_{i+1}}\}} + .. + \theta_{\mathbf{v},\{\mathbf{g_1},\mathbf{g_{k-1}},\mathbf{g_k}\}}\right)$$

Clearly, $w \in \mathrm{Stab}_v^J$, and in addition its restriction to v equals:

$$\begin{aligned} w|_v &= \left(\sum_{i=2}^{k-1} \partial^- \theta_{\mathbf{v},\{\mathbf{g_1},\mathbf{g_i},\mathbf{g_{i+1}}\}}\right)|_v = \sum_{i=2}^{k-1} \theta_{\mathbf{v},\{\mathbf{g_i},\mathbf{g_{i+1}}\}} + \theta_{\mathbf{v},\{\mathbf{g_1},\mathbf{g_i}\}} + \theta_{\mathbf{v},\{\mathbf{g_1},\mathbf{g_{i+1}}\}} \\ &= \left(\sum_{i=3}^{k-1} \theta_{\mathbf{v},\{\mathbf{g_i},\mathbf{g_{i+1}}\}}\right) + \theta_{\mathbf{v},\{\mathbf{g_1},\mathbf{g_2}\}} + \theta_{\mathbf{v},\{\mathbf{g_k},\mathbf{g_1}\}} = z \end{aligned}$$

2. Assume the correction of the claim for dimensions lower than d, d'. Consider a generators subset $J \subset S$ and let $z \in \mathcal{C}_v^J$. In addition, consider a generator $g \in J$.

   Decompose $z = z|_{gv} + z|_{/gv}$ where $z|_{gv}$ and $z|_{/gv}$ are the chains consisting of all the faces in z that have support on gv and its complement. Define z^* to be the $d' - 1$-dimensional chain obtained by excluding gv from any face of $z|_{gv}$. Notice that $(\partial^- z^*)|_v$ has to be zero since it contains all the faces of $\partial^- z$ that contain gv (up to the removal). In other words, $z^* \in \mathcal{C}_v^{J/g}$. Thus, we can use the induction assumption and infer the existence of $w^* \in \mathrm{Stab}_v^{J/g}$ such that $z^* = w^*|_v$.

   Now, since $w^* \in \mathrm{Stab}_v^{J/g}$, then $\partial^- w^* = 0$. And since the local complex is contractible, w^* can be decomposed into a sum of stabilizers rooted in v. Let $\theta_{\mathbf{v},\mathbf{S_1'}}, \theta_{\mathbf{v},\mathbf{S_2'}}, \theta_{\mathbf{v},\mathbf{S_3'}}, \ldots, \theta_{\mathbf{v},\mathbf{S_l'}}$ be the d'-dimensional chains corresponding to the stabilizers composing w^*. Namely:

$$w^* = \sum_i \partial^- \theta_{\mathbf{v},\mathbf{S_i'}}$$

   Since those stabilizers are basis elements of $\mathrm{Stab}_v^{J/g}$, we have that $g \notin S_i'$ for any i. Hence, we can define the extended d'-dimensional chain w to be the chain obtained by adding g to each of the stabilizers deriving w^*. Namely:

$$w = \sum_i \partial^- \theta_{\mathbf{v},\mathbf{S_i'} \cup \mathbf{g}}$$

However, $\theta_{\mathbf{v},\mathbf{S}'_1\cup\mathbf{g}},\ldots,\theta_{\mathbf{v},\mathbf{S}'_\mathbf{i}\cup\mathbf{g}}$ are $d'+1$-dimensional chains corresponding to the stabilizers, which are basis elements of Stab_v^J. Therefore, $w\in\mathrm{Stab}_v^J$. In addition, opening up the boundary operator yields:

$$\begin{aligned} w|_v &= \sum_i \big(\partial^- \theta_{\mathbf{v},\mathbf{S}'_\mathbf{i}\cup\mathbf{g}}\big)|_v = \sum_i \Big(\sum_{g'\in S'_i/g} \theta_{\mathbf{v},\mathbf{S}'_\mathbf{i}/\mathbf{g}'\cup\mathbf{g}} + \theta_{\mathbf{v},\mathbf{S}'_\mathbf{i}}\Big) \\ &= \sum_{\theta_{\mathbf{v},\mathbf{S}'}\in w^*|_v} \theta_{\mathbf{v},\mathbf{S}'\cup\mathbf{g}} + \sum_i \theta_{\mathbf{v},\mathbf{S}'_\mathbf{i}} = z|_{gv} + \sum_i \theta_{\mathbf{v},\mathbf{S}'_\mathbf{i}} \end{aligned}$$

Thus, adding $w|_v$ to z results in:

$$z + w|_v = z|_{/gv} + \sum_i \theta_{\mathbf{v},\mathbf{S}'_\mathbf{i}}$$

Which is a d'-dimensional chain, supported on vertices of the form $g'v$ for $g'\in J/g$, namely a d'-dimensional chain on the complex induced by the $d-1$-dimensional skeleton. In addition, since the boundary operator can only advance the root of a face forward, we have that $\partial^- w|_v = (\partial^- w)|_v$ (as w is rooted in v) and therefore $\partial^-(z+w|_v) = \partial^- z + (\partial^- w)|_v = 0$.

To summarize, we have both $z+w|_v\in\mathcal{C}_v^{J/g}$ and $z+w|_v$ sets on a $d-1$-dimensional skeleton. Therefore, we can repeat in a recursive manner, with each step resulting in a d'-dimensional chain in a skeleton with dimension decreased by one. We stop when $d=d'$; in that case, the resulting chain has to be trivial.

□

We would like to differentiate chains whose projection onto the finite Toric with side length n closes a non-trivial cycle from those which don't. For this, we fix a rough threshold via the size function. Chains which have size less than n will be called finite. Observe that a chain might have a projection that doesn't close a non-trivial loop on the Toric and still be marked, according to our definition, as infinite. So, the classification is rough, yet it will turn out to be sufficient.

Definition 4.22. *Let z be a d'-dimensional chain. We will say that a chain z is finite if:*

$$\mathbf{Size}(z)\leq n-1$$

Similarly, z is infinite if it's size is greater than or equal to n.

Remark 4.23. *Observe that if $\mathbf{Size}(z)<n$, then the projection of z on the n-side length Toric does not close a non-trivial loop.*

The next claim states that the translation by stabilizer that minimizes its size of a finite chain admits a boundary with size equal to itself.

Claim 4.24. *Let $x \in (\mathcal{F}_{d'}; \mathbb{F}_2)$ be a d'-dimensional finite chain, consider the stabilizer $z \in \mathcal{C}_Z^{\perp}$ which adding it to x minimize the size , Namely:*

$$z = \arg\min_{z \in \mathcal{C}_Z^{\perp}} \mathbf{Size}(x + z)$$

Then:

$$\mathbf{Size}(x + z) = \mathbf{Size}(\partial^-(x + z))$$

Proof. First, observe that since the vertex domain of $\partial^- x$ is contained in the vertex domain of x, it follows that for any x the inequality $\mathbf{Size}(x) \geq \mathbf{Size}(\partial^- x)$ holds. Combining this with the fact that $\partial^- z = 0$ for any $z \in \mathcal{C}_Z^{\perp}$, we have that $\mathbf{Size}(x+z) \geq \mathbf{Size}(\partial^-(x + z))$. Now suppose that $\mathbf{Size}(x) > \mathbf{Size}(\partial^- x)$; that implies the existence of $v \in x/\partial^- x$ such that v is a local maximum of one of the potential walls in x and $(\partial^- x)|_v = 0$. Let's split into cases, depending on the potential wall type.

1. If v is a maximum of L_{d+1}, then by Claim 4.21, there exists a stabilizer z rooted at v, such that $z|_v = z|_{v+} = x|_{v+} = x|_v$ (where we used that v is a local maximum of L_{d+1} in x to infer $x|_v = x|_{v+}$). Therefore, $(x + z)|_v = 0$, meaning adding z to x excludes v. In addition, since the rooted stabilizer z is supported only on the vertices in $\mathbf{Str}_{0+}(v, x)$, the addition cannot introduce a vertex with a higher L_g value than the maximal L_g value in $\mathbf{Str}_{0+}(v, x)$ into x. So, repeating the process until excluding all the vertices that get the maximal L_{d+1}-value in x, and summing the stabilizers, results in a stabilizer z for which $L_{d+1}(x+z) < L_{d+1}(x)$.

2. If, for some generator $g \in S$, v is a maximum of L_g, then without loss of generality, pick v among the maximum points of L_g with the highest value of $-L_{d+1}$ (maximize $\sum_{g \in S} L_g(v)$). Since stepping through the positive direction can only increase the value of $-L_{d+1}$, it follows that v is also a local max of $-L_{d+1}$. Now consider the skeleton system where the directions are flipped, and $\theta_{v,S'}$ is set to be the face containing the vertices that are reached by the selection of reverse steps from S' (i.e., $g^{-1}v \in \theta_{v,S'}$). Similarly, for the sake of the proof, let's present the following notation for the positive neighborhood (Definition 4.5) in the reversed system:

   $$\mathbf{Str}_{0-}(v, x) := \left\{u \in x \colon \exists g \in S \text{ such } u = g^{-1}v\right\}$$

   So, $\mathbf{Str}_{0-}$ plays the same role as $\mathbf{Str}_{0+}$ does in the original skeleton. In that system, we can apply Claim 4.21 (in the reversed direction) and get that there is a stabilizer z such that adding it to x excludes v from x. By the same argument as above, no vertex with a higher $L_{g'}$-value (although L_{d+1} can increase, that's fine[10]) than what it already has in $\mathbf{Str}_{0-}(v, x)$ is introduced into x. Repeating the process excludes all the vertices with the maximal value of L_g in x, so eventually, we end up with $x + z$, such that $L_g(x + z) < L_g(x)$.

We keep excluding vertices until we get the desired equality.

□

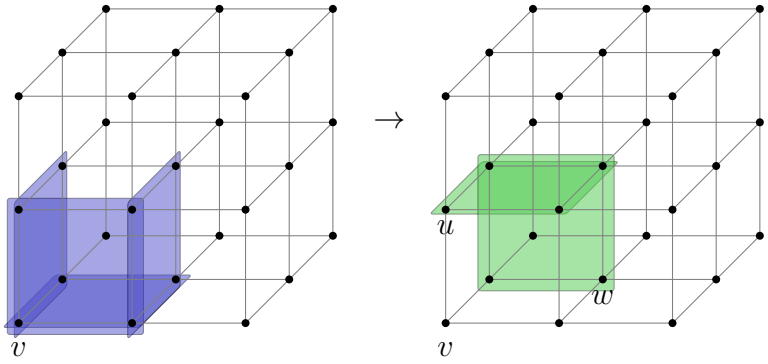


Figure 6: Consider the $(3, 2)$-Toric code, with qubits and X-checks set on the 2- and 1-dimensional faces. In blue, we have a 2-dimensional chain where the vertex achieving the maximum of L_{d+1}, denoted by v, does not see any unsatisfied checks in its positive local view. The green faces represent a 2-dimensional chain, which is the **reduced** form of the blue. In particular, any local maximum of L_{d+1} (vertices u and w) sees a non-trivial syndrome in its positive local view. Namely, the maximal points of L_{d+1} in the chain are also the maximal points in the boundary ∂^-.

We close the section by defining the *reduced* form of a finite d'-dimensional chain.

Definition 4.25. *Let $x \in (\mathcal{F}_{d'}; \mathbb{F}_2)$ be a d'-dimensional finite chain. We will say that x is reduced if for any vertex $v \in x$ that is a maximum point of a potential wall $L \in L_1, \ldots, L_{d+1}$, v is also in $\partial^- x$. In particular, we have that* $\mathbf{Size}(x) = \mathbf{Size}(\partial^- x)$ *and that the reduced form of a given chain is obtained by the construction presented in the proof of Claim 4.24.*

We emphasize that the reduced form is not unique. Throughout the paper, every time we choose a reduced form of a chain, we mean obtaining it from the given algorithm presented in Claim 4.24, so the reduced form is effectively unique.

4.4 Shrinking

We are about to prove that the sweep rule shrinks finite errors. The following claim relates the local maximum points of the potential walls $L_1, \ldots, L_d$ in an assignment z to the structure of the syndrome they observe. Later, we will use that relation to infer that the decoding step of the sweep rule can't spread the error to a vertex[11] with a higher value than the current global maximum of $L_1, \ldots, L_d$. In other words, the error does not spread in the positive directions.

Claim 4.26. *Let z and ξ be d'-dimensional finite chains such that ξ results from the application of the sweep rule on z. For any generator $g \in S$, the maximal values of L_g in ξ and $\partial^- \xi$ are lower than the maximal values in z and $\partial^- z$. Namely:*

$$\max\{L_g(u) : u \in \xi\} \leq \max\{L_g(u) : u \in z\}$$
$$\max\{L_g(u) : u \in \partial^- \xi\} \leq \max\{L_g(u) : u \in \partial^- z\}$$

[10] If it happens, we will apply the first case in the next exclusion.

[11] To faces whose vertices have higher values of $L_1, \ldots, L_d$ than the current global maximum values.

Proof. First, we argue that for a vertex v in $\partial^- z$, and a generator $g \in S$, such that v is a local maximum of L_g in either z or $\partial^- z$, it implies that $gv \notin \partial^- z$. The contradiction where v is a maximum of L_g in z follows immediately, since containment in $\partial^- z$ implies containment in z. Where v is a maximum in $\partial^- z$, suppose through contradiction that $gv \in \partial^- z$, then

$$L_g(gv) = (gv)_g = 1 + v_g = 1 + L_g(v)$$

In contradiction to v being a local maximum of L_g in $\partial^- z$.

For a vertex $v \in \partial^- z$ denote the correction it applies by $\varphi_v \in (\mathcal{F}_{d'}; \mathbb{F}_2)$, so

$$\xi = \sum_{v \in \partial^- z} \varphi_v + z \quad \text{and} \quad \partial^- \xi = \sum_{v \in \partial^- z} \partial^- \varphi_v + \partial^- z$$

So for any vertex $u \in \xi / z$, there is a vertex $v \in \partial^- z$ such that $u \in \varphi_v$. Moreover, since ϕ_v is a rooted chain, it follows that $L_g(u) \leq L_g(v) + 1$, where the equality holds when g belongs to the generator's subset of the face φ_v. Similarly for $u \in \partial^- \xi / \partial^- z$.

Now, combining with the sweep rule's first condition, which restricts the correction applied by a vertex to have a vertex support that is contained in the positive syndrome it sees, we have that for any vertex v which is a local maximum of L_g (in either z or $\partial^- z$), any $u \in \varphi_v$ satisfies $L_g(u) = L_g(v)$. In other words, ξ and $\partial^- \xi$ don't contain vertices of the form gv where v is a local maximum of L_g in z and $\partial^- z$, and the desired inequalities follow. □

Next, we prove that for a finite X-error, the sweep-cycle advances the corner (the maximum of L_{d+1}) of its reduced form. We assume we are given a finite error in its reduced form, and therefore whose corner sees a syndrome, and show that the sweep-rule application excludes the corner. In other words, the sweep-rule shrinks X-errors.

Claim 4.27. *Let z be a finite reduced d'-dimensional chain, and let ξ be the reduced result of applying the sweep rule to it. Then:*

1. *For any generator $g \in S$:* $\max\left\{L_g(v) : v \in z\right\} \geq \max\left\{L_g(v) : v \in \xi\right\}$
2. *And for L_{d+1} :* $\max\left\{L_{d+1}(v) : v \in z\right\} \geq \max\left\{L_{d+1}(v) : v \in \xi\right\} + 1$

In particular, $\mathbf{Size}(\xi) \leq \mathbf{Size}(z) - 1$.

Proof. The first inequality was proved in Claim 4.26, so it's left to prove the second. Denote by φ the applied correction and by v the vertex on which L_{d+1} gets the maximal value in z, if there are more then one vertex then choose v arbitrarily. Since z is reduced, then by Claim 4.24 it holds that $v \in \partial^- z$. Moreover, since v is a local max of L_{d+1} in z the two follows:

1. Claim 4.7 guarantees that $\partial^- \varphi|_v = \partial^- z|_v$. Thus, adding $\varphi|_v$ to z excludes v from the boundary.

2. For any other vertex $u \in z$, there is no face $\theta_{\mathbf{u},\mathbf{S}'} \in z$ such that $v \in \theta_{\mathbf{u},\mathbf{S}'}$. Otherwise, it would contradict v being a local maximum of L_{d+1} in z. Thus, $v \notin \mathbf{Str}_{0+}(u, \partial^- z)$, so by Claim 4.7, $v \notin \varphi|_u$.

Combining the two, we get that $v \notin \partial^-(z+\varphi)$, so the reduced form of $z+\varphi$ does not contain v. Namely, $v \notin \xi$. Repeating the above for all the other vertices which get the maximal L_{d+1}-value in z yields that $L_{d+1}(\xi) \leq L_{d+1}(z) - 1$.

□

Consider a vertex on the boundary of a chain resulting from an application of the sweep rule. That vertex did not come from nowhere; there must be vertices on the preceding chain, which sit on its boundary, and are the cause for the local view seen by the vertex. The following claim formulates that insight; moreover, it states a relation between the potential walls' values of the vertex and the (part of) vertices who caused it.

Claim 4.28. *Let z be a finite d'-dimensional chain, and let ξ be the result of applying the sweep rule to it. Let v be a vertex in $\partial^-\xi$, then there is a vertex $u \in \partial^- z$ such $L_i(u) \geq L_i(v)$ for $i \neq d+1$ and $L_{d+1}(u) > L_{d+1}(v)$ if $i = d+1$, and there is a d'-dimensional face f such $v, u \in f$.*

Proof. Without loss of generality, we can assume that z and ξ are reduced. We start by proving for $i = d+1$. Now, it splits upon whether $v \in z + \xi$:

1. If $v \in z+\xi$, then v belongs to a face that flipped during the correction, denote it's root by u and by definition of the sweep-rule $(\partial^- z)|_{u+} \neq 0$. Therefore, $u \in \partial^- z$, and u, ξ share a face which u is it's root. Since u it's the root of the face we get $L_{d+1}(u) > L_{d+1}(v)$.

2. Else, if $v \notin z + \xi$, then v has to be in z. Combining with the fact that $v \in \xi$, it follows that v is not a local maximum of L_{d+1} in $\partial^- z$, otherwise the application of the sweep-rule would exclude v (See the proof of Claim 4.27), so it wouldn't be in ξ. Pick u to be the neighbor of v in $\partial^- z$ such that $L_{d+1}(u) > L_{d+1}(v)$.

It remains to prove for $i \in [d]$, in this case if $v \in \partial^- z$ then take $u = v$. Otherwise, there must be a face f that was flipped during the sweep-rule containing v. Denote by g the corresponding generator for i and by u' the root of f. If $L_g(u') = L_g(v)$ then set $u = u'$. Otherwise, g is one of the generators that span f, the second condition of the sweep-rule (Definition 4.6) requires that for u' flipping a face in the g direction, it has to see the vertex gu' in $\partial^- z$, in that case, set $u = g'u$. □

We finished analyzing the first part of the sweep-cycle that acts in the computational basis. In the next subsection, we introduce the transformation into the dual code. Since the dual code is a $(d, d-d')$-Toric code, we also find that the sweep-cycle shrinks reduced phase errors.

4.4.1 Ideal decoder

Although the main role of the section is to show that a single sweep-cycle keeps error clusters under control during a noisy run (by admitting a modest shrinking), an immediate corollary of this is that the application of non-noisy multiple iterations of the sweep-cycle has the potential to erase errors, namely a decoder. Here, a classical measurement is given, we act on classical bits, and the sweep-cycle consists only of the first stage that works over the computational basis, without moving to the dual code. We formulate the corollary as follows:

Corollary 4.29. *Consider the (d, d')-Toric code, induced by a Toric with side length n. Given a finite connected reduced error $z \in (\mathcal{F}_{d'}; \mathbb{F}_2)$ on that Toric, with $\mathbf{Size}(z) \leq n-1$. Then an application of $\mathbf{Size}(z)$ sweep cycles (partial cycles consisting only of the first stage) erases the error.*

When z is not connected, disjoint components might merge, and it is no longer true that the size of the chain decreases in each iteration. However, we will see later (Claim 7.3, Claim 7.4, and Lemma 6.2) that performing $\Theta(n^{2d+1})$ sweep-cycles erases the error with high probability, if z is assumed to be distributed according to local stochastic noise.

4.5 Logical Gates

The $\theta_{\mathbf{v},\mathbf{S}'}$ notation gives a nice interpretation of the logical code words. Particularly, an X-type logical codeword is given as follows: Take a subset of generators $S' \subset S$ and consider the subgroup generated by it, $\langle S' \rangle$; let us denote the subgroup by K. Then the logical codeword is the X-product over all $\theta_{\mathbf{u},\mathbf{S}'}$ such that $u \in K$. Moreover, replacing K by a coset whose base subgroup is $\langle S' \rangle$, we get another presentation of the same logical codeword.

Claim 4.30. *For any generator subset S' of size d', denote by $x_{S'}$ and $z_{S'}$ the following d'-dimensional chains:*

$$x_{S'} := \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S}'} \qquad z_{S'} := \sum_{u \in \langle S/S' \rangle} \theta_{\mathbf{u},\mathbf{S}'}$$

Then $x_{S'}$ and $z_{S'}$ are binary representations of anticommuting logical codewords of the (d, d')-Toric code. In particular, $x_{S'} \in \mathcal{C}_X/\mathcal{C}_Z^{\perp}$, $z_{S'} \in \mathcal{C}_Z/\mathcal{C}_X^{\perp}$, and $x_{S'} z_{S'} = 1$. In addition, let $X_{S'}$ and $Z_{S'}$ be the Pauli matrices obtained by multiplying each non-trivial coordinate of them by X or Z, respectively.

Proof. By definition, $x_{S'}$ and $z_{S'}$ are binary assignments on the d'-dimensional faces. We start by showing that $x_{S'} \in \mathcal{C}_X$ and $z_{S'} \in \mathcal{C}_Z$; namely, that the boundaries $\partial^- x_{S'}$

and $\partial^+ z_{S'}$ vanish.

$$\begin{aligned}\partial^- \sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S}'} &= \sum_{u\in\langle S'\rangle}\sum_{g\in S'} \theta_{\mathbf{u},\mathbf{S}'/\mathbf{g}} + \theta_{\mathbf{gu},\mathbf{S}'/\mathbf{g}} \\ &= \sum_{g\in S'}\sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S}'/\mathbf{g}} + \sum_{u\in\langle S'\rangle} \theta_{\mathbf{gu},\mathbf{S}'/\mathbf{g}} \\ &= \sum_{g\in S'}\sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S}'/\mathbf{g}} + \sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S}'/\mathbf{g}} = 0\end{aligned}$$

$$\begin{aligned}\partial^+ \sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S}/\mathbf{S}'} &= \sum_{u\in\langle S'\rangle}\sum_{g\in S'} \theta_{\mathbf{u},\mathbf{S}/\mathbf{S}'\cup\mathbf{g}} + \theta_{\mathbf{g^{-1}u},\mathbf{S}/\mathbf{S}'\cup\mathbf{g}} \\ &= \sum_{g\in S'}\sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S}/\mathbf{S}'\cup\mathbf{g}} + \sum_{u\in\langle S'\rangle} \theta_{\mathbf{g^{-1}u},\mathbf{S}/\mathbf{S}'\cup\mathbf{g}} \\ &= \sum_{g\in S'}\sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S}/\mathbf{S}'\cup\mathbf{g}} + \sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S}/\mathbf{S}'\cup\mathbf{g}} = 0\end{aligned}$$

Thus $x_{S'} \in \mathcal{C}_X$ and $z_{S'} \in \mathcal{C}_Z$, now for showing that they are not binary representations of stabilizers, it's enough to show they admit no vanishing product.

$$x_{S'} z_{S'} = \sum_{u\in\langle S'\rangle} \theta_{\mathbf{u},\mathbf{S}'} \sum_{u\in\langle S/S'\rangle} \theta_{\mathbf{u},\mathbf{S}'} = \theta_{\mathbf{e},\mathbf{S}'}\theta_{\mathbf{e},\mathbf{S}'} = 1$$

Therefore, the Paulis $X_{S'}$ and $Z_{S'}$ are X- and Z-type anticommuting logical code words. □

The (d, d')-Toric code admits a fold-transversal realization of an isomorphism gate into the $(d, d-d')$-Toric code, for any dimensions d, d'. Particularly, the gate exchanges logical Pauli $X_{S'}$ (of the first code) with logical Pauli $Z_{S/S'}$ (of the second code) and vice versa. It is realized by an application of transversal Hadamards followed by permuting the qubits according to a transformation which we name the *ϕ-transpose map*. We define the transpose map as follows:

Definition 4.31 (The Transpose Map)**.** *Let d, d' be integers such that $d' < d$, and let $\mathcal{G}$ be a (d, d')-Toric complex generated by the generator set S. We define the transpose map $\phi : (\mathcal{F}^n_{d'}; \mathbb{F}_2) \to (\mathcal{F}^n_{d-d'}; \mathbb{F}_2)$ by acting on the basis elements as follows:*

$$\phi(\theta_{\mathbf{v},\mathbf{S}'}) \to \theta_{\mathbf{S}'\mathbf{v},\mathbf{S}/\mathbf{S}'}$$

Notice that the action of the transpose map over the qubits is a one-to-one map between the qubits of the (d, d')-Toric code and the qubits of the $(d, d-d')$-Toric code, and therefore can be realized at constant depth using ancillas.

Definition 4.32 (Hadamard-Type Gate $\mathbf{H}_\phi$)**.** *We define the logical Hadamard-type gate $\mathbf{H}_\phi$: $L(\mathcal{H}) \to L(\mathcal{H})$, to be a code isomorphism (d, d')-Toric $\to$ $(d, d-d')$-Toric, as follows, for any $S' \subset S$, of size d':*

$$\mathbf{H}_\phi(X_{S'}) \to Z_{S/S'} \qquad \mathbf{H}_\phi(Z_{S'}) \to X_{S/S'}$$

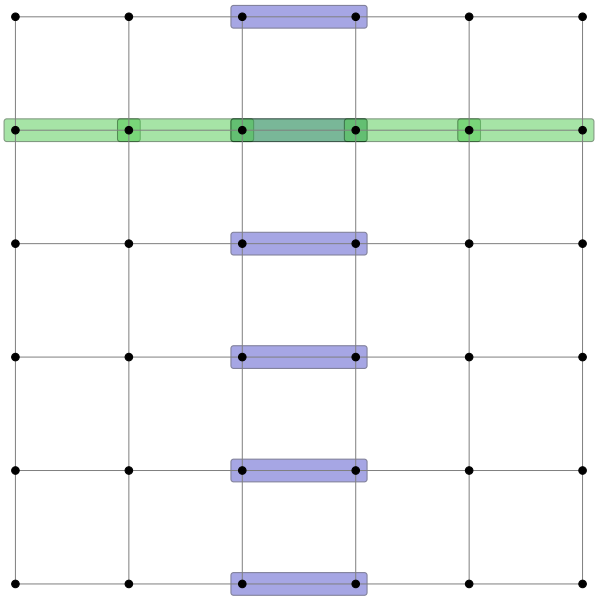


Figure 7: In blue and green are the binary presentations x_S and z_S of logical code words of the $(2,1)$-Toric code. The edge $\{(5,4),(0,4)\}$ is omitted for clarity (it should be a green edge that hovers over the green line).

Claim 4.33. *The Hadamard-type gate $\mathbf{H}_\phi$ has a fold-transversal implementation by, first applying $\mathbf{H}^{\otimes \mathcal{F}^n_{d'}}$ over all the physical qubits, and then permuting them by the transpose map ϕ, namely $\hat{\mathbf{H}}_\phi = \phi \circ \mathbf{H}^{\otimes \mathcal{F}^n_{d'}}$.*

Proof. Clearly, for any subset of generators $S' \subset S$, the map ϕ sends $x_{S'}$ to $z_{S/S'}$:

$$\phi(x_{S'}) = \phi(\sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S}'}) = \sum_{u \in \langle S' \rangle} \theta_{\mathbf{S}'\mathbf{u},\mathbf{S}/\mathbf{S}'} = \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S}/\mathbf{S}'} = z_{S/S'}$$

Moreover, ϕ sends binary representations of Z-stabilizers of the (d,d')-Toric code to binary representations of X-stabilizers of the $(d,d-d')$-Toric code, and vice versa. To see this, consider a $d'+1$ dimensional face, and consider the image of the stabilizer it defines.

$$\begin{aligned}
\phi\left(\partial^- \theta_{\mathbf{v},\mathbf{S}'}\right) &= \phi\left(\sum_{g \in S'} \theta_{\mathbf{v},\mathbf{S}'/\mathbf{g}} + \theta_{\mathbf{gv},\mathbf{S}'/\mathbf{g}}\right) \\
&= \sum_{g \in S'} \theta_{\mathbf{g}^{-1}\mathbf{S}'\mathbf{v},(\mathbf{S}/\mathbf{S}')\cup\mathbf{g}} + \theta_{\mathbf{g}^{-1}\mathbf{S}'\mathbf{gv},(\mathbf{S}/\mathbf{S}')\cup\mathbf{g}} \\
&= \partial^+ \theta_{\mathbf{S}'\mathbf{v},\mathbf{S}/\mathbf{S}'}
\end{aligned}$$

And for the checks set on the $d'-1$-dimensional face we have:

$$\begin{aligned}\phi\left(\partial^+\theta_{\mathbf{v},\mathbf{S}/\mathbf{S}'}\right) &= \phi\left(\sum_{g\in S'}\theta_{\mathbf{g}^{-1}\mathbf{v},(\mathbf{S}/\mathbf{S}')\cup\mathbf{g}} + \theta_{\mathbf{v},(\mathbf{S}/\mathbf{S}')\cup\mathbf{g}}\right)\\ &= \sum_{g\in S'}\theta_{(\mathbf{S}/\mathbf{S}')\mathbf{v},\mathbf{S}'/\mathbf{g}} + \theta_{\mathbf{g}(\mathbf{S}/\mathbf{S}')\mathbf{v},\mathbf{S}'/\mathbf{g}}\\ &= \partial^-\theta_{(\mathbf{S}/\mathbf{S}')\mathbf{v},\mathbf{S}'}\end{aligned}$$

Thus, $\phi\circ\mathbf{H}^{\otimes\mathcal{F}^n_{d'}}$ sends Z-stabilizers (at dimension d') to X-stabilizers (at dimension $d-d'$) and vice versa, thereby sending the satisfied checks to satisfied checks. Combining this with the above, $\phi\circ\mathbf{H}^{\otimes\mathcal{F}^n_{d'}}$ realizes the logical isomorphism $\mathbf{H}_\phi$ from the (d,d')-Toric code into the $(d,d-d')$-Toric code. □

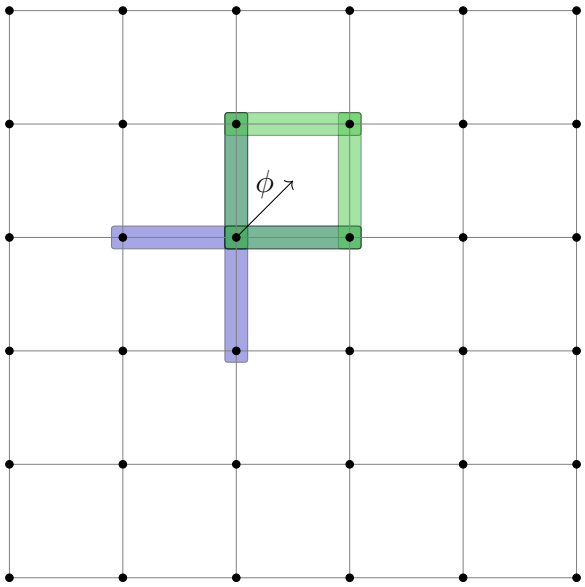


Figure 8: An outline of Example 4.34. The ϕ-transpose sends Z-stabilizers to X-stabilizers and vice versa. Here we consider the $(2,1)$ Toric code; namely, the Z-stabilizers (parity checks) are set on the vertices, and the X-stabilizers (phase checks) are set on the squares. In blue and green, we have the edges corresponding to checks $\partial^+\theta_{(\mathbf{2},\mathbf{3}),\{\}}$ and $\partial^-\theta_{(\mathbf{2},\mathbf{3}),\{(\mathbf{0},\mathbf{1}),(\mathbf{1},\mathbf{0})\}}$. The example shows that indeed ϕ maps between the two.

Example 4.34. *Consider the* $(2,1)$*-Toric code. Name the generators in* S *by* $g_1 = (1,0)$ *and* $g_2 = (0,1)$*. Consider the check corresponding to the* Z*-stabilizer* $\partial^+\theta_{(\mathbf{2},\mathbf{3}),\{\}}$*.*

$$\partial^+\theta_{(\mathbf{2},\mathbf{3}),\{\}} = \{\theta_{(\mathbf{2},\mathbf{3}),\{(\mathbf{1},\mathbf{0})\}}, \theta_{(\mathbf{2},\mathbf{3}),\{(\mathbf{0},\mathbf{1})\}}, \theta_{(\mathbf{1},\mathbf{3}),\{(\mathbf{1},\mathbf{0})\}}, \theta_{(\mathbf{2},\mathbf{2}),\{(\mathbf{0},\mathbf{1})\}}\}$$

The ϕ-transpose map acts on it as follows:

$$\phi(\theta_{(\mathbf{2},\mathbf{3}),\{(\mathbf{1},\mathbf{0})\}}) = \theta_{(\mathbf{3},\mathbf{3}),\{(\mathbf{0},\mathbf{1})\}}$$
$$\phi(\theta_{(\mathbf{2},\mathbf{3}),\{(\mathbf{0},\mathbf{1})\}}) = \theta_{(\mathbf{2},\mathbf{4}),\{(\mathbf{1},\mathbf{0})\}}$$
$$\phi(\theta_{(\mathbf{1},\mathbf{3}),\{(\mathbf{1},\mathbf{0})\}}) = \theta_{(\mathbf{2},\mathbf{3}),\{(\mathbf{0},\mathbf{1})\}}$$
$$\phi(\theta_{(\mathbf{2},\mathbf{2}),\{(\mathbf{0},\mathbf{1})\}}) = \theta_{(\mathbf{2},\mathbf{3}),\{(\mathbf{1},\mathbf{0})\}}$$

What gives edges on a rectangle that converges with the bits on the support of the X-stabilizer $\theta_{(\mathbf{4},\mathbf{3}),\{(\mathbf{0},\mathbf{1}),(\mathbf{1},\mathbf{0})\}}$.

Equipped with $\mathbf{H}_\phi$, we gain a mapping into the dual code that both keeps the Toric skeleton and sends the Z-checks to $d'-1$ dimensional faces. This enables the use of the same local decoder to act on both $\mathcal{C}_X$ and its dual. We summarize it in the following immediate corollary:

Corollary 4.35. *Consider an Pauli $P \in \mathcal{P}$, acting on the infinite complex, whose X- and Z-components are set on the reduced chains $x, z \in (\mathcal{F}_{d'}; \mathbb{F}_2)$. Denote by $\tilde{P}$ the Pauli operator result by an application of the sweep-rule on P and by $\tilde{x}, \tilde{z} \in (\mathcal{F}_{d'}; \mathbb{F}_2)$ the chains corresponding to the X- and Z-components of $\tilde{P}$. If both $x, \phi(z)$ are finite chains, Then:*

1. *For any $i \in [d]$:*

$$\max_{v \in x} L_i(v) \geq \max_{v \in \tilde{x}} L_i(v) \qquad \max_{v \in \phi(z)} L_i(v) \geq \max_{v \in \phi(\tilde{z})} L_i(v)$$

2. *And for $i = d+1$:*

$$\max_{v \in x} L_{d+1}(v) \geq \max_{v \in \tilde{x}} L_{d+1}(v) + 1 \qquad \max_{v \in \phi(z)} L_{d+1}(v) \geq \max_{v \in \phi(\tilde{z})} L_{d+1}(v) + 1$$

In particular:

$$\mathbf{Size}(\tilde{x}) \leq \mathbf{Size}(x) - 1 \qquad \mathbf{Size}(\phi(\tilde{z})) \leq \mathbf{Size}(\phi(z)) - 1$$

Throughout the construction, we encode each qubit in a code block; namely, we restrict ourselves to using only a single codeword of the (d, d')-Toric code. From now on, we will think of $S' \subset S$ as a fixed d'-size subset of generators that were picked from the head. And we will decode the logical $|1\rangle$ state by $x_{S'}$. To simplify terminology, we call such usage 'nice encoding'.

Definition 4.36 (The Nice Encoding)**.** *Consider (d, d')-Toric code and fix a subset $S' \subset S$, at size d'. We define the nice encoding to be the subsystem code, which uses $x_{S'}$ to encode a single logical qubit. Namely, the logical codewords are:*

$$|\bar{0}\rangle = \left|\mathcal{C}_Z^{\perp}\right\rangle$$
$$|\bar{1}\rangle = \left|x_{S'} + \mathcal{C}_Z^{\perp}\right\rangle$$

We are now ready to give a full description of the realization of the logical operators. The logical Paulis $\mathbf{X}$ and $\mathbf{Z}$ are realized by the application of $X_{S'}$ and $Z_{S'}$. For the gates $\mathbf{S}$ and $\mathbf{T}$, we use single-qubit gate teleportation as presented in Figure 12. In single-qubit gate teleportation, the resource $|\mathbf{S}\rangle$ is initialized as $\mathbf{S}|+\rangle$, similarly for $|\mathbf{T}\rangle$. The gate $\mathbf{1}_{S'}$ stands for the classical procedure that, given the measurement results, checks whether the logical bit $|1\rangle$ is turned on. We will return to this gate with a higher resolution depiction of it after the description of the Hadamard teleportation.



Figure 9: Simulating the $\mathbf{S}$ and $\mathbf{T}$ gates, using the $|\mathbf{S}\rangle$, $|\mathbf{T}\rangle$ magic states.

For the Hadamard gate, we use a modification of the standard two-qubit gate teleportation as shown in Figure 13. The realization differs from the standard realization as the measurement in the phase basis is done by first moving to the dual space via the application of physical $\mathbf{H}^{\otimes \mathcal{F}^n_{d'}}$, measuring, reorganizing the qubits through ϕ, and then checking for the $x_{S/S'}$ outcome (thus, it can be thought of as the application of $\mathbf{H}_\phi$ and then measurement). In addition, unlike the single-qubit gate teleportation, here $|\mathbf{H}\rangle$ is initialized as $I \otimes \mathbf{H}(|00\rangle + |11\rangle)$ (up to normalization).

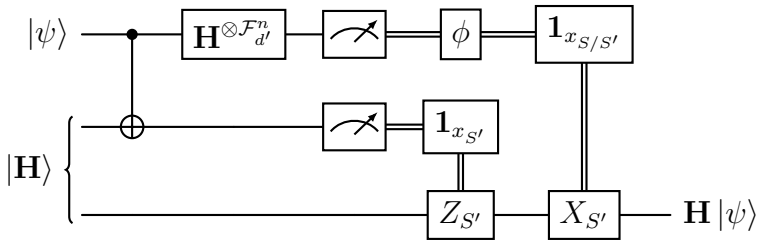


Figure 10: Gate teleportation of the Hadamard gate, using the resource $|\mathbf{H}\rangle$. We emphasize that although the Hadamard (at least Hadamard-type) has an $O(1)$-depth gate realization (at least when $d' = d/2$), we still need all the operations on the quantum register to be strictly transversal for the sake of the proof. Thus, we realize the Hadamard via gate teleportation.

The classical procedure of logical codeword checking, namely $\mathbf{1}_{x_{S'}}$ / $\mathbf{1}_{x_{S/S'}}$, is divided into five stages. In the first, we find all the connected components of the ∂^--boundary. Finding connected components can be done in $O(\log^2(n))$ depth (see [SV82] and [AS87] for logarithmic time connected components in the PRAM model and [Lev80] for the simulation of PRAM machines using bounded-fan circuits). Then, we broadcast the connected components. Conceptually, each connected component is sent to its own Toric; technically, we copy components of the syndromes into clean registers. The

broadcast also requires at most $O(\log(n))$ depth since there can be at most #checks $= O(n^d)$ connected components and therefore at most $O(n^d)$ copies. Then, we decode each of the copies in parallel. The separation of the components guarantees that during decoding, components will not merge with each other. For the decoding, we iterate sweep-rule cycles (defined in Definition 4.6) until erasing the syndrome. Later, we will see that n cycles succeed to erase the syndrome with high probability[12]. We erase the syndrome by combining the results of all the decoding sessions together by Xoring their correction suggestions. Each of the Xor operations is between at most $O(n^d)$ qubits and therefore requires at most $O(\log(n))$ depth. At this point, we assume that the combined register stores the measurement without errors. We finish by getting back into the logical space, namely a multiplication of binary matrix, which can be done at logarithmic depth $O(\log(n))$ (at the cost of broadcasting into $O(n^{2d})$-wide again). So the overall depth for performing a logical measurement is $O(n)$. The additional classical width cost is due to the broadcasting which introduce at most $O(n^{2d})$ additional classical bits.

[12]In the proof of the Theorem 6.3 we show that $\Theta(n)$-size connected component of the syndrome is not likely to occur.

4.6 Propagation Through Gate-Teleportation

Having an explicit description of the gate's realizations enables tracking how faults propagate through the computation. The next claim formulates the transitions of faults via the gate-teleportation gadgets.

Claim 4.37. *The propagation of a Pauli through the (conditioned) gate teleportation gadgets has a well-defined structure.*

Proof. For chains $x, z \in (\mathcal{F}_d^n; \mathbb{F}_2)$, we denote by $P_{x,z}$ the Pauli consisting of X's and Z's on the faces of x and z. We prove the claim by propagating Paulis manually. We start by propagating Pauli through the **S**-teleportation, let P_{x_1,z_1} and P_{x_2,z_2} be the Paulis that act on the first and the second registers. Then:

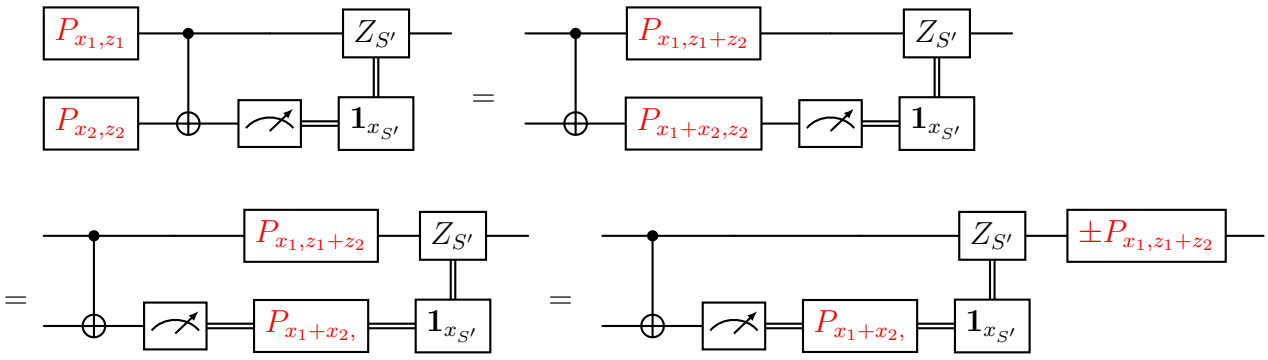


Figure 11: Pauli propagation trough **S** teleportation.

For the **T**-teleportation, we denote by P_{x_3,z_3} the Pauli act on the third register containing the resource state $|\mathbf{S}\rangle$, which might be used through the gadget to 'correct' the teleportation. Let $\mathbf{1}^2_{x_{S'}}$ be the indicator for the event in which measuring the resource register (that prior to the entangle, holds the state $P_{x_2,z_2}|\mathbf{T}\rangle$) results in $|\bar{1}\rangle$. Then:

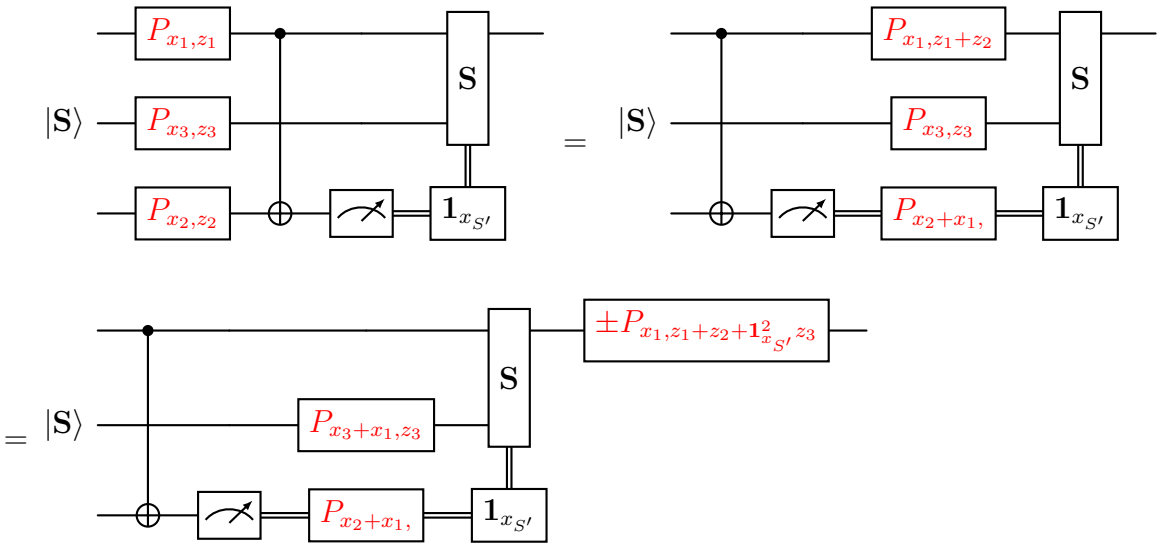


Figure 12: Pauli propagation trough **T** teleportation.

Where to infer the last passage we used to the Pauli-propagation through **S**-teleportation. Finally, for the **H**-teleportation we have:

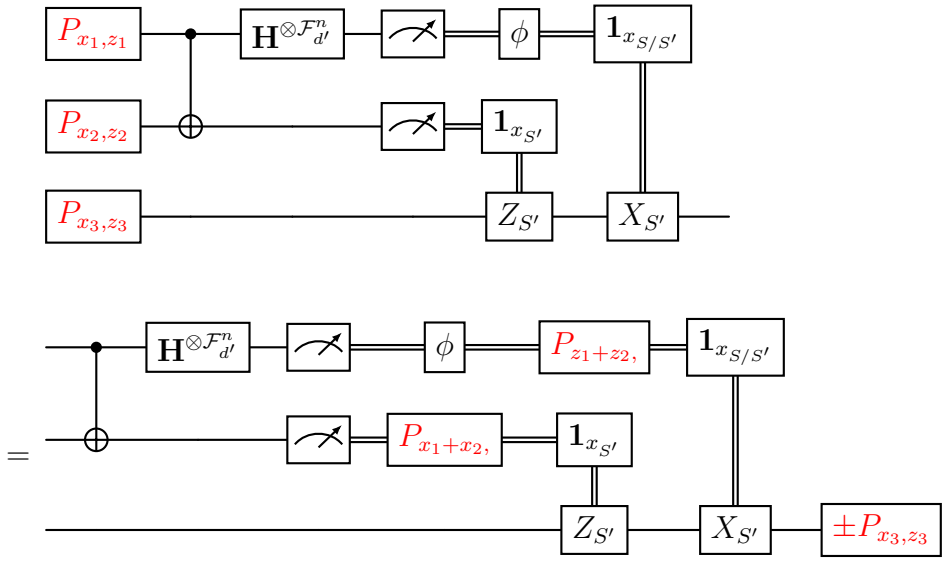


Figure 13: Pauli propagation trough **H** teleportation.

Notice that the Pauli P_{z_3,x_3} consists of applying X's on the d'-dimensional faces of the chain z_3, and applying Z's on the d'-dimensional faces of x_3. □

We summarize by restating the claim above for Clifford + measurement gates, which also use sweep-cycles.

Corollary 4.38. *Let C be a quantum circuit, such that all its registers are encoded via the nice encoding, and the gates of C are either logical Pauli, transversal* **CX**, *sweep-cycles, or* $\mathbf{H}, \mathbf{S}$, *and* $\mathbf{T}$ *realized via gate-teleportation. Then the propagation of Pauli through a (conditioned) subjection of C has a well defined structure. In particular, the logical operators propagate Paulis in transversal manner.*

5 Toom's Contours

5.1 The Barrier Argument

We are going to move between the infinite and the finite back and forth, so we will need to extend the size function, and the reduced notion to chains on the finite-complex.

Definition 5.1. *Consider a d'-dimensional chain on the d-dimensional Toric $x \in (\mathcal{F}^n_{d'}; \mathbb{F}_2)$. We extend the notation of size to x as follows. Let x' be the ∞-lift of x, so $x' \in (\mathcal{F}_{d'}; \mathbb{F}_2)$. Let the chain $y \in (\mathcal{F}_{d'}; \mathbb{F}_2)$ be a subset chain of x' such:*

1. *The projection of y back to finite complex covers x.*
2. *The size of y is minimal.*

Then, we set:

$$\mathbf{Size}(x) := \mathbf{Size}(y)$$

The chain y will be referred to as the minimizing cover of x. We extend the notation of reduced form to a d'-dimensional chain on the finite complex $x \in (\mathcal{F}^n_{d'}; \mathbb{F}_2)$, to be its translation by stabilizers that gets the minimal size. Namely, for $z \in Im\ \partial^- : (\mathcal{F}^n_{d'+1}; \mathbb{F}_2) \to (\mathcal{F}^n_{d'}; \mathbb{F}_2)$, the chain $x+z$ is a reduced form of x if $\mathbf{Size}(x+z)$ is minimal among the choices of z.

In the following, we prove that the definition of the size of chains in the finite complex respects the triangle inequality.

Claim 5.2. *Let the chains $x, y \in (\mathcal{F}^n_{d'}; \mathbb{F}_2)$ such that $V(x) \cap V(y) \neq \emptyset$, then:*

$$\mathbf{Size}(x+y) \leq \mathbf{Size}(x) + \mathbf{Size}(y)$$

Proof. First, notice that if the ∞-lifted chains $x', y' \in (\mathcal{F}_{d'}; \mathbb{F}_2)$ satisfy that the projection of x' covers x and the projection of y' covers y, then the projection of $x' \cup y'$ covers $x + y$. Denote the d'-dimensional faces in x, y that intersect by f, g, and pick x', y' such that f, g are also in x', y'. Thus, $V(x') \cap V(y') \neq \emptyset$, and by Claim 4.18, we have:

$$\mathbf{Size}(x+y) \leq \mathbf{Size}(x' \cup y') \leq \mathbf{Size}(x') + \mathbf{Size}(y')$$

To finish, we argue that we can choose x', y' such that $\mathbf{Size}(x) = \mathbf{Size}(x')$ and $\mathbf{Size}(y) = \mathbf{Size}(y')$. Let $\tilde{x} \in (\mathcal{F}_{d'}; \mathbb{F}_2)$ be the d'-dimensional chain on the infinite skeleton such that $\mathbf{Size}(x) = \mathbf{Size}(\tilde{x})$. Split into cases depending on whether $f \in \tilde{x}$.

- If $f \in \tilde{x}$, then we can just choose $x' = \tilde{x}$.
- If not, then since the projection of $\tilde{x}$ covers x, there must be a vertex $v \in \mathcal{G}_d$, such that v is a product of the integer powers of $g_1^n, \ldots, g_d^n$, and a d'-dimensional face $\tilde{f}$ in $\tilde{x}$ such that $\tilde{f}$ is a translation of f by v. Then, construct x' by taking the translations of $\tilde{x}$ faces by v^{-1}. The size function is invariant to translations, so:

  $$\mathbf{Size}(x) = \mathbf{Size}(\tilde{x}) = \mathbf{Size}(v^{-1}x') = \mathbf{Size}(x')$$

  and in addition, $f = v^{-1}\tilde{f} \in x'$. Moreover, since the projecting is invariant to translation by integer powers of $g_1^n, \ldots, g_d^n$, the projection of $\tilde{x}$ equals the projection of x', namely, covers x.

So,

$$\mathbf{Size}(x + y) \leq \mathbf{Size}(x') + \mathbf{Size}(y') = \mathbf{Size}(x) + \mathbf{Size}(y)$$

□

We will use the following two elementary lifting facts and omit their proofs. They are discrete instances of the standard local relation between a Torus and its universal cover: below the periodicity scale, the covering map is injective on the relevant region, so chains may be treated as chains in the infinite lattice. In the Toric-code literature, the same principle appears in the identification of non-trivial logical operators with homologically non-trivial cycles of linear extent [Den+02], and in the restriction of Toric arguments to local regions whose diameter is smaller than the system size [KP19]. The claims below merely formulate this observation for the size function and minimizing covers used here.

Claim 5.3. *Let $x \in (\mathcal{F}^n_{d'-1}; \mathbb{F}_2)$ be a $d'-1$ dimensional chain of the finite complex, and let $y \in (\mathcal{F}_{d'-1}; \mathbb{F}_2)$ be the $d'-1$ dimensional chain on the infinite complex such that y is the minimizing cover of x. If $\mathbf{Size}(x) < n$, and:*

$$x \in Im\ \partial^- : (\mathcal{F}^n_{d'}; \mathbb{F}_2) \to (\mathcal{F}^n_{d'-1}; \mathbb{F}_2)$$

Then, also:

$$y \in Im\ \partial^- : (\mathcal{F}_{d'}; \mathbb{F}_2) \to (\mathcal{F}_{d'-1}; \mathbb{F}_2)$$

Claim 5.4. *Let $x \in (\mathcal{F}^n_{d'}; \mathbb{F}_2)$ be a d'-dimensional chain of the finite complex. Suppose that $\mathbf{Size}(x) < n$. Then there is a translation x' of x (on the finite complex), such that x' has a minimizing cover $y \in (\mathcal{F}_{d'}; \mathbb{F}_2)$, such that the faces of y align with the finite complex. In other words, '$x' = y$'.*

Next, we translate the fact that a chain enclosing a non-trivial loop has a size which is at least the side of the Toric.

Claim 5.5. *Suppose that the chain $x \in (\mathcal{F}^n_{d'}; \mathbb{F}_2)$ closes a non-trivial loop on the Toric, then* $\mathbf{Size}(x) \geq n-1$.

Proof. Let $x' \in (\mathcal{F}_{d'}; \mathbb{F}_2)$ be a minimizing cover for x. In particular, there are vertices $v_1, \ldots, v_l$ in x' such that their projection corresponds to the non-trivial loop in x. Assume the non-trivial cycle is in the g-direction, thus the L_g values of the projections of $v_1, \ldots, v_l$ cover all the integer range $\{0, 1, \ldots, n-1\}$. Thus, there are two vertices $v, u \in v_1, \ldots, v_l$ such that their L_g-difference is at least $L_g(u) - L_g(v) \geq n-1$. Thus,

$$\mathbf{Size}(x) = \mathbf{Size}(x') \geq \mathbf{Size}(\{u, v\}) \geq n-1$$

□

Using the claim above, we are able to introduce the notion of filling boundaries. The following claim states that if the connected components of the boundary are all of size lower than n, then we can match each of them to a chain such that its boundary agrees with the component and its size is lower than the component size. That will lead us to define the filling operator (Definition 5.7).

Claim 5.6. *Let $x \in (\mathcal{F}^n_{d'}; \mathbb{F}_2)$ be a d'-dimensional chain. Suppose that there is an integer $l \leq n-1$ such each connected component of $\partial^- x$ has size $< l$. Then there are d'-dimensional chains $y_1, y_2, \ldots$ such that each $\partial^- y_i$ is a connected component of $\partial^- x$ and* $\mathbf{Size}(y_i) \leq \mathbf{Size}(\partial^- y_i)$.

Proof. Consider a d'-dimensional chain $x \in (\mathcal{F}^n_{d'}; \mathbb{F}_2)$ such that $\partial^- x$ has a decomposition $\partial^- x = \tilde{x}_1 + \tilde{x}_2$, where $\tilde{x}_1, \tilde{x}_2$ are $d'-1$-dimensional chains, and $\tilde{x}_1$ is a connected component of $\partial^- x$. In particular $\tilde{x}_1$ is not connected to $\tilde{x}_2$. By disconnectivity, $\tilde{x}_1$ does not share a $d'-2$ dimensional face with $\tilde{x}_2$. Thus, $\partial^- \tilde{x}_1$ and $\partial^- \tilde{x}_2$ are disjoint. On the other hand, since $\partial^{-\circ 2} x = 0$, then combining with disjointness, it follows that:

$$\partial^-(\tilde{x}_1 + \tilde{x}_2) = 0 \Rightarrow \partial^- \tilde{x}_1 = \partial^- \tilde{x}_2 = 0$$

Namely, $\tilde{x}_1 \in \ker \partial^- : (\mathcal{F}^n_{d'-1}; \mathbb{F}_2) \to (\mathcal{F}^n_{d'-2}; \mathbb{F}_2)$. Suppose $\tilde{x}_1 \notin \text{Im } \partial^- : (\mathcal{F}^n_{d'}; \mathbb{F}_2) \to (\mathcal{F}^n_{d'-1}; \mathbb{F}_2)$, then we have that $\tilde{x}_1$ is a non-trivial codeword of the $(d, d'-1)$-Toric code. Since $d' \geq 2$, and $\tilde{x}_1$ has to close a non-trivial loop, so by Claim 5.5, the size of $\tilde{x}$ is at least $n-1$, which is a contradiction. So we get that $\tilde{x}_1 \in \text{Im } \partial^- : (\mathcal{F}^n_{d'}; \mathbb{F}_2) \to (\mathcal{F}^n_{d'-1}; \mathbb{F}_2)$. In particular, there is a d'-dimensional chain x_1 such that $\partial^- x_1 = \tilde{x}_1$. We argue that x_1 can be chosen such $\mathbf{Size}(x_1) \leq \mathbf{Size}(\tilde{x}_1)$.

By Claim 5.4, since $\mathbf{Size}(\tilde{x}_1) < n$, there is a translation of the finite complex for which the minimizing cover of the translation of $\tilde{x}_1$ aligns with the finite complex. Since the size function is invariant to translations, we can assume, without loss of generality, that the given finite complex is already translated. Consider the minimizing cover of $\tilde{x}_1$, and denote it by $\tilde{y}_1$ Since $\mathbf{Size}(\tilde{x}_1) < n$, then by Claim 5.3 we have that $\tilde{y}$ can be chosen such:

$$\tilde{y} \in \text{Im } \partial^- : (\mathcal{F}_{d'}; \mathbb{F}_2) \to (\mathcal{F}_{d'-1}; \mathbb{F}_2)$$

Let $y \in (\mathcal{F}_{d'}; \mathbb{F}_2)$ be a d'-dimensional reduced chain such that $\partial^- y = \tilde{y}$. By the definition of the algorithm in Claim 4.24, from which y is obtained, it follows that the potential walls that bound $\tilde{y}$ also bound y, and therefore y is also aligned with the finite complex. Thus, we can take x_1 to be equal to y. By Claim 4.24:

$$\mathbf{Size}(x_1) = \mathbf{Size}(y) = \mathbf{Size}(\partial^- y) = \mathbf{Size}(\tilde{y}) = \mathbf{Size}(\tilde{x}_1)$$

So x_1 satisfies the desired condition for the connected component $\tilde{x}_1$. To complete, we repeat recursively on the connected components of $\tilde{x}_2$. □

Definition 5.7. *Let $x \in (\mathcal{F}^n_{d'}; \mathbb{F}_2)$ be a d'-dimensional chain. x will be said to be a fillable chain if each of the connected components of its ∂^--boundary, namely $\partial^- x$, has size $< n-1$. By Claim 5.6, there are d'-dimensional reduced chains $x_1, \ldots,$ such that each $\partial^- x_i$ is a connected component of $\partial^- x$. Thus:*

$$\partial^-(x + \sum_i x_i) = 0 \quad \Rightarrow x + \sum_i x_i \in \ker \partial^- \colon (\mathcal{F}^n_{d'}; \mathbb{F}_2) \to (\mathcal{F}^n_{d'-1}; \mathbb{F}_2)$$

Namely, $x + \sum_i x_i$ is a codeword in $\mathcal{C}_X / \mathcal{C}_Z^{\perp}$. We define the filling operator, denoted by $[x]$, which for a given fillable chain x, returns the homology class (logical operator) of $x + \sum_i x_i$. Lastly, to ensure that the filling operator is canonical, fix for each $d'-1$-dimensional chain, which is a connected component, a unique reduced d'-dimensional chain. With the notations above, for each ∂x_i fix a unique x_i.

The following claim tracks the evolution of the connected component of the boundary (in terms of the maximal size), where the process is restricted such that in each step the chain is allowed to differ by at most a single face.

Claim 5.8. *Let $x_1, x_2 \in (\mathcal{F}^n_{d'}; \mathbb{F}_2)$ be d'-dimensional chains, and let l be an integer such the Hamming distance between x_1 and x_2 is at most l. Denote by r the size of the maximal connected component of $\partial^- x_1$. Then the maximal size of a connected component of $\partial^- x_2$ is at most $4\binom{d'}{2} l(r+1)$*

Proof. Consider the adjacency graph (definition 4.2) with respect to the $d'-1$-dimensional chain $\partial^- x_1$, where each vertex is a $d'-1$-dimensional face of $\partial^- x_1$, and each edge in the graph is a $d'-2$ face in the intersection of two $d'-1$-dimensional faces in $\partial^- x_1$. A d'-dimensional chain contains $4\binom{d'}{2}$. Hence, introducing a new d'-dimensional face to x_1 adds at most $4\binom{d'}{2}$ edges to the adjacency graph, each corresponding to a $d'-2$ face contained in the new d'-dimensional face. So introducing l faces into x'_i adds at most $4\binom{d'}{2} l$ new edges, and therefore connects at most the same number of connected components in $\partial^- x'_1$, or a connected component induced by the additional l faces. Using the triangle inequality for intersecting sets (Claim 5.2), we have that any connected component of $\partial^- x'_2$ is at most $4\binom{d'}{2} l$ times the maximal size of a connected component of $\partial^- x_1$, namely $4\binom{d'}{2} l(r+1)$ (Where the 1 in the closure stands for the connected components of a single face, each corresponding to one of the additional l faces.) □

Next, we present the main claim of the section, which argues that for a sequence of chains $x_1, \ldots, x_t$, such that any two sequential chains differ by at most a single face, if $[x_1] \neq [x_t]$, then there must be an intermediate step i such that $\partial^- x_i$ has a connected component with sufficiently large size. After that, we present claim 5.10, which generalizes the claim to allow sequential elements to be at a distance greater than one.

Claim 5.9. *Let $x_1, \ldots, x_t \in (\mathcal{F}^n_{d'}; \mathbb{F}_2)$ be d'-dimensional chains, such that x_1 is a trivial chain and for any $i \in [t-1]$, the Hamming distance between x_i and x_{i+1} is one. There is a constant $c \in (0,1)$, depending only on d, d', such that if n is sufficiently large, and for any $x_i \in x_1, \ldots, x_t$, the size of each of the connected components of $\partial^- x_i$ is bounded by cn, then:*

$$[0] = [x_1] = [x_2] = ... = [x_t]$$

Proof. Assume through contradiction that there is i such that $0 = [x_i] \neq [x_{i+1}]$. Denote by $x_i^1, x_i^2, \ldots$ and $x_{i+1}^1, x_{i+2}^2, \ldots$ the reduced d'-dimensional chains such that $\partial^- x_i^j$ is a connected component of $\partial^- x_i$, as defined in Definition 5.7. Hence:

$$[x_{i+1}] \sim_{\mathcal{C}_{Z^\perp}} [x_{i+1}] + [x_i] \sim_{\mathcal{C}_{Z^\perp}} x_i + x_{i+1} + \sum_j x_i^j + \sum_j x_{i+1}^{j+1}$$

Denote by w the chain:

$$w := x_i + x_{i+1} + \sum_j x_i^j + \sum_j x_{i+1}^{j+1}$$

Since the Hamming distance between x_i and x_{i+1} is at most one, and since each d'-dimensional face contributes a constant support over $d'-1$-dimensional chains, the ∂^--boundaries of x_i and x_{i+1} are at a Hamming distance that is at most constant. Therefore, there are only constant connected components of $\partial^- x_i$ which are not connected components in $\partial^- x_{i+1}$ and vice versa. Denote by $y_1, \ldots, y_m$ the chains in $x_i^1, \ldots$ and $x_{i+1}^1, \ldots$ such that they belong to only one of the collections. Where m is bounded by a function of d, d', namely $m \leq M(d, d')$. So:

$$\sum_j x_i^j + \sum_j x_{i+1}^{j+1} = \sum_j^m y_j$$

Moreover, the vertices of any of $y_1, \ldots, y_m$ and $x_{i+1} + x_i$ have a nontrivial intersection. To see this, suppose that the vertices of y_j do not intersect the face $x_{i+1} + x_i$, and assume without loss of generality that $\partial^- y_j$ is a connected component of $\partial^- x_i$. Then, the vertices of $\partial^- y_j$ also do not intersect the vertices of $\partial^-(x_i + x_{i+1})$. So $\partial^- y_j$ is also a connected component of

$$\partial^- x_i + \partial^-(x_{i+1} + x_i) = \partial^- x_{i+1}$$

which is contradiction to the choices of the y_is. Therefore:

$$\mathbf{Size}(w) \leq \mathbf{Size}(x_i + x_{i+1}) + \sum_{j}^{m} \mathbf{Size}(y_j) \leq d + M(d,d')cn$$

Thus, for sufficiently large n ,if $c < \frac{1}{M(d,d')+1}$, we have that $\mathbf{Size}(w) < n-1$, a contradiction for w being a non-trivial codeword. □

Claim 5.10. *Let $x_1, \ldots, x_t \in (\mathcal{F}_{d'}^n; \mathbb{F}_2)$ be d'-dimensional chains defined as in Claim 5.9, with the change that any two sequential chains x_i, x_{i+1} are at a Hamming distance of at most Δ, and setting the constant $c' \leftarrow c/(8\binom{d'}{2}\Delta)$. Then the claim still holds.*

Proof. Let $x'_1, \ldots, x'_{\Delta t+1}$ be the chains obtained from $x_1, \ldots, x_t$ by taking the stacking between any two sequential elements $\Delta - 1$ chains, such that any two sequential chains of the resulted set are at Hamming distance at most one. Namely, for any $i \in [t]$, we have that $x'_{1+i\Delta} = x_{i+i}$ and x'_{i+1} differs from x'_i by at most one coordinate. By Claim 5.8, each connected component of $\partial^- x'_i$ has a size of at most $4\binom{d'}{2}\Delta(c'n+1) \leq cn$.

Thus, by Claim 5.9,we have that:

$$[0] = [x'_1] = [x'_2] = \ldots = [x'_{\Delta t+1}]$$

In particular:

$$\begin{aligned} &[0] = [x'_1] = [x'_{1+\Delta}] = \ldots = [x'_{1+i\Delta}] = \ldots = [x'_{\Delta t+1}] \\ \Rightarrow &[0] = [x_1] = [x_2] = \ldots = [x_t] \end{aligned}$$

□

We summarize the section with the following corollary that incorporates the above insight into an ongoing computation. Namely, changing the logical codeword implies an intermediate step with a sufficiently large connected component of the syndrome. In the computation context, the sequence of the chains is derived from the Paulis in the step-propagation of a conditioned subject circuit.

Corollary 5.11 (Barrier argument)**.** *Let $c' \in (0,1)$ be the constant as defined in Claim 5.10. Let $P_1, \ldots, P_{\tau_f}$ be a step-propagation of a conditioned subjected circuit $C[\mathcal{E}, \alpha]$ (Definition 3.10). Assume that the arity of the gates of C is bounded by Δ, and that each of the registers in C is encoded in the (d,d')-Toric code. Denote by R the number of quantum registers in C. For a Pauli P_τ in the step-propagation, denote by $x_\tau \in (\mathcal{F}_{d'}^n; \mathbb{F}_2)^{\otimes R}$ the X-component of P_τ and by $x_\tau|_{R_j}$ the restriction of x_τ to register R_j. Then, if for any step τ and for any register R_j, there is not a connected component of $\partial^- x_\tau|_{R_j}$ of size greater than $c'n$, then:*

$$[x_{\tau_f}|_{R_j}] = [0]$$

A similar result holds for the Z-components, where we define $z_\tau|_{R_j}$ to be the $d-d'$-dimensional chain obtained by applying the ϕ-transpose on the chain supporting the Z-component of P_τ restricted to the register R_j.

Equivalently, if the subjected computation introduces a logical error, then at some intermediate step one of the X- or Z-syndromes has a connected component of size at least $c'n$.

Proof. The proof follows immediately from Claim 5.10, where we use the fact that the arity is bounded by Δ, and therefore sequential Paulis in the step-propagation are at a Hamming distance of at most Δ. □

Remark 5.12. *Consider the application of an iterative sweep rule over a chain whose boundary is at a size lower than $c'n$. Since the sweep rule only decreases the size of the boundary, it is easy to see that along the application, the size of the connected component of the boundary is kept below $c'n$. Thus, each time, the filling operator will accomplish the chain to the same codeword. In particular, we get that decoding a measurement as presented in Definition 4.36 doesn't result in a logical error, provided that each of the connected components of the boundary, prior to the broadcast, does not have a size exceeding $c'n$.*

5.2 Toom's Contours

We are about to zoom out from a single register encoded in the Toric code to describe a fully fault-tolerant computation. For a given quantum circuit, assumed to consist of gates $\mathbf{X}, \mathbf{Z}, \mathbf{CX}, \mathbf{H}, \mathbf{S}, \mathbf{T}$ and measurements, we define its Toric encoded circuit as its version where each qubit is encoded by the Toric code. Formally:

Definition 5.13 (Troic Encoded Circuit)**.** *Let C be a quantum circuit. We say that C is a Toric encoded circuit if the following holds:*

1. *Any layer of the circuit consists of classical and quantum registers, such that there is a Toric code $\mathcal{C}$ such that the quantum registers are encoded by the nice encoding of $\mathcal{C}$. At time $t = 0$, the quantum registers are initialized to be either $|\bar{0}\rangle$, encoded $|\mathbf{H}\rangle$, encoded $|\mathbf{S}\rangle$, or encoded $|\mathbf{T}\rangle$. The classical registers are initialized to be zeros.*

2. *The layers of C on the quantum registers at even times are sweep-cycles, and the layers at odd times are realizations of either logical Paulis, $\mathbf{CX}$, or measurements in the computational and parity bases. On the classical registers, including the measured register, we allow any two-bit gates.*

For Toric encoded circuits, we define their *base graph*, which can be thought of as a recording of a running from the Toric's point of view. The *base graph* is obtained by taking the Toric's vertices at different times as the vertices and connecting any two sequential in time if they are support faces — qubits — which are either the same or are entangled by a quantum gate.

Definition 5.14 (Base graph G_o)**.** *Let C be a Toric encoded circuit at depth μ. Notice that measurements take quantum registers into classical, Thus any quantum register has a maximal depth, which we denote by $\{\mu_i\}$. Let's denote by $\{\mathcal{G}_i\}$ the skeleton of the ∞-lift of the Toric complexes that induce the codes. We define the base graph of C, denoted by $G_o = (V_o, E_o)$, as follows:*

1. *The vertices V_o are tuples, where the first coordinates correspond to a vertex in the disjoin union over the vertices in $\{\mathcal{G}_i\}$, and the second coordinate is associated with a time. Namely:*

$$V_o := \bigcup_i \big(V(\mathcal{G}_i) \times [\mu_i]\big)$$

2. *The edges E_o are derived from the original edges in $\mathcal{G}_1, \ldots, \mathcal{G}_k$, with the addition that any two vertices in preceding times t and $t+1$ that support qubits (faces) such that C^∞ acts non-trivially on those qubits at time t are also connected by an edge. Namely:*

$$E_o := \big\{\{(v,t),(u,t')\} : \text{ if}$$
$$\text{either there exists } i \text{ such } \{v,u\} \in \mathcal{G}_i \text{ and } t' = t+1$$
$$\text{or there exists } u_i = (g_i, l_i) \in C^\infty \text{ such } v,u \in l_{i,2} \text{ and } t = l_{i,1}, t' = l_{i,1}+1\big\}$$

Where we define a vertex v to belong to location l if there is a qubit-face f in l such that v belongs to its associated face.

Notice that since the gates' locations are invariant to the conditioning of outcomes measurement (Remark 3.9), we have that for any outcomes vector $\alpha \in \mathbb{F}_2^{(\cdot)}$, the base graph of the conditioned circuit $C[\mathcal{E}, \alpha]$ is the same as for $C[\mathcal{E}]$.

As a concept, if the base graph is a recording of a run from the Torices' point of view, then a *Toom's contour* is a recording of the deviation's evolution in the eyes of the Torices' vertices. We use the notion of x, z-component of a lifted Pauli and the Pauli-path to describe a deviation:

Definition 5.15. *Let $P \in \mathcal{P}^{(\cdot)}$ be a Pauli, and let $Q \in \mathcal{P}^{(\cdot)}$ be its ∞-lift. Assume that P acts on a Toric encoded circuit, namely it has a decomposition such that each element of it acts on a register encoded in the (d, d')-Toric code. We denote by $x(Q)$ the d'-dimensional chain in the infinite complex that supports the X-component of Q, and by $z(Q)$ the $d-d'$-dimensional chain, such that the inversion of the ϕ-transpose sends it to the d'-dimensional chain that supports the Z-component of Q, namely $\phi^{-1}(z(Q))$ is the chain that supports the Z-component.*

Definition 5.16. *Consider the propagation $(\mathbf{P}, \mathbf{a})$ of a conditioned subject circuit $C[\mathcal{E}, \alpha]$ (Definition 3.10). A Pauli-path of $C[\mathcal{E}, \alpha]$ is the ∞-lift of the Paulis in $\mathbf{P}$.*

For a Pauli-path $P_1, .., P_{\mu+1}$, the Toom's contour is the restriction of the base graph to vertices that witness a deviation, namely, those that belong to qubits supporting $P_1, P_2, .., P_{\mu+1}$.

Definition 5.17 (Toom's Contour Graph of $C(\mathcal{E}, \alpha)$, (Standard form))**.** *For a Toric encoded quantum circuit C, set of faults $\mathcal{E}$, measurements outcomes α, and a Pauli-path $P_1, ..$ of $C[\mathcal{E}, \alpha]$. We define the Toom's contour graph $\mathcal{H} = (V_\mathcal{H}, E_\mathcal{H})$ as follows:*

1. *The vertices* $V_{\mathcal{H}} = V_{\mathcal{H},X} \cup V_{\mathcal{H},Z}$ *correspond to vertices of faces supporting the Pauli-path. They consist of the triple* $(v,t,\cdot)$ *where* (v,t) *is a vertex of the base graph, and the third is a symbolic coordinate that is either* $\{X\}$ *or* $\{Z\}$ *corresponding to the component of the Pauli that* v *supports at time* t*.*

$$\begin{aligned} V_{\mathcal{H},X} :=& \big\{ v_t \times \{X\} : v_t = (v,t) \in V_o \\ & \qquad \text{and there exists a face } f \in \partial^- x(P_t) \text{ such } v \in f \big\} \\ V_{\mathcal{H},Z} :=& \big\{ v_t \times \{Z\} : v_t = (v,t) \in V_o \\ & \qquad \text{and there exists a face } f \in \partial^- z(P_t) \text{ such } v \in f \big\} \end{aligned}$$

2. *The edges* $E_{\mathcal{H}}$ *is partitioned into two types* $A_{\mathcal{H}}$ *and* $F_{\mathcal{H}}$*. For any realization of logical gate* g *which applied in time* t *in* $C[\mathcal{E}]^\infty$*, we denote by* $E_o|_g$ *the edges in* E_o *which induced by* g*. Then, we add edge to* $A_{\mathcal{H}}$ *as follow:*

   (a) *If* g *does not belong to the faults, namely* $g \notin \mathcal{E}$*, and* g *is not a measurement in the phases-basis, then we add the arrows:*

   $$\big\{ \{\tilde{u} = (u,t,\cdot), \tilde{v} = (v,t+1,\cdot)\} : \{\{u,t\},\{v,t+1\}\} \in E_o|_g \text{ and } \tilde{v}, \tilde{u} \in V_{\mathcal{H},X} \\ \text{or } (v,t+1,\cdot)\} : \{\{u,t\},\{v,t+1\}\} \in E_o|_g \text{ and } \tilde{v}, \tilde{u} \in V_{\mathcal{H},Z} \big\}$$

   Where if g *is a measurement in the phase-basis, then we add type-alternating arrows as follows:*

   $$\big\{ \{\tilde{u} = (u,t,\{Z\}), \tilde{v} = (v,t+1,\{X\})\} : \{\{u,t\},\{v,t+1\}\} \in E_o|_g \\ \text{and } \tilde{v} \in V_{\mathcal{H},X}, \tilde{u} \in V_{\mathcal{H},Z} \big\}$$

   (b) *Else, if* $g \in \mathcal{E}$ *then, we add:*

   $$\big\{ \{\tilde{u} = (v,t,\cdot), \tilde{v} = (v,t+1,\cdot)\} : \text{ such either } \tilde{u} \in V_{\mathcal{H},X} \text{ or } \tilde{u} \in V_{\mathcal{H},Z} \big\}$$

   To construct $F_{\mathcal{H}}$*, we connect any pair of vertices* (v,t)*,* (u,t) *if there exists a vertex* $(w,t+1)$ *such both of them are connected to it trough* $A_{\mathcal{H}}$*. Namely:*

$$F_{\mathcal{H}} = \big\{ e = \{u,v\} : \text{ exists } w \in V_{\mathcal{H}} \text{ such } \{v,w\},\{u,w\} \in A_{\mathcal{H}} \big\}$$

We call $A_{\mathcal{H}}$ *and* $F_{\mathcal{H}}$ *the arrows and forks edges respectively.*

For the analyses, we would like to extend Toom's contour to record in their final layer a partial time-step, namely a computation up to a certain step. For this, we first consider Toom's contour of the partial computation circuit up to a step (Definition 3.2). Then, since we would like to refer to the connected chain at the final layer as the 'final error clusters', we add at the end a layer whose purpose is to connect in time (by arrows) the connected components of Toom's contour in the prior layer, which are connected by checks (their union is identified with a connected chain).

Definition 5.18 (Analysis Toom's Contour Up to Step τ)**.** *For a Toric encoded quantum circuit C, a set of faults $\mathcal{E}$, outcomes α, and a step τ, let $C[\mathcal{E},\alpha]_\tau$ be the partial computation circuit of $C[\mathcal{E},\alpha]$ up to step τ. We define the analysis Toom's contour at step τ $\mathcal{H}_\tau = (V_\mathcal{H}, E_\mathcal{H})$ to be the graph obtained by taking the Toom's contour of $C[\mathcal{E},\alpha]_\tau$ and adding edges as follows:*

1. *Let t be the depth of $C[\mathcal{E},\alpha]_\tau$. We add the following additional arrows:*

$$\Big\{\big\{\tilde{u} = (u, t-1, \cdot), \tilde{v} = (v, t, \cdot)\big\} \text{ if either there exists}$$
$$\text{a } d'-2 \text{ dimensional face } f \text{ such } \tilde{u}, \tilde{v} \in V_{\mathcal{H},X} \text{ and } u, v \in f$$
$$\text{or there exists a } d-d'-2 \text{ dimensional face } f \text{ such } \tilde{u}, \tilde{v} \in V_{\mathcal{H},Z} \text{ and } v, u \in f\Big\}$$

2. *Then, we add forks to preserve the property that any two vertices with the same time coordinate, which are connected to a third (with a $+1$ higher time coordinate) via arrows, are also connected by a fork.*

5.3 Space-Time Graph

Not a few of the results about Tooms contours are derived from their 'directed' structure, regardless of their connection to Torices. Thus, we split the discussion to talk about *space-time graphs*, which are graphs whose edges can be partitioned into arrows and forks.

Here, we present the results and proofs first given by Berman and Simon [BS88] and later by Gacs [Gac21] about *space-time grahps*. Although the proofs we give are almost identical, we chose to present them rather than referring to the sources. The reason for this is that in the language appearing in [BS88] and [Gac21], vertices are bits[13], so conceptually their Toom's contours track different creatures. Thus, jumping into their text while keeping our image in mind might cause confusion, which we would like to avoid.

Definition 5.19. *Consider a graph $G = (V, A, F)$, where V is a set of vertices, and A and F are sets of undirected edges. Let $T\colon V \to \mathbb{N}$, We say that (G, T) is a space-time graph if and only if:*

1. $\{x, y\} \in A \Rightarrow T(y) = T(x) \pm 1$
2. *For $x, y \in V$ $\{x, y\} \in F$ if and only if there exists $z \in V$ such both $\{x, z\}, \{y, z\} \in A$, and $T(x) = T(y) = T(z) - 1$.*

Additionally, for any vertex $v \in V$ of the space-time graph, denote by $pre(v)$ the set of vertices defined as follow:

$$pre(v) := \{u \in V\colon \{u, v\} \in A \text{ and } T(v) = T(u) + 1\}$$

[13] It is also true that the gates are different, yet that difference alone does not constitute a conceptual change.

We extend the notion for subsets of vertices $X \subset V$ by $pre(X) = \bigcup_{v \in X} pre(v)$. We will call T the time coordinate, A arrows, and F forks. Observe that from the second requirement, it follows that forks can connect only vertices that have the same time value (set by T).

Remark 5.20. *The Toom's contour graph defined in Definition 5.17 is, by definition of it's construction, a time-graph.*

The history of a vertex consists of all the vertices that are connected to it by arrows. Here we allow passing trough arrows in both directions, so it might be possible that the history of vertex u contains vertices with the same time coordinate as u. That motivates to define *slices*. A *slice* is a (maximal) subset of vertices with the same time coordinate such they share the same history. In [BS88], slices take the role of error droplets.

Definition 5.21 (History, Slice)**.** *For a time graph (G, T) and a vertex $u \in V$, let the history of u, denoted by H_u, be the restriction of G to vertices that both precede u in time (namely, vertices whose time $\leq T(u)$) and connect to u via a path of arrows. A slice is a maximal subset of vertices $K \subset V$ such that for any two vertices $u, v \in K$, $H_u = H_v$ is satisfied. As a consequence, the time coordinate in K has to be constant. Thus, we can extend the time coordinate and the history to slices and write for a slice K, $T(K)$ and H_K.*

The following claim ensures us an 'independent' argument about slices. Here we state that two slices with the same time coordinate have to hold disjoint histories. Another way to think about this is that if the first and the second slices are 'induced by n_1 and n_2 faults,' then the slice obtained by merging them 'is induced by $n_1 + n_2$ faults.'

Claim 5.22. *Let K_1, K_2 be two slices such that $T(K_1) = T(K_2)$, then either $K_1 = K_2$ or H_{K_1}, H_{K_2} are disjoint.*

Proof. Assume a non-empty intersection, so there exists $z \in H_{K_1} \cap H_{K_2}$ such that for any $x \in K_1$ and $y \in K_2$, there exist arrow-paths $z \to x$ and $z \to y$. Thus, for any $w \in H_{K_1}$, one can concatenate a path $w \to x \to z \to y$ and conclude that $y \in H_{K_2}$. □

Next, we would like to study how slices look a moment before they merge. For this, we define the *graph of a slice*. We construct this graph by taking a slice and considering all the slices that precede it by a single time unit. We take those slices as vertices and connect any two of them that have vertices sharing a fork.

Definition 5.23 (The Graph of Slice)**.** *Let K be a slice. The graph of K, denoted by $G_K = (V_K, E_K)$, is the graph whose vertices are slices $R \subset H_K$ with $T(R) = T(K) - 1$. Two vertices are connected by an edge if and only if there is a fork whose ends belong to the slices associated with them. Namely, $S, R \in V_K$ are connected in G_K if and only if there is $\{s, r\} = f \in F$ such that $s \in S$ and $r \in R$.*

The graph of slices is connected. That fact means that a spanning tree of a (subgraph of) a slice's history can be found by combining the spanning trees of its processors. The following gives the proof:

Claim 5.24. *G_K is connected.*

Proof. Let $R, S \in V_K$ and consider $r, s \in V$ such that $r \in R, s \in S$. Recall that $R, S \subset H_K$ and therefore $r, s \in H_K$, Thus, by definition of the 'History' there exists a path, passing trough arrows only, from r to s in H_K. Consider a path that is minimal in the number of vertices with a time coordinate equal to $T(K)$; denote that number by k. We prove by induction on k that R and S are connected in G_K.

1. Base. $k = 0$, In this case, any vertex on the path between r and s has an equal or lower time coordinate than r and is also connected to r via a path of arrows, namely belongs to H_r. In particular, s is also a vertex on the path, and therefore $s \in H_r \Rightarrow H_r = H_s$, so $R = S$.

2. Induction step. Since $k > 0$, there exists a vertex y on the path from r to s such that $T(y) = T(K)$. Let x be the vertex which precedes y on the path, and z be the vertex that follows y. Since the values of T at the two ends of an arrow differ by 1, and since y gets the maximal T value in H_K it follows that:

   (a) $\{x, z\} \subset \mathrm{pre}(y) \Rightarrow \{x, z\} \in F$.

   (b) The slices of x and z, say K_x and K_z has time value $T(K_x) = T(K_z) = T(K) - 1$, Namely they are both members of V_K.

   Since the path connecting r, x in H_K has $k - 1$ vertices with $T(\cdot) = T(K)$, then by the induction hypothesis, R and K_x are connected in G_K, and by similar argument so are K_z and S. On the other hand, $\{x, z\} \Rightarrow \{K_x, K_z\} \in E_K$. Thus, R and S are connected in G_K.

□

Now, we would like to plug in the fact that the vertices originate from the infinite plane. To do so, we equip slices with poles. The poles are the vertices in the slice, which are the maximal points of potential walls.

Definition 5.25 (Poles Selection)**.** *Let (G, T) be a space-time, whose vertices are associated with points in $\mathbb{R}^d$. Let A be either a slice, arrow, or a fork in (G, T). We define a poles selection for A to be a selection of $d+1$ vertices, $v_1, .., v_{d+1}$, not necessary different, from A. The size of A is defined to be:*

$$\mathbf{Size}(A) := \sum_{i}^{d+1} \max_{v \in A} L_i(v)$$

Thus, for any selection of poles $v_1, .., v_{d+1}$ we have that:

$$\mathbf{Size}(A) \geq \max_{v \in A} L_i(v_i)$$

For a vertex u, we define the ith predecessor of a u, denoted by $pre_i(u)$, to be it's predecessor which achieves the maximal value of L_i, namely:

$$pre_i(u) := \arg \max_{v \in pre(u)} L_i(v)$$

The following claim relates the size of a given slice to the sizes of its predecessor slices.

Claim 5.26. *Consider a time graph (G, T) whose vertices are associated with points in $\mathbb{R}^d$. Let K be a slice in G, such that $K \neq H_K$. Let $v_1, \ldots, v_{d+1}$ be a pole selection for K. Then for some integer k, there exist $k+1$ slices $K_0, K_1, \ldots, K_k$ and k forks $F_1, \ldots, F_k$ with pole selections for them that satisfy:*

1. *For any $i \in [d+1]$, the ith predecessor $pre_i(v_i)$ belongs to the selected poles of $K_0, \ldots, K_k$.*
2. *Consider the graph whose vertices are associated with $K_0, \ldots, K_{k+1}$, and there is an edge between K_i and K_j if and only if there is a fork in $F_1, \ldots, F_k$ that intersects both K_i's and K_j's selected poles. Then the graph formed from $K_0, \ldots, K_k$ is connected.*
3. *For any $j \in [0, k]$, denote by $v_j^1, .., v_j^{d+1}$ the selected poles for K_j, then:*

$$\sum_{j=0}^{k} \sum_{i=1}^{d+1} L_i(v_j^i) + \sum_{i=1}^{k} \mathbf{Size}(F_i) \geq \sum_{i=1}^{d+1} L_i\left(pre_i(v_i)\right)$$

Proof. First, notice that K doesn't contain leaves in (V, A), since a leaf is connected only to itself via arrows, and therefore its 'slice' equals its 'History'. Thus, $\text{pre}_i(v_i)$ is not empty for any $i \in [d+1]$. Next, consider the graph $G_K = (V_K, E_K)$ from Definition 5.23. Since $\{v_i, \text{pre}_i(v_i)\}$ is an arrow, $\text{pre}_i(v_i)$ belongs to a slice in V_K. For $i = 1, \ldots, d+1$, denote by R_i the slice in V_K that contains $\text{pre}_i(v_i)$.

By Claim 5.24 G_K is connected and therefore there exist a minimal collection $K_0, .., K_k$ of slices form V_K which contains $R_1, R_2, ..R_{d+1}$, and which is connected in G_K. We pick a set of forks $\mathbf{F} = \{F_1, .., F_k\}$ which realizes a spanning tree for this collection of slices.

Let us denote by $J_0, .., J_{2k}$ the enumerated union $K_0, ..K_k, F_1, .., F_k$. Observes that for any choice of $(d+1)(2k+1)$ elements $z_0^1, z_0^2, z_0^3, .., z_0^{d+1} .., z_{2k}^{d+1}$ in $\mathbb{R}^d$, such that $z_j^i \in J_j$ it holds that:

$$\sum_{j=0}^{k} \sum_{i=1}^{d+1} L_i(z_j^i) + \sum_{i=1}^{k} \mathbf{Size}(F_i) \geq \sum_{j,i} L_i(z_j^i) \tag{1}$$

Later, we are going to take the points $\{z_j^i\}$ from the intersection of J_j's. For convenience, let us assume that for the index subset $X \subset [2k] \cup \{0\}$, the intersection $\bigcap_{j \in X} J_j$ returns a single point. If there is more than one, then decide on the point

in a way that all the chosen points simultaneously satisfy that for any index subsets Y, X, such that $X \cap Y$ and they both introduce a non-trivial intersection $\bigcap_{j\in Z} J_j : Z \in \{X, Y\}$, then:

$$\bigcap_{j\in X} J_j = \bigcap_{j\in Y} J_j$$

Now consider a non-trivial maximal intersection $\bigcap_{j\in X} J_j$. The intersection is maximal in the sense that for any $j' \notin X$, it holds that $\bigcap_{j\in X\cup\{j'\}} J_j = \emptyset$. We claim that for any i and for any such X, there is $t \in X$ such that J_t is the successor of J_j on the path to R_i for any $j \in X/\{t\}$. Assume not. Let $x, y \in X$, and denote the paths from J_x, J_y to the R_i by $J_x, J_{x_2}, \ldots, R_i$, and $J_y, J_{y_2}, \ldots, R_i$, such that $J_y \neq J_{x_2}$. If $y = x_2$, $y_2 = x$, or $x_2 = y_2 \in X$, then pick other x, y (if there is no such, we get an immediate contradiction). In that case, $J_x, J_y, J_{y_2}, \ldots, R_i$ is a path from J_x to R_i which is different from $J_x, J_{x_2}, \ldots, R_i$. Therefore, there are at least two paths from J_x to the R_i, namely, there is a cycle in contradiction to the that the collection $K_0, \ldots, K_k, F_1, .., F_k$ is a tree.

We will choose the z_j^i by the following process. Pick a z_j^i that has not been set yet. If $J_j = R_i$, then set $z_j^i \leftarrow \mathrm{pre}_i(v_i)$ - we call this a type-1 assignment. Otherwise, consider the path from J_j to R_i (by connectivity, it has to be such a one, and by minimality, there is exactly one). Let J_t be the successor on the path, then pick $z_j^i \in J_j \cap J_t$ - similarly, we call this assignment a type-2 assignment. Since any type-2 assignment is associated with a non-trivial intersection, and any non-trivial intersection induces a type-2 assignment for all i's (since for all i's, one of the J_j on its support is a successor of the rest on their path to the R_i), we have that:

$$\begin{aligned}
\sum_{j,i} L_i(z_j^i) &= \sum_{j,i,z_j^i\in\text{type-1}} L_i(z_j^i) + \sum_{j,i,z_j^i\in\text{type-2}} L_i(z_j^i) \\
&= \sum_{i}^{d+1} L_i\left(\mathrm{pre}_i(v_i)\right) + \sum_{X\colon \bigcap_{j\in X} J_j\neq\emptyset} (|X|-1) \sum_{i}^{d+1} L_i\left(\bigcap_{j\in X} J_j\right) \qquad (2) \\
&= \sum_{i}^{d+1} L_i\left(\mathrm{pre}_i(v_i)\right)
\end{aligned}$$

To complete, we select the ith pole of K_j to be z_j^i. Then, for $i = 1, \ldots, d+1$, the ith predecessor $\mathrm{pre}_i(v_i)$ is the ith pole of R_i. This assures (1). Since for any intersection $K_i \cap F_j$ a type-2 assignment is picked from the intersection to K_i, we have that the type-2 selected poles of $K_0, K_1, \ldots, K_k$ are covered by the vertices of $F_1, F_2, \ldots, F_k$. This assures (ii). Plugging Equation (2) into Equation (1) assures (3). □

Until now, the claims stated were true for any kind of space-time graph (where their vertices can be associated with coordinates in $\mathbb{R}^d$). The next claims assume that the given time graph is a Toom's contour induced by the noisy running of a Toric encoded circuit.

Claim 5.27. *For Toom's contours, the maximal size of a fork is at most* $2d$.

Proof. Consider the rules of forks connecting in Definition 5.17. For forks obtained by rules (a), (b), or as a result of a fault, it's clear that their vertices at their ends are on intersecting faces (when projecting the vertices on $\mathcal{G}$). The maximal fork, in those cases, has the form of:

$$F = \{v, (\prod_{g \in S'_1} g)(\prod_{g \in S'_2} g)v\}$$

Where S'_1 and S'_2 are subsets of S of size either d' or $d - d'$, and therefore the size of such forks is at most $\mathbf{Size}(F) = 2\max\{d', d - d'\} \leq 2d$. □

While in the construction presented in [BS88], all the layers were the same (consisting of the same gates up to wiring), here different layers might differ both in realization and role. Thus, we would like to distinguish between them. We will define the *predecessor decomposition* of a slice to be the collection of slices and forks which satisfy the condition in Claim 5.26. The order of a slice is a number in $\{0, 1, 2, 3\}$ that matches the time modulo four of the layer that results in the slice. Therefore, it encodes the layer type.

Definition 5.28. *Let K be a slice, and let $v_1, \ldots, v_{d+1}$ be a selection of poles for it. Then, let $K_0, \ldots, K_k$ be slices in H_K, and let $F_1, \ldots, F_k$ be forks, such that $K_0, \ldots, K_k$, $F_1, \ldots, F_k$ satisfy the conditions in Claim 5.26 with respect to the poles selection. We call $K_0, \ldots, K_k$, $F_1, \ldots, F_k$ the predecessor decomposition of K with respect to the pole selection $v_1, \ldots, v_{d+1}$. When the pole selection is either clear from the context or can be chosen arbitrarily, we will omit it.*

Furthermore, we say that K is a result of a sweep-cycle, faults, or a logical operation if the layer at time $T(K) - 1$ consists of sweep-cycle gates, fault gates (whose source is $\mathcal{E}$), or either measurement or logical Pauli. The order of the slice is its time plus 3 modulo 4, namely $order(K) = T(K) + 3 \mod 4$. So the orders of slices that result from a sweep-cycle, faults, logical operation, and again faults are $0, 1, 2, 3$ respectively.

The distinction between slices resulting from different layers allows us to adjust Claim 5.26 by conditioning on the slice type. Yet, before doing so, we will have to extend the infinity characterization to slices.

Corollary 5.29. *Consider a slice K in it, and a selection of poles $v_1, \ldots, v_{d+1}$, such that $K \neq H_K$. Since $K \neq H_K$, it follows that K has a predecessor decomposition with respect to $v_1, \ldots, v_{d+1}$, denoted by $K_0, \ldots, K_k, F_1, \ldots, F_k$. For each slice K_j, denote its selected poles by $z_j^1, \ldots, z_j^{d+1}$. The inequality in Claim 5.26 becomes:*

1. *If K is a result of a sweep-cycle, then:*

$$\sum_{j=0}^{k} \sum_{i}^{d+1} L_i(z_j^i) + \sum_{i=1}^{k} \mathbf{Size}(F_i) \geq \sum_{i}^{d+1} L_i(v) + 1$$

2. *If K is a result of either faults or a logical operation, then:*

$$\sum_{j=0}^{k} \sum_{i}^{d+1} L_i(z_j^i) + \sum_{i=1}^{k} \mathbf{Size}(F_i) \geq \sum_{i}^{d+1} L_i(v)$$

Proof. Let us split upon what K results from:

1. For the sweep-cycle part, we use Corollary 4.35 and Claim 4.28. The first states that within an application of the sweep-cycle over a chain, the maximal L_{d+1} value of the ∂^--boundary of the resulting chain decreases by at least one, while the values of each L_g do not increase. The second assures that for any local maximum of L_i in the resulting ∂^--boundary, there is a vertex in the ∂^--boundary of the original such that they (1) share a d'-dimensional face and (2) the L_i-value of the vertex in the result is at most the L_i-value of the vertex in the original, where if $i = d+1$, the relation becomes strictly less. Thus, prior to forming the spanning tree of the predecessor slices, for every v_i in $v_1, .., v_{d+1}$ we select a vertex $\mathrm{pre}_i(v_i)$ that has either at most the same L_i-value as v_i if $i \neq d+1$ or an L_{d+1}-value that is bigger by at least 1. That gives the inequality.

2. If K is a result of faults, then it cannot contain a vertex that supports a fault. That is because no arrow connection is made in that case, and hence, if it were to contain a vertex that supports a fault, that vertex would be the only one in K, namely $K = H_K$. So, we can imagine that all the vertices in K support the identity operator and handle that case as a result of a logical operator.

3. Where K is the result of a logical operator, we use the observation that a pure logical operator doesn't change the slices (with the exception that the measurement in the phase-basis switches the vertex types, namely $\{X\} \leftrightarrow \{Z\}$, of part of the vertices in the slice, see Claim 4.37). So we have that the support of K is contained (up to vertices' types) in the support of $\cup_0^k K_i$, and therefore for any $i \in [d+1]$:

$$\max\{L_i(v) : v \in K\} \leq \max\{L_i(v) : v \in \bigcup_{i=0}^{k} K_i\}$$

□

The following claim extends the corollary to slices at the final layer of an *analysis Toom's contour up to step* τ.

Claim 5.30. *For a step τ let $\mathcal{H}_\tau$ be Toom's contour analysis up to step τ. Denote by t the time at which the gate at step τ is applied. Then, for a slice K and selection for its poles $v_1, \ldots, v_{d+1}$ such that $T(K) = t+1$ and $K \neq H_K$, the inequality in Claim 5.26 becomes:*

$$\sum_{j=0}^{k} \sum_{i}^{d+1} L_i(z_j^i) + \sum_{i=1}^{k} \mathbf{Size}(F_i) \geq \sum_{i}^{d+1} L_i(v) - 2d$$

Proof. We allow vertices of $V_{\mathcal{H},X}$ at the final time layer to be connected by an arrow if their counterpart on the skeleton shares a $d'-2$ dimensional face. Similarly, two vertices of $V_{\mathcal{H},Z}$, such that one of them is at the top layer, are connected if their images

in the map ϕ^{-1} share a $d-d'-2$ dimensional face. Therefore, for any $i \in [d]$, the value of L_i for v_i might be greater by at most 1 than the value of $\text{pre}_i(v_i)$, and the L_{d+1} for v_{d+1} might be greater by at most $\max\{d'-2, d-d'-2\}$ than the value of the $d+1$ pole of K. Hence, overall, we get that:

$$\begin{aligned}\sum_{i=1}^{d+1} L_i\left(\text{pre}_i(v_i)\right) &\geq \sum_{i=1}^{d}\big(L_i\left(v_i\right)-1\big) + L_{d+1}(v_{d+1}) - \max\{d'-2, d-d'-2\} \\ &\geq \textbf{Size}(K) - 2d\end{aligned}$$

□

5.4 Minimal Explantation Argument

We are about to prove the main lemma stating that the number of faults causing an occurrence of slice is linear in both the slice's size and the minimal number of edges required to span a Toom's contour that induces it. First, we define the explanation tree of a slice to be the subtree of its history that connects its poles. The relation between the leaves of the subtree and its arrows, forks, and the size of its origin slice is embedded as a requirement in the definition. Then, we prove that, assuming no time admits an infinite error, any slice has an explanation tree.

Definition 5.31. *Consider a Tooms contour graph induced by Toric encoded circuit, with sweep-cycle parameter $\xi = 1$, and let α, β be constants and $\mathcal{X}_0, \mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3 \in \mathbb{R}$ be quadruple, defining such:*

$$\alpha := 40(d+1)^2, \quad \beta := 5(d+1)$$
$$\mathcal{X}_r := 1 - r\frac{d+1}{\alpha} = 1 - \frac{r}{40(d+1)}$$

In addition, we would like to refer to $\mathcal{X}_0, \mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3$ in a cyclic manner, so we define:

$$\mathcal{X}_{-1} := \mathcal{X}_3$$

Given sets W of points, A of arrows, and F of forks. The sets W, A, and F form an explanation of a slice K, with poles selection $v_1, \ldots, v_{d+1}$, and order $order(K) = r$, if and only if:

1. *$W \cup \{v_1, .., v_{d+1}\} \subset H_K$ and the graph $(W \cup \{v_1, .., v_{d+1}\}, A \cup F)$ is connected.*
2. *All elements of W are leaves.*
3. *For $m = |W|$ and $s = \sum_i^{d+1} L_i(v_i)$ we have $|A| \leq \alpha(m - \mathcal{X}_r) - \beta s$ and $|F| \leq m-1$.*

Furthermore, for a step τ, and a slice K in the analysis of Toom's contour up to step τ, with pole selection $v_1, .., v_{d+1}$, we extend the definition of explanation as above, but require the following weaker inequality instead: For $m = |W|$ and $s = \textbf{Size}(K)$, we have $|A| \leq \alpha(m+1) - \beta s$ and $|F| \leq m-1$.

Claim 5.32. *Let $P_1, \ldots, P_{\mu+1}$ be the Pauli-path that induces Toom's contour. For every poles selection $v_1, \ldots, v_{d+1}$ of K, there is an explanation. Moreover, any slice in the analysis of Toom's contour up to step τ, and any poles selection for it, there is an explanation.*

Proof. Let $h = T(K) - \min_{u \in H_K} T(u)$. For slices in a standard Toom's contour, we prove by induction on h.

1. Base: $h = 0$, meaning T is constant on H_K, so no arrows connect points of H_K. Thus, H_K, as well as K, consists only of a single leaf, namely K is a result of a fault, and $r = \text{order}(\mathcal{X}_r) \in \{1, 3\}$. In that case, there is a single vertex $u \in V_{\mathcal{H}}$ such that $W = \{u\}$, so $m = 1$, $s = 0$, and $|A|, |F| = 0$. It is easy to check that those values satisfy the requirement.

2. Induction Step: $h \neq 0$ implies $K \neq H_K$. By Claim 5.26, we know that for some k there exist slices $K_0, \ldots, K_k$, pole selections for them, and forks $F_1, \ldots, F_k$ that satisfy Claim 5.26 with respect to the pole selection. In particular, denote by $z_j^1, .., z_j^{d+1}$ the pole selection for K_j. Then, for $r \neq 0$ it holds that:

$$\sum_{j=0}^{k} \sum_{i=1}^{d+1} L_i(z_j^i) + \sum_{i=1}^{k} \mathbf{Size}(F_i) \geq s$$

   Whereas, for $r = 0$:

$$\sum_{j=0}^{k} \sum_{i=1}^{d+1} L_i(z_j^i) + \sum_{i=1}^{k} \mathbf{Size}(F_i) \geq s + \xi = s + 1$$

   Consider $i \in \{0, \ldots, k\}$, and denote $s_j = \sum_i^{d+1} L_i(z_j^i)$. By the inductive hypothesis and since $T(K_i) = T(K) - 1$, we know that for K_i and its selected poles $z_i^1, \ldots, z_i^{d+1}$, there is an explanation formed by a set of leaves $\mathbf{W}_i$, a set of arrows $\mathbf{A}_i$, and a set of forks $\mathbf{F}_i$. Let $m_i = |\mathbf{W}_i|$. Thus, the sets $\mathbf{A}_i$ and $\mathbf{F}_i$ have at most $\alpha(m_i - \mathcal{X}_{r-1}) - \beta s_i$ arrows and $m_i - 1$ forks. We define:

   (a) $\mathbf{W} = \cup_{i=0}^{k} \mathbf{W}_i$, Notice that $\mathbf{W}_i \subset K_i$ and that $T(K_i) = T(K_j)$ for all i, j, thus by Claim 5.22 $\mathbf{W}$ is a union of disjoint set, Hence W contains $m = \sum_{i=0}^{k} m_i$ leaves.

   (b) $\mathbf{F} = \{F_1, .., F_k\} \cup_{i=0}^{k} \mathbf{F}_i$ with at most $k + \sum_{i=0}^{k} m_i - 1 = m - 1$ elements.

   (c) $\mathbf{A} = \{\{v_i, \text{pre}_i(v_i)\}\} \cup_{i=0}^{k} \mathbf{A}_i$. For bounding the number of arrows in $\mathbf{A}$ we condition upon whether $r = 0$, For convenient, let us denote by Y the indicator, which indicate if $r = 0$.

$$\begin{aligned}
&d + 1 + \sum_{i=0}^{k} \alpha(m_i - \mathcal{X}_{r-1}) - \beta s_i \\
\leq &d + 1 + \alpha m - \mathcal{X}_{r-1}\alpha(k+1) - \beta\left(s + Y\xi - k \cdot \mathbf{Size}(\text{fork})\right) \\
= &\alpha\left(m - \mathcal{X}_r\right) - \beta s + \left(\alpha(\mathcal{X}_r - \mathcal{X}_{r-1}) + d + 1 - \mathcal{X}_{r-1}k\alpha - \beta Y\xi + \beta k \cdot \mathbf{Size}(\text{fork})\right)
\end{aligned}$$

So it's left to show that the term in the closure is negative for all integers k:

$$\alpha(\mathcal{X}_r - \mathcal{X}_{r-1}) + d + 1 - \mathcal{X}_{r-1}k\alpha - \beta Y\xi + \beta k \cdot \textbf{Size}(\text{fork})$$

We do that by showing that for the choice of the parameter values, $\alpha, \beta, \mathcal{X}_0, \ldots, \mathcal{X}_3$, the slope is negative, and then the value of the term at k equals zero is negative. For the slope to be negative, we require:

$$-\alpha\mathcal{X}_{r-1} + \beta\textbf{Size}(\text{fork}) \leq 0 \Rightarrow \textbf{Size}(\text{fork}) \leq \frac{\alpha}{\beta}\mathcal{X}_{r-1}$$

For negativity at the origin, $k = 0$, we require for $r \neq 0$:

$$\begin{aligned}\alpha(\mathcal{X}_r - \mathcal{X}_{r-1}) + d + 1 &\leq 0 \\ \Rightarrow \mathcal{X}_r &\leq \mathcal{X}_{r-1} - \frac{d+1}{\alpha}\end{aligned}$$

And for $r = 0$:

$$\begin{aligned}\alpha(\mathcal{X}_0 - \mathcal{X}_3) + d + 1 - \beta\xi &\leq 0 \\ \Rightarrow \mathcal{X}_0 &\leq \mathcal{X}_3 - \frac{d+1}{\alpha} + \frac{\beta}{\alpha}\xi\end{aligned}$$

So in total:

$$\begin{aligned}\mathcal{X}_1 &\leq \mathcal{X}_0 - \frac{d+1}{\alpha} \\ \mathcal{X}_2 &\leq \mathcal{X}_0 - 2\frac{d+1}{\alpha} \\ \mathcal{X}_3 &\leq \mathcal{X}_0 - 3\frac{d+1}{\alpha} \\ \mathcal{X}_0 &\leq \mathcal{X}_0 - 4\frac{d+1}{\alpha} + \frac{\beta}{\alpha}\xi \\ \textbf{Size}(\text{fork}) &\leq \frac{\alpha}{\beta}\mathcal{X}_r\end{aligned}$$

The first three inequalities hold by the definitions of the $\mathcal{X}_r$, the fourth is satisfied since $\frac{\beta}{\alpha}\xi = 5\frac{d+1}{\alpha}$, the last is obtained due to Claim 5.27, that states that the maximal size of a fork is at most $2d$, and that $\frac{\alpha}{\beta}\mathcal{X}_r$ is greater than $4d$.

Finally, by Claim 5.26, the graphs induced by $\mathbf{A}_i \cup \mathbf{F}_i$ are connected to each other trough $F_1, .., F_k$, thus the graph induced by $\mathbf{A} \cup \mathbf{F}$ is also connected. Therefore, $\mathbf{W}, \mathbf{A}, \mathbf{F}$ form an explanation of K.

It's left to prove for slices in the analysis of Toom's contours. Notice that the 'forming rules' of the analysis of Toom's contour are different from the standard Toom's contour only at the last layer. Therefore, the induction proof captures all the slices not in the

last layer. For any slice at the last layer, we repeat the induction step while using Claim 5.30 to bound the expansion in size. So the bounding over $\mathbf{A}$'s size becomes:

$$\begin{aligned}|\mathbf{A}| &\leq \alpha\,(m+1) - \beta s \\ &\quad + (-\alpha(1+\mathcal{X}_{r-1}) + d + 1 - \mathcal{X}_{r-1}k\alpha + 2d + \beta k \cdot \mathbf{Size}(\text{fork})) \\ &\leq \alpha\,(m+1) - \beta s \\ &\quad + (-\alpha + d + 1 - \mathcal{X}_{r-1}k\alpha + 2d + \beta k \cdot \mathbf{Size}(\text{fork}))\end{aligned}$$

Where we used the fact that $\mathcal{X}_{r-1} \in (0,1)$ for $r \in \{0,1,2,3\}$. Clearly, the bound is satisfied since $\alpha \geq 3d$ and $\mathcal{X}_{r-1}\alpha \geq \beta \cdot \mathbf{Size}(\text{fork})$. □

We summarize by using the linear relation between the leaves of a Toom's contour and the faults that induce it. That gives an upper bound on the probability of a slice occurring when the noise is local stochastic.

Corollary 5.33. *Let K be a slice with size s, and explanation with $|A|$ arrows, and $|F|$ forks. Then it is induced by at least the following number of faults:*

$$\frac{1}{2^{\max\{d',d-d'\}}}\frac{1}{2}\left(|F| + 1 + \alpha^{-1}(|A| + \beta s - 1)\right)$$

Furthermore, the ratio of leaves to edges in an explanation is greater than $\frac{1}{2\alpha} = \frac{1}{80(d+1)^2}$.

Proof. We bound Claim 5.32 by the number of leaves in the explanation and use the fact that any d'-face has $2^{d'}$ vertices, so each X-fault and Z-fault contributes at most $2^{d'}$ and $2^{d-d'}$ leaves to the explanation. □

6 Proof of the Theorem

We are ready to prove the main theorem (Theorem 6.3). We start by giving a formal definition of a simulation.

Definition 6.1 (Simulation and Logical Errors)**.** *Consider a quantum code $\mathcal{C}$ of length n and dimension k. A quantum circuit $D\colon L(\mathcal{H}_2^{\otimes n}) \to L(\mathcal{H}_2^{\otimes n})$ will be said to be a referee for $\mathcal{C}$ if, when given a codeword of $\mathcal{C}$, it results in the first k output qubits, the logical state of the given. We say that a Pauli operator $P \in \mathcal{P}^{\otimes n}$ is a logical error with respect to D if $DP \neq D$.*

For quantum circuits C and $\tilde{C}$, a noise channel $N_{\mathcal{E}}$, and $p \in (0,1)$, we say that $\tilde{C}$ simulates C with probability at least $1-p$ if

1. *There is a mapping between the qubits of C and the registers of $\tilde{C}$, along with a referee such that, at the end of running $\tilde{C}$ (on the zero state), decoding the registers back to the logical space results in the same quantum states as those produced by running C.*

2. *Let $\mathcal{E} \sim N_{\mathcal{E}}$ be a set of faults. The probability induced by $N_{\mathcal{E}}$ that at the end of the running of $\tilde{C}[\mathcal{E}]$, one of its quantum registers introduces a logical error (with respect to the referee) is less than p.*

The next lemma is the core of the theorem; it analyzes the probability of a logical error occurring after the initialization stage, namely when assuming the given input is encoded.

Lemma 6.2. *For any d, d', there are constants $a, b \in \mathbb{R}^+$ and $p_0 \in (0,1)$ that depend only on d, d' and satisfy the following. Consider a quantum circuit C with depth μ and width w, and let $\mathcal{N}_\mathcal{E}$ be a local stochastic noise channel with a noise rate $p < p_0$. There is a Toric encoded circuit $\tilde{C}$ such that each of its quantum registers is encoded by an n-side length, (d, d')-Toric code, and it simulates C with probability:*

$$q = 1 - O\left(poly(w, \mu, n)\right)\left(ap^b\right)^{\frac{n}{2d}}$$

To prove the lemma, we will need the following two claims. The first bounds, for a given regular graph, the number of its subgraphs having at most k edges. This claim will be used to infer the number of possible explanation trees for a given slice. The second states that each tree has a subtree with fewer than half of the edges and a leaves-to-edge ratio that is at least half of the ratio of the original tree. This claim will be used to infer that any explanation has a subtree where, under the projection to the finite Toric, its faults are mapped uniquely, meaning no fault is counted twice.

Claim (Claim 8.1)**.** *Let $G = (V, E)$ be a $\Delta + 1$-regular graph, and let k be an integer less than $|E|$. Fix a vertex $r \in V$. There is a constant number σ, depending only on Δ, such that the number of subgraphs rooted at r with fewer than k edges is at most σ^k.*

Claim (Claim 8.4)**.** *Let $\gamma, \beta \in \mathbb{R}_{>0}$ and n be a positive integer. For any tree T with $\tau \geq \beta n$ edges and m leaves such that $\tau \leq \gamma m$, there is a subtree T' of T with a number of edges τ' less than $\frac{\beta}{2}n$ and a number of leaves greater than $\frac{\beta}{4\gamma}n$.*

The proofs of the claims are given in the appendix. We continue with the proof of the lemma.

Proof. We construct $\tilde{C}$ as follows: for each input wire of C, we initialize the logical zero $|\mathbf{0}\rangle$ of the (d, d')-Toric code. For each of the gates **H** and **S**, we initialize the encoded $|\mathbf{H}\rangle$ and $|\mathbf{S}\rangle$ states, and for each **T**, we initialize copies of both $|\mathbf{T}\rangle$ and $|\mathbf{S}\rangle$ states. The even layers of $\tilde{C}$ are sweep cycles, while on the odd layers, we set the realization of the logical gates. For **CX**, we set the transversal realization, and for **H**, **S**, and **T**, we use gate teleportation. Thus the width of $\tilde{C}$ expands to at most $(w + 2\mu)n^{2d}$.

We realize the gate teleportation as described in Definition 4.36, so the gate teleportation admits $O(n + \log n)$ and $O(n^{2d})$ depth and width overhead. Thus, the depth of $\tilde{C}$ expands to $\Theta(\mu n)$[14]. We will denote the procedure of inferring the logical measurement in the gate teleportation[15] by D, and we also use D as a realization of measurement in the computational basis and as our referee (Definition 6.1). Denote by S the number of locations, so $S = O(w\mu^2 n^{2d+1})$.

[14]The constant inside the Θ also accounts for the fact that the realization of a logical gate might contain multiple gate teleportations.

[15]This is the codeword checking procedure - $\mathbf{1}_{x_{S'}}$ / $\mathbf{1}_{x_{S/S'}}$ as appears in Definition 4.36

Consider a set of faults $\mathcal{E}$, and consider the representation of the subjected circuit $C[\mathcal{E}]$ as a distribution over conditioned measured circuits:

$$C[\mathcal{E}] = \sum_{\alpha} \mathbf{Pr}\left[\alpha|\mathcal{E}\right] C[\mathcal{E}, \alpha]$$

Denote by τ_f the last step of $C[\mathcal{E}, \alpha]$, and let $(\mathbf{P}, \mathbf{a})$ be a step-propagation of $C[\mathcal{E}, \alpha]$, where $\mathbf{P} := P_1, P_2, .., P_{\tau_f+1}$ and $\mathbf{a} := \alpha^1, \alpha^2, .., \alpha^{\tau_f+1}$ (Definition 3.10). Thus, if $D \circ C[\mathcal{E}] \neq C$, then there is at least one outcome α for which $D \circ C[\mathcal{E}, \alpha] \neq C$, namely:

$$D \circ C[\mathcal{E}, \alpha] = D \circ P_{\tau_f+1} C[, \alpha^{\tau_f+1}] \neq C$$

For this, one of the following two has to occur: either the decoder D fails to decode P_{τ_f+1}, or the conditioned circuit $C[, \alpha^{\tau_f+1}]$ deviates from the ideal circuit, namely $C[, \alpha^{\tau_f+1}] \neq C$. The second occurs only if, along the running, a gate-teleportation fails, meaning the decoding procedure that erases errors from the measured register failed, and therefore the gate applied following the measurement was incorrect.

By the barrier argument (Corollary 5.11), occurrence of both options implies that there is a connected component of either the X-syndrome or Z-syndrome of at least one of the Paulis in the propagation $P_1, .., P_{\tau_f+1}$ at a size greater than cn (where c is the constant c' defined in Claim 5.10), and hence, the Pauli-path induced by the propagation has a connected component of the syndrome at a size at least cn.

Since connectivity by faces in the analysis of Toom's contour implies connectivity by arrows, it follows that an occurrence of that event implies the existence of a step τ such that the analysis of Toom's contour $\mathcal{H}_\tau$ of $C[\mathcal{E}, \alpha]_\tau$ has a slice of size at least cn. So it's sufficient to prove that with probability at least q, for the sampled faults $\mathcal{E}$, there are no possible outcomes α and a step τ such that the analysis of Toom's contour $\mathcal{H}_\tau$ of $C[\mathcal{E}, \alpha]_\tau$ has a slice of size at least cn.

Denote by B_τ the event where τ is the first step such that $\mathcal{H}_\tau$ has a slice at size greater than cn, and by A_τ the event where all explanations of slices (and any of their pole selections) in $\mathcal{H}_\tau$ have fewer than $\frac{n}{2d}$ edges. Using the conditional probability formula, we have:

$$\begin{aligned}\mathbf{Pr}_{\sim\mathcal{N}_\mathcal{E}}\left[B_\tau\right] &= \mathbf{Pr}_{\sim\mathcal{N}_\mathcal{E}}\left[B_\tau \cap A_\tau\right] + \mathbf{Pr}_{\sim\mathcal{N}_\mathcal{E}}\left[B_\tau \cap \bar{A}_t\right] \\ &\leq \mathbf{Pr}_{\sim\mathcal{N}_\mathcal{E}}\left[B_\tau \cap A_\tau\right] + \mathbf{Pr}_{\sim\mathcal{N}_\mathcal{E}}\left[\bar{A}_t\right]\end{aligned}$$

We begin by bounding the contribution of the right term. According to Corollary 5.33, there exists a constant c_1, depending only on d, such that the ratio of leaves to edges in the explanation is at least c_1. By Claim 8.4, it follows that any such explanation has more than $\frac{n}{2d}$ edges containing a subtree with fewer than $\frac{n}{4d}$ edges, with a leaves-to-edges ratio greater than $\frac{1}{2}c_1$. Using the union bound, we bound the probability that there is a vertex in the base graph that is a root of such a subtree. Since the

construction of the ∞-lift sets periodic conditions, it's sufficient to ask if there is such a vertex in the restriction to the finite n-side length Toric.

$$\mathbf{Pr}_{\sim\mathcal{N}_\mathcal{E}}\left[\bar{A}_\tau\right] \leq \bigcup_{\substack{v\in V_o,\tau'\leq\tau \\ L_g(v)\in[0,n]}} \mathbf{Pr}_{\sim\mathcal{N}_\mathcal{E}}\Big[v \text{ is a root of subtree in } \mathcal{H}_{\tau'} \text{ with less than } \frac{n}{4d} \text{ edges and more than } \frac{1}{2}c_1\frac{n}{4d} \text{ leaves}\Big]$$

Now, since any two connected vertices are at a space distance of at most

$$\max\{\mathbf{Size}(\text{arrow}), \mathbf{Size}(\text{fork})\} \leq 2d$$

It follows that all the vertices in the subtree are at a space distance of at most $\frac{1}{2}n$ from each other. Thus, no two of them have the same space coordinate modulo n. In particular, projecting the infinite plane back to the finite Toric sends each of the leaves of the subtree to a unique target, namely no fault is counted twice. Therefore the number of faults that induce the explanation is at least:

$$\frac{1}{2^{\max\{d',d-d'\}}}\frac{1}{2}\frac{1}{2}c_1\frac{n}{4d} \geq \frac{1}{2^{d+4}}c_1\frac{n}{d}$$

Plugging into the union bounds and combining Claim 8.1 yields:

$$\mathbf{Pr}_{\sim\mathcal{N}_\mathcal{E}}\left[\bar{A}_\tau\right] \leq O(S)\sigma^{\frac{n}{4d}}p^{\frac{c_1 n}{2^{d+4}d}} \leq O(S)\left(\sigma p^{\frac{1}{2^{d+2}}c_1}\right)^{\frac{n}{4d}}$$

It's left to bound the probability of $B_\tau \cap A_\tau$. By Claim 5.32 any slice, and any pole selection of it, in $\mathcal{H}_\tau$, have an explanation. Confining to the event A_τ, any explanation implies that any explanation has fewer than $\frac{n}{2d}$ edges and therefore, by the same argument as above, the projection of each explanation tree into the finite Toric sends its leaves to unique targets.

Any slice of size at least cn, has poles selection $v_1..,v_{d+1}$ such:

$$\sum_{i}^{d+1} L_i(v_i) \geq cn \tag{3}$$

Consider a slice with size at least cn and a pole selection for it satisfying Equation (3). Claim 5.32 and Corollary 5.33 state that there are constants $c_2, c_3 > 0$ such that any explanation for the slice and the selection, with x edges, has at least $c_2 cn + c_3 x$ leaves. Therefore, we obtain:

$$\mathbf{Pr}_{\sim\mathcal{N}_\mathcal{E}}\left[B_\tau \cap A_\tau\right] \leq \bigcup_{\substack{v\in V_o,x\geq 1 \\ L_g(v)\in[0,n]}} \mathbf{Pr}_{\sim\mathcal{N}_\mathcal{E}}\Big[v \text{ is a root of explanation with } x \text{ edges and at least } c_2cn + c_3x \text{ leaves, which are projected to different targets }\Big]$$

So, for sufficiently small p such that $\sigma p^{c_3} < 1$, the probability is bounded by:

$$\sum_{x=1}^{\infty} O(S) p^{c_2 cn} \left(\sigma p^{c_3}\right)^x \leq O(S) p^{c_2 cn}$$

Combing the bounds we have, that there is constant c_4 depending only on d, d' such:

$$\mathbf{Pr}_{\sim \mathcal{N}_\mathcal{E}}\left[B_\tau\right] \leq O(S) p^{c_4 n}$$

Now, by the union bound over the time steps, with a probability greater than:

$$1 - O\left(S^2\right) p^{c_4 n}$$

No step admits a slice of size at least cn in its corresponding analysis of Toom's contour, so the barrier argument holds; in other words, the propagated errors do not accumulate into a logical error. □

The lemma gives the theorem almost immediately. To complete the proof, we need to plug in the initialization phase, which initializes encoded zeros and resource states subjected to local stochastic noise. We emphasize that the resulting states are subjected to local stochastic noise (in addition to the computation). We neither prove nor describe that phase here; instead, we refer to [Gro19], [Got14], [FGL18], and [NP25].

Theorem 6.3. *There exists a constant $p_0 \in (0,1)$ such that for any local stochastic noise channel $\mathcal{N}_\mathcal{E}$ with rate $p < p_0$, and a quantum circuit C with depth μ and width w, there exists a Toric encoded circuit $\tilde{C}$, with depth $\Theta(\mu n^8 + \ Polylog(n))$ and width $\Theta((w+\mu)n^8 \ Polylog(n))$ that simulates C with probability:*

$$1 - O\left(poly(w, \mu, n)\right) p^{\Omega(n)}$$

and in addition:

1. *After a short initialization stage (at depth $\Theta(poly(\log n))$), each of the qubits is encoded in the n-length 4D Toric code.*
2. *An error correction cycle takes $O(1)$ depth, and the gates in each correction cycle don't depend on time. In particular, if C is the identity circuit, then after the initialization stage, the layers in $\tilde{C}$ are the same.*

7 The Sweep-rule is Itself a Decoder

The role of the sweep-rule in the fault tolerance construction is to restrain the errors in the system rather than erase them. Here, we show that it can also be used as a decoder. Concurrently, if the initial error is distributed according to local stochastic noise, then the excitation of noiseless sweep-cycles for $2dn^{2d}$ times - the number of vertices times the side's length - erases the errors.

To prove it, we first need to introduce the notation of two-translations, which refer to a pair of vertices on the ∞-lift that share the same source on the finite Toric.

Definition 7.1. *Consider a Toom's contour $\mathcal{H}$. We say that vertices $u, v \in V_{\mathcal{H}}$ are two-translations if there is a vertex v' on the finite Toric, such that both v and u are translations of v'.*

Claim 7.2. *Let t_0 be a time, such there are two translations $\tilde{v} = (v, t_0, \{\cdot\})$, $\tilde{u} = (u, t_0, \{\cdot\})$ and $\tilde{v}, \tilde{u}$ are connected via arrows. Then, there is a slice in $\mathcal{H}$ at size greater than n.*

Proof. Denote by K the slice for which $\tilde{v}, \tilde{u} \in H_K$. Observe that we can assume that $T(K) - t_0$ is a non-negative integer, since if it is, then $K \in H_{\tilde{u}}, H_{\tilde{v}}$ and therefore $H_{\tilde{u}} = H_{\tilde{v}}$, namely $\tilde{u}$ and $\tilde{v}$ are contained in the same slice. Thus, we prove by induction on $T(K) - t_0$.

1. Base. If $t_0 = T(K)$ then $\tilde{v}, \tilde{u} \in K$ so clearly $\mathbf{Size}(K) \geq n$.

2. Step. We argue that at time $t_0 + 1$ there must be two translations that belong to H_K. Let u' be a vertex, such that $\tilde{u} \in \mathrm{pre}(u')$. Notice that if there is no such vertex, then $\tilde{u}$ does not appear in the history of any vertex with a time coordinate greater than t_0, in contradiction to $\tilde{u} \in H_K$. Since the vertices of $\mathcal{H}$ are sets defined by the ∂^--boundary of a Pauli-path, and a Pauli-path is an ∞-lift of propagation, we have that there has to be a vertex v' in $\mathcal{H}$ which is a translation of u'. If $\{\tilde{u}, u'\} \in E_{\mathcal{H}}$, then we are finished. Else, by the arrows connection rules (Definition 5.17), it has to be that $v' \in \mathcal{E}^\infty$, but then it has to be that u' is also in $\mathcal{E}^\infty$, which is a contradiction to $\{\tilde{u}, u'\}$ being an arrow.

□

The following claim states that an ideal run (not subject to noise) of $2dn^{2d}$ sweep-cycles, with the additional assumption that no two translations are connected by the arrows, succeeds in erasing the syndrome.

Claim 7.3. *Let $|\psi\rangle$ be a logical codeword of the (d, d')-Toric code with side length n, and let $P \in \mathcal{P}^{(\cdot)}$. Let C be the quantum circuit that takes $P|\psi\rangle$ as an input and performs over it $2dn^{2d}$ sweep-cycles. For a possible measurement outcome $\alpha \in \mathbb{F}_2^{(\cdot)}$ (of stabilizer measurements), consider the Toom's contour $\mathcal{H}$ induced by $C[, \alpha]$. Assuming that any two vertices u, w, which are two different translations of the same vertex on the finite Toric (a vertex in $\mathcal{G}_n^d$), are not connected by arrows in $\mathcal{H}$. Then, at the end of the running of $C[, \alpha]$, the syndrome is erased.*

Proof. We will prove that the decoder erases the ∂^--boundary corresponding to the X-component of P, and the proof for the Z-component is identical. Let P^∞ be the ∞-lift of P and let $v_1, v_2, \ldots$ be the vertices supporting the faces of $\partial^- x(P^\infty)$. During the ideal computation, no new faults are introduced, and therefore $v_1, v_2 \ldots$ are the only leaves in the induced Toom's contour. Without loss of generality, assume that the first m vertices $v_1, .., v_m$ align with the vertices of the finite Toric (namely non-translated vertices), denote them by V'.

They set a natural relation over its slices as follows: two slices K_1 and K_2, not necessarily with the same time coordinate, will be said to be *history-related*, $K_1 \sim K_2$,

if there are leaves u, w such that both u and w are translations of a vertex $v_i \in V'$ and u, w belong to the histories of K_1, K_2 respectively. Thus, for a time t and a leaf $v \in V'$, we can define the notation $K_{v,t}$ to be the slice at time t that is reached from leaf v by arrows. We emphasize that $\sim$ is not an equivalence relation, since it doesn't satisfy transitivity. Yet, let us match slices to 'classes' when the map is many-to-many. Here, a class is a grouping of translations of slices which are all history-related. For a vertex $v \in V'$, the delegate of the slices reached from vertex v and its translations, namely the translations of slices $\{K_{v,t} : t\}$, will be denoted by $\mathbf{K}_v$. Proving that at the end of time $2dn^{2d}$, for any leaf v, the slice $K_{v,t}$ is empty, or equivalently is of size zero, and therefore its translations, implies the correctness of the claim.

For proving this, let us present the following terminology. For leaves v, u, w, and a time coordinate t, consider the following:

1. Suppose that $K_{v,t}$ and $K_{u,t+1}$ are slices at times t and $t+1$, such $K_{v,t} \sim K_{u,t+1}$. If $\mathbf{Size}(K_{u,t+1}) > \mathbf{Size}(K_{v,t})$, we will say that the class $\mathbf{K}_v$ grows at time t. Similarly, if $\mathbf{Size}(K_{u,t+1}) < \mathbf{Size}(K_{v,t})$, then we will say that class $\mathbf{K}_v$ shrank at time t.

2. Suppose that $K_{v,t}$ and $K_{u,t}$ are slices at time t, and $K_{w,t+1}$ is a slice at time $t+1$, such that $K_{v,t} \not\sim K_{u,t}$, $K_{w,t+1} \sim K_{v,t}$, and $K_{w,t+1} \sim K_{u,t}$. Then we will say that $\mathbf{K}_v$ and $\mathbf{K}_u$ merged at time t.

3. The size of a class $\mathbf{K}_v$ at time t is $\mathbf{Size}(K_{v,t})$.

Since no new faults are introduced during the computation, we have that in each step, the only option for a class to grow is by merging. Notice that, by definition, classes cannot 'split'. Namely, if $K_{v,t} \sim \mathbf{K}_u$, then for any time $t' \geq t$, it holds that $K_{v,t'} \sim \mathbf{K}_u$. Specifically, a correction by the sweep rule that disconnects two connected components of the syndrome that were connected, prior to correction, belonged to the same slice and therefore to the same class, would still belong to the same class after the correction.

Therefore, there can be at most $m-1$ merges. And since the number of classes is bounded by the number of vertices on the Toric, we have that $m \leq n^d$, namely at most $n^d - 1$ merges can occur. Hence, during the $2dn^{2d}$ sweep-cycles, there has to be a continuous sequence of $2dn^d$ cycles for which no merge occurs.

Moreover, we argue that the maximum size of a slice is bounded by $2dn^d$. For this, for a time t we define $V'_t \subset V'$, to be the maximal subset such for any $v, u \in V'_t$ it holds $K_{v,t} \not\sim K_{u,t}$. Then, we define the following potential function:

$$\Phi_t = \sum_{v \in V'_t} (\mathbf{Size}(K_{v,t}) + 1)$$

Observe that $\Phi_1 = m \leq n^d$. Now, for a vertex $v \in V'_{t+1}$, denote the predecessor of $K_{v,t+1}$ by $K_{u'_0,t}, K_{u'_1,t}, \ldots, K_{u'_{k_v},t}$ and their translation into the finite Toric by $K_{u_0,t}, K_{u_1,t}, \ldots, K_{u_{k_v},t}$, so $u_i \in V'_t$. Using Corollary 5.29 and Claim 5.27, where the

poles for $K_{v,t+1}$ are selected to be the extremis of $L_1, .., L_{d+1}$, we have:

$$\begin{aligned}\Phi_{t+1} &= \sum_{v\in V'_{t+1}} \big(\mathbf{Size}(K_{v,t+1})+1\big) \le \sum_{v\in V'_{t+1}} \big(\big(\sum_{i=0}^{k_v} \mathbf{Size}(K_{u'_i,t})\big) + 2dk_v + 1\big) \\ &= \sum_{v\in V'_{t+1}} \big(\big(\sum_{i=0}^{k_v} \mathbf{Size}(K_{u_i,t})\big) + (2d-1)k_v + 1 + k_v\big) \\ &= \sum_{v\in V'_{t+1}} \big(\big(\sum_{i=0}^{k_v} \mathbf{Size}(K_{u_i,t}) + 1\big) + (2d-1)k_v\big)\end{aligned}$$

Where, in the last transition, we used the fact that the size function is invariant to translations. Now, any vertex of V'_t has to appear at most once. To see this, assume towards contradiction the existence of two predecessor slices $K_{u'_i,t}$, $K_{w'_j,t}$ which are translations of the same slice. Denote by $K_{u,t+1}$ and $K_{w,t+1}$ the slices whose $K_{u'_i,t}$, $K_{w'_j,t}$ are their predecessors. Let's split to cases, whether $u = w$.

- If $u \neq w$, then u'_i, w'_j belong to the histories of different slices $K_{u,t+1}, K_{w,t+1}$, and they are a translation of the same vertex in $V'_t \subset V'$. Hence, $K_{u,t+1} \sim K_{w,t+1}$, in contradiction to $u, w \in V'_{t+1}$.
- Otherwise, if $u = w$, then we found two vertices u'_i and w'_j, which are translations of the same origin vertex of the finite Toric, such that they belong to the same history $H_{K_{w,t+1}}$, namely connected by an arrow, in contradiction to the given of.

Now, observe that each fork connecting two slices is associated with a merge; therefore, $\sum_t \sum_{v\in V'_t} k_v$ is lower than the number of merges, and the bound on the potential function becomes:

$$\begin{aligned}\Phi_{t+1} &\le \sum_{v\in V'_t} \mathbf{Size}(K_{v,t}) + 1 + \sum_{v\in V'_{t+1}} (2d-1)k_v \\ &\le \Phi_t + (2d-1)k = \Phi_1 + (2d-1)\cdot \ \#\text{ merges } \ \le 2dn^d\end{aligned}$$

Thus, since $\mathbf{Size}(K_{v,t}) \le \Phi_t$, we have that during the computation, no slice reaches size $2dn^d$. During the continuous sequence of $2dn^d$ cycles, each sweep cycle decreases the size of any slice by at least 1. This is followed by Corollary 5.29, combining that no merging occurred, namely there is only one predecessor slice. Hence, at the end of that sequence of $2dn^d$ cycles, all the slices' sizes shrink to zero; namely, the connected components are erased after those cycles. □

Claim 7.3 guarantees only that the syndrome is erased; in particular, it says nothing about whether a logical error has or hasn't occurred. To complete the correctness of the decoder, we would like to use the barrier argument. Yet, instead of showing that during the ideal correction no connected component of the syndrome passes the barrier, it will be more convenient to reduce the passing question to the subjected-to-noise setting. We introduce a replacement argument, stating that the probability of an event

occurring during a computation where part of its layers are non-noisy is lower than the probability of having the same event in a computation where all its layers are noisy. More precisely, any bound that assumes only that the noise channel is local stochastic captures both cases.

Claim 7.4 (Replacement argument)**.** *Let C, $|\psi\rangle$, and P be the quantum circuit, the quantum state, and a Pauli as defined in Claim 7.3, with the change that P is distributed according to local stochastic noise $\mathcal{N}_\mathcal{E}$ with an error rate $p \in (0,1)$ rather than being fixed.*

*Denote by A the event in which, at the end of the running of C, namely after the $2dn^{2d}$ **ideal** (non-noisy) iterations of the sweep-cycle, the error is not erased, and denote by B the event that, during the application of a subjection of C to $\mathcal{N}_\mathcal{E}$, either the Toom's contour admits a slice of size at least cn, or two translated vertices become arrow-connected.*

Let $f : (n, d, d', p) \to \mathbb{R}$ be an upper bound on $\mathbf{Pr}\big[B\big]$. Then, we have:

$$\mathbf{Pr}\big[A\big] \leq f(n, d, d', p) \tag{4}$$

Proof. A corollary of combining Claim 7.3 with Corollary 5.11 (The barrier argument) is that if during the $2dn^{2d}$ ideal sweep-cycles, there is no step for which its propagation admits a slice of size at least cn (and therefore no step admits a syndrome's connected component that passes the barrier), and no two translated vertices become arrow-connected, then the syndrome is erased and no logical error occurred. So it's sufficient to show that the probability that at least one of those conditions does not hold is lower than $f(n, d, d', p)$. To do that, we show that there is a local stochastic noise channel $\mathcal{N}'_\mathcal{E}$ with an error rate p such that $C[\mathcal{E}] = C$ where $\mathcal{E} \sim \mathcal{N}'_\mathcal{E}$. We do that by defining $\mathcal{N}'_\mathcal{E}$ to be the identity channel. Clearly, it's satisfying the local stochastic noise inequality Definition 3.7, and therefore Equation (4) has to holds. □

We now prove that the sweep rule is itself a decoder.

Theorem 7.5. *There exists a constant $p_0 \in (0,1)$ such that for any local stochastic noise channel $\mathcal{N}_\mathcal{E}$ with rate $p < p_0$, and a quantum state $\rho = \mathcal{N}_\mathcal{E}(|\psi\rangle)$, where $|\psi\rangle$ is a codeword of the (d, d')-Toric code, the application of $2dn^{2d}$ (ideal) sweep-cycles on ρ results in $|\psi\rangle$ with probability at least:*

$$1 - O\left(poly(n)\right) p^{\Omega(n)}$$

Proof. By the replacement argument, it is sufficient to consider the application of $2dn^{2d}$ sweep-cycles subjected to $\mathcal{N}_\mathcal{E}$, and bound the probability that during the subjected application there is a step τ and an analysis of Toom's counter $\mathcal{H}_\tau$ such that either $\mathcal{H}_\tau$ contains a slice of size at least cn, or $\mathcal{H}_\tau$ contains two translated vertices connected by arrows. Claim 7.2 states that the second event is contained in the event that there is a slice of size at least n, and therefore contained in the first event. Denote by B_τ the event in which step τ admits a slice of size at least cn. The proof of Lemma 6.2 bounds the probability of B_τ by the location's number multiple $p^{\Omega(n)}$ for $p < p_0$. Taking the union bound over the steps gives the desired result. □

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8 Appendix

8.1 Counting Subtrees.

Claim 8.1. *Let $G = (V, E)$ be a $\Delta+1$-regular graph, and let k be an integer less than $|E|$. Fix a vertex $r \in V$. There is a constant number α, depending only on Δ, such that the number of subgraphs rooted at r with fewer than k edges is at most α^k.*

Proof. Denote by T_Δ and T_2 the infinity Δ and binary tress. There is an injection between the subtrees of G, rooted at v, with at most k edges, to the subtrees of T_Δ rooted at v, with at most k edges. Moreover, there is an injection from the subtrees of T_Δ rooted at v, with at most k edges to the subtrees of T_2, rooted at v, with at most $k \log \Delta$ edges. The injection is obtained by replacing each of the vertex with a (complete) binary tree over $\Delta - 1$-leaves, each leaf is associated with a outgoing edge of the original vertex.

Thus, it's enough to bound the number of of subtrees in T_2 rooted at v, with at most $k \log \Delta$ edges, that number is less than the number of binary tress with $k \log \Delta + 1$ leaves. For exact leaves number we get the $k \log \Delta$ Catalan number, which its limit is:

$$\sim \frac{4^{k \log \Delta}}{(k \log \Delta)^{\frac{3}{2}} \sqrt{\pi}} \leq \frac{\Delta^{2k}}{(k \log \Delta)^{\frac{3}{2}} \sqrt{\pi}}$$

Therefore we can bound by:

$$\sum_{j=1}^{k} \frac{\Delta^{2j}}{(j \log \Delta)^{\frac{3}{2}} \sqrt{\pi}} \leq \frac{\Delta^{2k}}{\sqrt{\pi} \log^{\frac{3}{2}} \Delta} \sum_{j=1}^{k} \frac{1}{j^{\frac{3}{2}}} \leq \zeta(\frac{3}{2}) \cdot \frac{\Delta^{2k}}{\sqrt{\pi} \log^{\frac{3}{2}} \Delta}$$

□

8.2 Tree Separator Lemma

For a tree $T = (V, E)$ and a vertex $v \in V$, we call the subtree obtained by taking the descendants of v the subtree rooted at v.

Claim 8.2. *Let $T = (V, E)$ be a tree with $m > 3$ leaves and τ edges, then it has a subtree with $m' \in (\frac{1}{3}m, \frac{2}{3}m)$ leaves and at most $\frac{m'}{m}\tau$ edges.*

Proof. Order the vertices according to their heights. Denote by $v \in V$ the lowest vertex for which the subtree rooted at v has at least $\frac{1}{3}m$ leaves. Now, denote by T_1 the subtree obtained through the following iterative process over v's children:

1. Initialize T_1 as the subtree rooted at v.
2. For a child u of v:
   (a) Check if erasing from T_1 the subtree rooted at u doesn't decrease the number of leaves below $\frac{1}{3}m$. If so, then we update T_1 to be the result of the erasure.

Clearly, the algorithm is defined such that T_1 has to have at least $\frac{m}{3}$ leaves. Furthermore, since any child of v is a root of a subtree with fewer than $\frac{m}{3}$ leaves (otherwise we get a contradiction to the minimality of v's height), we have that if T_1 would have more than $\frac{2}{3}m$ leaves, then the subtraction of any of its children would yield a tree with at least $\frac{2}{3}m - \frac{1}{3}m = \frac{1}{3}m$ leaves. Namely, v has in T_1 a child u such that the subtraction of the subtree rooted at u from T_1 results in a subtree with more than $\frac{1}{3}m$ leaves. But if indeed there were such a child, then the algorithm would erase it, so there is not.

Now let $T_2 = T/T_1 \cup v$, namly the tree induced by the subtraction of T_1's edges from T. Observes that since the summation of leaves in T_1 and T_2 is the leaves in T, and T_1 has no more than $\frac{2}{3}m$ leaves, and no less than $\frac{1}{3}$ leaves, it holds that the number of leaves in T_2 is also in the range $(\frac{1}{3}m, \frac{2}{3}m)$. We claim that at least one of the subtrees has fewer than $\frac{m'}{m}\tau$ edges.

To see this, denote by m_1, m_2 and by τ_1, τ_2 the leaves and the edges in T_1, T_2 respectively. Now:

$$\frac{\tau}{m} = \frac{\tau_1 + \tau_2}{m_1 + m_2} = \frac{m_1}{m_1 + m_2}\frac{\tau_1}{m_1} + \frac{m_2}{m_1 + m_2}\frac{\tau_2}{m_2}$$

Namely, $\frac{\tau}{m}$ is a weighted average of $\frac{\tau_1}{m_1}$ and $\frac{\tau_2}{m_2}$, and therefore it is greater than their minimum. Peak the subtree with the minimal quotient. □

By repeating recursively on the above construction, we get the following corollary:

Corollary 8.3. *Let $T = (V, E)$ be a tree with m leaves and τ edges. Then, for any $\alpha > 0$, it has a subtree with $m' \in \left(\frac{1}{3}\alpha m, \frac{2}{3}\alpha m\right)$ leaves and at most $\frac{m'}{m}\tau$ edges.*

Claim 8.4. *Let $\gamma, \beta \in \mathbb{R}_{>0}$ and n be a positive integer. For any tree T with $\tau \geq \beta n$ edges and m leaves such that $\tau \leq \gamma m$, there is a subtree T' of T with a number of edges τ' less than $\frac{\beta}{2}n$ and a number of leaves greater than $\frac{\beta}{4\gamma}n$.*

Proof. Since the construction of claim 8.2 preserves the ratio of edges to leaves, we have that for a T' obtained using the construction, the number of edges satisfies $\tau' \leq \gamma m'$. So, it's enough to look for the maximal α such that:

$$\gamma m' \leq \gamma \frac{2}{3}\alpha m \leq \frac{\beta}{2}n \Rightarrow \alpha = \frac{3\beta}{4\gamma}\frac{n}{m}$$

Which gives $m' \geq \frac{1}{3}\alpha m = \frac{\beta}{4\gamma}n$.

□